

EC200A Series

Hardware Design

LTE Standard Module Series

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Safety Information

The following safety precautions must be observed during all phases of operation, such as usage, service or repair of any terminal or mobile incorporating the module. Manufacturers of the terminal should notify users and operating personnel of the following safety information by incorporating these guidelines into all manuals of the product. Otherwise, Quectel assumes no liability for customers' failure to comply with these precautions.



Full attention must be paid to driving at all times in order to reduce the risk of an accident. Using a mobile while driving (even with a handsfree kit) causes distraction and can lead to an accident. Please comply with laws and regulations restricting the use of wireless devices while driving.



Switch off the terminal or mobile before boarding an aircraft. The operation of wireless appliances in an aircraft is forbidden to prevent interference with communication systems. If there is an Airplane Mode, it should be enabled prior to boarding an aircraft. Please consult the airline staff for more restrictions on the use of wireless devices on an aircraft.



Wireless devices may cause interference on sensitive medical equipment, so please be aware of the restrictions on the use of wireless devices when in hospitals, clinics or other healthcare facilities.



Terminals or mobiles operating over radio signal and cellular network cannot be guaranteed to connect in certain conditions, such as when the mobile bill is unpaid or the (U)SIM card is invalid. When emergency help is needed in such conditions, use emergency call if the device supports it. In order to make or receive a call, the terminal or mobile must be switched on in a service area with adequate cellular signal strength. In an emergency, the device with emergency call function cannot be used as the only contact method considering network connection cannot be guaranteed under all circumstances.



The terminal or mobile contains a transceiver. When it is ON, it receives and transmits radio frequency signals. RF interference can occur if it is used close to TV sets, radios, computers or other electric equipment.



In locations with explosive or potentially explosive atmospheres, obey all posted signs and turn off wireless devices such as mobile phone or other terminals. Areas with explosive or potentially explosive atmospheres include fueling areas, below decks on boats, fuel or chemical transfer or storage facilities, and areas where the air contains chemicals or particles such as grain, dust or metal powders.

About the Document

Revision History

Version	Date	Author	Description
-	2022-01-05	Anthony LIU/ Kexiang ZHANG	Creation of the document
1.0	2022-01-11	Anthony LIU/ Kexiang ZHANG	First official release
1.1	2022-12-01	Stephen HE/ Kexiang ZHANG	- Added the applicable module EC200A-EL and related information. - Updated the information about USB serial drivers (Chapter 2.2). - Opened WLAN interface (Figures 1, 2, Tables 4, 6 and Chapter 4.13). - Updated the pin 3 from RESERVED to SLEEP_IND and added related description (Figure 2, Tables 6 and 24). - Added the DC characteristics of the pins in RGMII/RMII (Table 6). - Updated the comment of RGMII/RMII_RST_N pins (Tables 6 and 23). - Updated the VBAT_RF current capability requirement from 1.8 A to 2.0 A (Tables 6, 8 and 39). - Added the USIM_GND pin and related description (Tables 6 and 13). - Updated the conditions for enabling the module into sleep mode in the UART application scenario (Chapter 3.2.1). - Updated current requirements for power supply capabilities (Chapter 3.4.2). - Updated the time difference between powering up VBAT and pulling down PWRKEY (Chapter

			3.5).
			12. Updated the turn-off timing diagram and added related information (Chapter 3.6.1).
			13. Updated the data of power consumption (Chapter 6.3).
			14. Updated the data of digital I/O characteristic (Chapter 6.4).
			15. Updated the recommended thermal profile parameters (Figure 45 and Table 52).
			1. Added the GNSS function and related information.
			2. Added frequency range supported by B41 (Chapters 2.1 and 5.1.1).
			3. Updated pin 43 from RESERVED to GNSS_1PPS; Updated pins 143 and 144 from RESERVED to GRFC1 and GRFC2 respectively.
			4. Added a note on avoiding abnormal RF functions caused by current sink on the module's pins (Chapter 2.4).
			5. Updated the DC characteristics of SD_SDIO__VDD (Table 6).
			6. Added a footnote for VBAT_BB (Table 6 and Table 8).
			7. Added notes on modules' pins (Chapter 2.5).
1.2	2024-01-29	Shiye ZHU/ John QIN/ Jerry LIN	8. Updated the power-up timing of turning on the module with PWRKEY (Figure 13).
			9. Updated the trace width of VBAT_BB (Chapter 3.4.3).
			10. Updated the logout time from network in the timing of turning off the module by PWRKEY (Chapter 3.6.1).
			11. Updated the figure of timing sequence for entering forced download mode (Figure 20).
			12. Updated MICBIAS pin as POUT channel (Chapter 4.6).
			13. Added a note on adding UART hardware flow control design (Chapter 4.7).
			14. Added related description of adding a resistor in series on SD_SDIO_CLK in order to improve signal quality (Chapter 4.8).
			15. Updated the value of the voltage dividing resistor used in the ADC interface (Chapter 4.9).
			16. Updated the length difference of

			RGMII/RMII_TX_[0:1], RGMII_TX_[2:3], RGMII/RMII_CTL_TX, RGMII_CK_TX, and length difference of RGMII/RMII_RX_[0:1], RGMII_RX_[2:3], RGMII/RMII_CTL_RX, RGMII/RMII_CLK and added the length difference between signal lines when using the RMII function; Added a note on pin 119 (Chapter 4.10). 17. Added the principle that should be complied with when designing the reserved pull-up resistor circuit for the WLAN SDIO interface (Chapter 4.13). 18. Updated the operating frequency range of B1, B5, B8 and B41 of EC200A-CN; Updated the operating frequency range of B1, B5 and B8 of EC200A-EU/EC200A-EL; Updated the operating frequency range of B1, B2, B4, B5 and B8 of EC200A-AU (Chapter 5.1.1). 19. Updated the conducted RF receiving sensitivity (Chapter 5.1.4). 20. Updated the power consumption (Chapter 6.3). 21. Updated recommended stencil thickness; Added a note on avoiding using materials that contain mercury (Hg) (Chapter 8.2).
1.3	2025-05-12	Shiye ZHU/ Leon LIANG/ Jerry LIN	1. Added AGNSS description and AGNSS parameters (Table 4 and 41). 2. Added a note about SD_DET pin (Table 6 and 20). 3. Updated the recommended time interval between powering up VBAT and pulling down PWRKEY pin low to not less than 700 ms (Chapters 3.5 and 4.2). 4. Added a note about PWRKEY and USB_BOOT power-on timing diagram applying to soft versions released after 2023 (Chapter 3.5 and 4.2). 5. Added a new chapter on SLEEP_IND (Chapter 4.11.3). 6. Updated reference circuit for RF antenna interfaces and added relevant note (Chapter 5.1.5). 7. Added a note about the impedance condition for power consumption test (Chapter 6.3).

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8. Updated reference circuit of GNSS active antenna and added relevant note (Chapter 5.2.3).
 9. Updated the bottom dimension of the module (Figure 45).
 10. Updated the module coplanarity requirement (Chapter 7.1).
 11. Updated the pre-baking time to 24 h (Chapter 8.1).
 12. Updated the reference document for recommended module stencil thickness;
Added a note prohibiting storage or use of unprotected modules in environments containing corrosive gases (Chapter 8.2).
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Contents

Safety Information.....	3
About the Document.....	4
Contents	8
Table Index.....	11
Figure Index	13
1 Introduction	15
1.1. Special Marks.....	15
2 Product Overview	16
2.1. Frequency Bands and Functions	16
2.2. Key Features	17
2.3. Functional Diagram	20
2.4. Pin Assignment	21
2.5. Pin Description	22
2.6. EVB Kit	30
3 Operating Characteristics	31
3.1. Operating Modes.....	31
3.2. Sleep Mode	32
3.2.1. UART Application.....	32
3.2.2. USB Application with USB Remote Wakeup Function	33
3.2.3. USB Application Without USB Remote Wakeup Function	34
3.2.4. USB Application Without USB Suspend Function	34
3.3. Airplane Mode	35
3.4. Power Supply	36
3.4.1. Power Supply Pins	36
3.4.2. Reference Design for Power Supply.....	36
3.4.3. Requirements for Voltage Stability	37
3.5. Turn-on	38
3.5.1. Turn-on with PWRKEY.....	38
3.6. Turn-off.....	41
3.6.1. Turn-off with PWRKEY.....	41
3.6.2. Turn-off with AT Command.....	41
3.7. RESET_N.....	42
4 Application Interfaces	44
4.1. USB Interface	44
4.2. USB_BOOT	45
4.3. (U)SIM Interface	47
4.4. PCM and I2C Interfaces.....	49
4.5. SPI Interface.....	51
4.6. Analog Audio Interfaces	52
4.6.1. Audio Interface Design Considerations	52

4.6.2.	Microphone Interface Circuit	53
4.6.3.	Earpiece Interface Circuit.....	53
4.7.	UARTs	54
4.8.	SD Card Interface.....	56
4.9.	ADC Interfaces	58
4.10.	RGMII*/RMII Interface	59
4.11.	Indication Signals	61
4.11.1.	Network Status Indication	62
4.11.2.	STATUS	62
4.11.3.	SLEEP_IND	63
4.12.	MAIN_RI.....	64
4.13.	WLAN SDIO Interface	64
5	RF Specifications	66
5.1.	Cellular Network	66
5.1.1.	Antenna Interface & Frequency Bands.....	66
5.1.2.	Antenna Tuner Control Interfaces.....	69
5.1.3.	Transmitting Power	70
5.1.4.	Receiver Sensitivity.....	71
5.1.5.	Reference Design	74
5.2.	GNSS	75
5.2.1.	Antenna Interface & Frequency Bands.....	75
5.2.2.	GNSS Performance	76
5.2.3.	Reference Design	77
5.3.	RF Routing Guidelines	78
5.4.	Requirements for Antenna Design.....	80
5.5.	RF Connector Recommendation	81
6	Electrical Characteristics and Reliability	83
6.1.	Absolute Maximum Ratings	83
6.2.	Power Supply Ratings.....	84
6.3.	Power Consumption.....	84
6.4.	Digital I/O Characteristics.....	97
6.5.	ESD Protection	98
6.6.	Operating and Storage Temperatures	99
7	Mechanical Information.....	100
7.1.	Mechanical Dimensions	100
7.2.	Recommended Footprint.....	102
7.3.	Top and Bottom Views	103
8	Storage, Manufacturing and Packaging	104
8.1.	Storage Conditions.....	104
8.2.	Manufacturing and Soldering	105
8.3.	Packaging Specification	107
8.3.1.	Carrier Tape	107
8.3.2.	Plastic Reel	108

8.3.3.	Mounting Direction	108
8.3.4.	Packaging Process	109
9	Appendix References	110

Table Index

Table 1: Special Marks.....	15
Table 2: Brief Introduction of the Module.....	16
Table 3: Wireless Network Types	16
Table 4: Key Features.....	17
Table 5: Parameter Definition	22
Table 6: Pin Description	22
Table 7: Overview of Operating Modes	31
Table 8: Pin Definition of Power Supply.....	36
Table 9: Pin Definition of PWRKEY	38
Table 10: Pin Definition of RESET_N	42
Table 11: Pin Definition of USB Interface	44
Table 12: Pin Definition of USB_BOOT	46
Table 13: Pin Definition of (U)SIM Interface	47
Table 14: Pin Definition of PCM Interface.....	50
Table 15: Pin Definition of I2C Interface	50
Table 16: Pin Definition of SPI Interface	51
Table 17: Pin Definition of Analog Audio Interfaces	52
Table 18: Pin Definition of Main UART	54
Table 19: Pin Definition of Debug UART	54
Table 20: Pin Definition of SD Card Interface.....	56
Table 21: Pin Definition of ADC Interface	58
Table 22: Characteristics of ADC Interfaces.....	58
Table 23: Pin Definition of RGMII/RMII Interface.....	59
Table 24: Pin Definition of Indication Signals	61
Table 25: Network Status Indication Pin Level and Module Network Status	62
Table 26: MAIN_RI.....	64
Table 27: Pin Definition of WLAN Interface	64
Table 28: Pin Definition of Cellular Network Interface	66
Table 29: Operating Frequency of EC200A-CN	66
Table 30: Operating Frequency of EC200A-AU	67
Table 31: Operating Frequency of EC200A-EU	68
Table 32: Operating Frequency of EC200A-EL	69
Table 33: Pin Definition of GRFC Interfaces.....	69
Table 34: Transmitting Power	70
Table 35: Conducted RF Receiver Sensitivity of EC200A-CN (Unit: dBm).....	71
Table 36: Conducted RF Receiver Sensitivity of EC200A-AU (Unit: dBm).....	72
Table 37: Conducted RF Receiver Sensitivity of EC200A-EU (Unit: dBm).....	73
Table 38: Conducted RF Receiver Sensitivity of EC200A-EL (Unit: dBm)	73
Table 39: Pin Definition of GNSS Antenna Interface	75
Table 40: GNSS Frequency (Unit: MHz).....	76
Table 41: GNSS Frequency (Unit: MHz).....	76

Table 42: Requirements for Antenna Design.....	80
Table 43: Absolute Maximum Ratings	83
Table 44: Power Supply Ratings.....	84
Table 45: EC200A-CN Power Consumption.....	84
Table 46: EC200A-AU Power Consumption	87
Table 47: EC200A-EU Power Consumption	91
Table 48: EC200A-EL Power Consumption	95
Table 49: VDD_EXT I/O Requirements	97
Table 50: (U)SIM High/Low-voltage I/O Requirements	97
Table 51: SDIO Low-voltage I/O Requirements	97
Table 52: SDIO High-voltage I/O Requirements.....	98
Table 53: Electrostatic Discharge Characteristics (Temperature: 25–30 °C, Humidity: 40 ±5 %).....	98
Table 54: Operating and Storage Temperatures	99
Table 55: Recommended Thermal Profile Parameters	106
Table 56: Carrier Tape Dimension Table (Unit: mm).....	107
Table 57: Plastic Reel Dimension Table (Unit: mm)	108
Table 58: Related Documents.....	110
Table 59: Terms and Abbreviations	110

Figure Index

Figure 1: Functional Diagram.....	20
Figure 2: Pin Assignment (Top View)	21
Figure 3: Module Power Consumption in Sleep Mode	32
Figure 4: Sleep Mode Application via UART Interface	33
Figure 5: Sleep Mode Application with USB Remote Wakeup.....	33
Figure 6: Sleep Mode Application with MAIN_RI.....	34
Figure 7: Sleep Mode Application Without Suspend Function	35
Figure 8: Reference Design of Power Supply.....	37
Figure 9: Power Supply Limits During Burst Transmission	37
Figure 10: Star Structure of the Power Supply	38
Figure 11: Reference Circuit of Turn-on with Driving Circuit	39
Figure 12: Reference Circuit of Turn-on with Button	39
Figure 13: Turn-on Timing.....	40
Figure 14: Timing of Turn-off with PWRKEY	41
Figure 15: Reference Circuit of RESET_N with Driving Circuit	42
Figure 16: Reference Circuit of RESET_N with Button	43
Figure 17: Timing of Resetting the Module	43
Figure 18: Reference Circuit of USB Interface	45
Figure 19: Reference Circuit of USB_BOOT	46
Figure 20: Timing Sequence for Entering Forced Download Mode	46
Figure 21: Reference Circuit of (U)SIM Interface with an 8-pin (U)SIM Card Connector	48
Figure 22: Reference Circuit of (U)SIM Interface with a 6-pin (U)SIM Card Connector	48
Figure 23: Timing Sequence for Short Frame Mode	50
Figure 24: Reference Design of PCM and I2C Interfaces	51
Figure 25: Microphone Interface Reference Circuit.....	53
Figure 26: Earpiece Interface Reference Circuit.....	53
Figure 27: Reference Design of UART Interface with a Voltage-level Translator (Main UART).....	55
Figure 28: Reference Design of UART Interface with Transistor Circuit (Main UART).....	55
Figure 29: Reference Circuit of SD Card Interface	57
Figure 30: Reference Circuit of RMII to PHY Interface.....	60
Figure 31: Reference Circuit of RGMII to PHY Interface.....	60
Figure 32: Reference Circuit of the Network Status Indication.....	62
Figure 33: Reference Circuits of STATUS.....	63
Figure 34: Reference Design of SLEEP_IND	63
Figure 35: Reference Circuit for RF Antenna Interfaces	74
Figure 36: Reference Circuit of GNSS Active Antenna	77
Figure 37: Microstrip Design on a 2-layer PCB	78
Figure 38: Coplanar Waveguide Design on a 2-layer PCB	78
Figure 39: Coplanar Waveguide Design on a 4-layer PCB (Layer 3 as Reference Ground).....	79
Figure 40: Coplanar Waveguide Design on a 4-layer PCB (Layer 4 as Reference Ground).....	79
Figure 41: Dimensions of the Receptacle (Unit: mm).....	81

Figure 42: Specifications of Mated Plugs	81
Figure 43: Space Factor of Mated Connector (Unit: mm).....	82
Figure 44: Module Top and Side Dimensions.....	100
Figure 45: Bottom Dimension (Bottom View).....	101
Figure 46: Recommended Footprint	102
Figure 47: Top and Bottom Views of the Module.....	103
Figure 48: Recommended Reflow Soldering Thermal Profile	105
Figure 49: Carrier Tape Dimension Drawing (Unit: mm)	107
Figure 50: Plastic Reel Dimension Drawing	108
Figure 51: Mounting Direction	108
Figure 52: Packaging Process	109

1 Introduction

This document defines the EC200A series module and describes its air interfaces and hardware interfaces which are connected with your applications.

It can help you quickly understand interface specifications, electrical and mechanical details, as well as other related information of the module. Associated with application notes and user guides, you can use this module to design and to set up mobile applications easily.

1.1. Special Marks

Table 1: Special Marks

Mark	Definition
*	Unless otherwise specified, an asterisk (*) after a function, feature, interface, pin name, command, argument, and so on indicates that it is under development and currently not supported; and the asterisk (*) after a model indicates that the model sample is currently unavailable.
[...]	Brackets ([...]) used after a pin enclosing a range of numbers indicate all pins of the same type. For example, SDIO_DATA[0:3] refers to all four SDIO pins: SDIO_DATA0, SDIO_DATA1, SDIO_DATA2, and SDIO_DATA3.

2 Product Overview

EC200A series is an LTE/WCDMA/GSM wireless communication module with receiving diversity, which provides data connectivity on LTE-FDD, LTE-TDD, HSDPA, HSUPA, HSPA+, WCDMA, EDGE and GPRS network data connection. It also provides voice functionality for your specific applications. You can choose a dedicated type based on the region or operator. The following table shows the frequency bands of EC200A series module.

Table 2: Brief Introduction of the Module

Categories	
Packaging and pins number	80 LCC pins; 64 LGA pins
Dimensions	(29.0 ±0.15) mm × (32.0 ±0.15) mm × (2.4 ±0.2) mm
Weight	approx. 4.4 g
Wireless network functions	LTE/WCDMA/GSM
Variants	EC200A-CN, EC200A-AU, EC200A-EU, EC200A-EL

2.1. Frequency Bands and Functions

Table 3: Wireless Network Types

Wireless Network Types	EC200A-CN	EC200A-AU	EC200A-EU	EC200A-EL
LTE-FDD	B1/B3/B5/B8	B1/B2/B3/B4/B5/B7/ B8/B28/B66	B1/B3/B5/B7/ B8/B20/B28	B1/B3/B5/B7/ B8/B20/B28
LTE-TDD	B34/B38/B39/B40/B41	B40	B38/B40/B41	B38/B40/B41
WCDMA	B1/B5/B8	B1/B2/B4/B5/B8	B1/B5/B8	B1/B5/B8

GSM ¹	EGSM900/DCS1800	GSM850/EGSM900/ DCS1800/PCS1900	EGSM900/ DCS1800	-
GNSS ²	GPS/BDS	-	-	-

NOTE

B41 supports 120 MHz (2535–2655) by default for EC200A-CN, and supports 140 MHz (2535–2675 MHz) by default for EC200A-EU/-EL.

2.2. Key Features

Table 4: Key Features

Features	Details
Power Supply	- Supply voltage: 3.4–4.5 V - Typical supply voltage: 3.8 V
SMS	- Text and PDU modes - Point-to-point MO and MT - SMS cell broadcast - SMS storage: ME by default
USB Interface	- Compliant with USB 2.0 specification (slave only) - Data transmission rate can reach up to 480 Mbps - Used for AT command communication, data transmission, software debugging and firmware upgrade - Supports USB serial driver for Windows 10/11, Linux 2.6–6.7 and Android 4.x–14.x systems
(U)SIM Interface	Supports (U)SIM card: 1.8/3.0 V
Audio Features	- Supports one digital audio interface: PCM interface - Supports one analog audio input interface (MIC) and one analog audio output interface (SPK) - GSM: HR/FR/EFR/AMR/AMR-WB - WCDMA: AMR/AMR-WB - LTE: AMR/AMR-WB - Supports echo cancellation and noise suppression
PCM Interface	- Used for audio function with external codec - Supports 16-bit linear data format

¹ For EC200A-CN, GSM is optional.

² GNSS is only supported by EC200A-CN and is optional.

	● Supports short frame synchronization ● Supports master and slave* modes
I2C Interface	● Supports one I2C interface ● Complies with <i>The I2C-bus Specification</i> (100/400 kHz)
SPI Interface	● Supports one SPI interface ● Maximum clock frequency: 52 MHz ● Supports master mode
Analog audio Interfaces	● Supports one analog audio input and one analog audio output channel
UART Interfaces	Main UART interface: ● Used for AT command communication and data transmission ● Baud rate: 115200 bps by default, Max. 921600 bps ● Supports RTS and CTS hardware flow control Debug UART interface: ● Used for the output of partial logs ● Baud rate: 115200 bps
SD Card Interface	● Supports SD 3.0 protocol
ADC Interfaces	● Supports two ADC interfaces ● Voltage range: 0 V–VBAT_BB
RGMII*/RMII Interface ³	● Supports one RGMII/RMII interface ● Supports RMII: 1.8/3.3 V ● Supports RGMII: 1.8 V
Network Indication	● NET_MODE and NET_STATUS to indicate network connectivity status
WLAN Interface	● Supports SDIO interface for WLAN function
AT Commands	● Compliant with 3GPP TS 27.007, 3GPP TS 27.005 and Quectel enhanced AT commands
Rx-diversity	● Supports LTE Rx-diversity
Antenna Interfaces	● Main antenna interface (ANT_MAIN) ● Rx-diversity antenna interface (ANT_DRX) ● GNSS antenna interface (ANT_GNSS) ⁴ ● 50 Ω characteristic impedance
Transmitting Power	● GSM850: Class 4 (33 dBm ±2 dB) ● EGSM900: Class 4 (33 dBm ±2 dB) ● DCS1800: Class 1 (30 dBm ±2 dB) ● PCS1900: Class 1 (30 dBm ±2 dB) ● GSM850 (8-PSK): Class E2 (27 dBm ±3 dB) ● EGSM900 (8-PSK): Class E2 (27 dBm ±3 dB)

³ EC200A-CN and EC200A-EL do not support RGMII function, so pins 79–80 and pins 82–84 are reserved. For more information, contact Quectel Technical Support.

⁴ GNSS is only supported by EC200A-CN and is optional.

	● DCS1800 (8-PSK): Class E2 (26 dBm $\pm$3 dB) ● PCS1900 (8-PSK): Class E2 (26 dBm $\pm$3 dB) ● WCDMA: Class 3 (23 dBm $\pm$2 dB) ● LTE-FDD: Class 3 (23 dBm $\pm$2 dB) ● LTE-TDD: Class 3 (23 dBm $\pm$2 dB)
LTE Features	● Supports 3GPP Rel-9 non-CA Cat 4 FDD and TDD ● Supports 1.4/3/5/10/15/20 MHz RF bandwidth ● Supports MIMO in DL direction ● Supports uplink QPSK and 16QAM modulations ● Supports downlink QPSK, 16QAM and 64QAM modulations ● FDD: Max. 150 Mbps (DL)/50 Mbps (UL) ● TDD: Max. 130 Mbps (DL)/30 Mbps (UL)
UMTS Features	● Supports 3GPP Rel-7 HSPA+/HSDPA/HSUPA and WCDMA ● Supports uplink QPSK and 16QAM modulations ● Supports downlink QPSK, 16QAM and 64QAM modulations ● HSPA+: Max. 21 Mbps (DL) ● HSUPA: Max. 5.76 Mbps (UL) ● WCDMA: Max. 384 kbps (DL)/384 kbps (UL)
GSM Features ⁵	GPRS: ● Supports GPRS multi-slot class 12 ● Coding scheme: CS 1–4 ● Max. 85.6 kbps (DL)/85.6 kbps (UL) EDGE: ● Supports EDGE multi-slot class 12 ● Supports GMSK and 8-PSK for different MCS (Modulation and Coding Scheme) ● Downlink coding schemes: MCS 1–9 ● Uplink coding schemes: MCS 1–9 ● Max. 236.8 kbps (DL)/236.8 kbps (UL)
GNSS Features ⁴	● Supports GPS and BDS ● Complies with NMEA 0183 protocol ● Data update rate: 1 Hz ● Supports AGNSS. For more details, see document [1].
Internet Protocol Features	● Supports TCP/UDP/PPP/NTP/NITZ/FTP/HTTP/PING/CMUX/HTTPS/FTPS/SSL/FILE/MQTT/MMS/SMTP/SMTPS protocols ● Supports PAP and CHAP for PPP connection
Temperature Range	● Normal operating temperature range ⁶: -35 to +75 °C ● Extended operating temperature range ⁷: -40 to +85 °C

⁵ For EC200A-CN, GSM is optional.

⁶ Within this range, the module's indicators comply with 3GPP specification requirements.

⁷ Within this range, the module retains the ability to establish and maintain functions such as voice, SMS, data transmission and emergency call, without any unrecoverable malfunction. Radio spectrum and radio network remain uninfluenced, whereas the value of one or more parameters, such as P_{out} , may decrease and fall below the range of the 3GPP specified tolerances. When the temperature returns to the normal operating temperature range, the module's indicators will comply with 3GPP specification requirements again.

	Storage temperature range: -40 to +90 °C
Firmware Upgrade	USB interface or DFOTA
RoHS	All hardware components are fully compliant with EU RoHS directive.

2.3. Functional Diagram

The following figure shows a block diagram of the module and illustrates the major functional parts.

- Power management
- Baseband
- DDR + NAND flash
- Radio frequency
- Peripheral interface

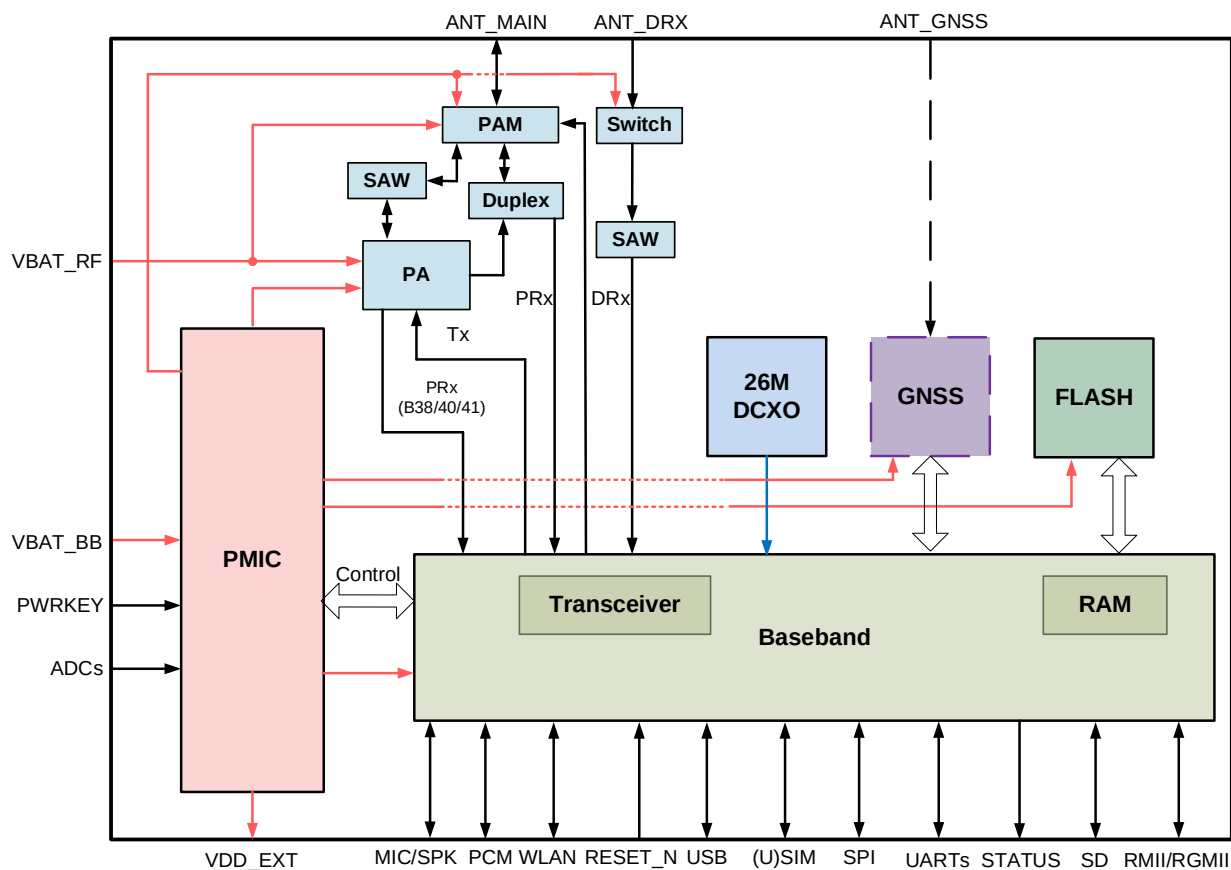


Figure 1: Functional Diagram

2.4. Pin Assignment

The following figure illustrates the pin assignment of the module.

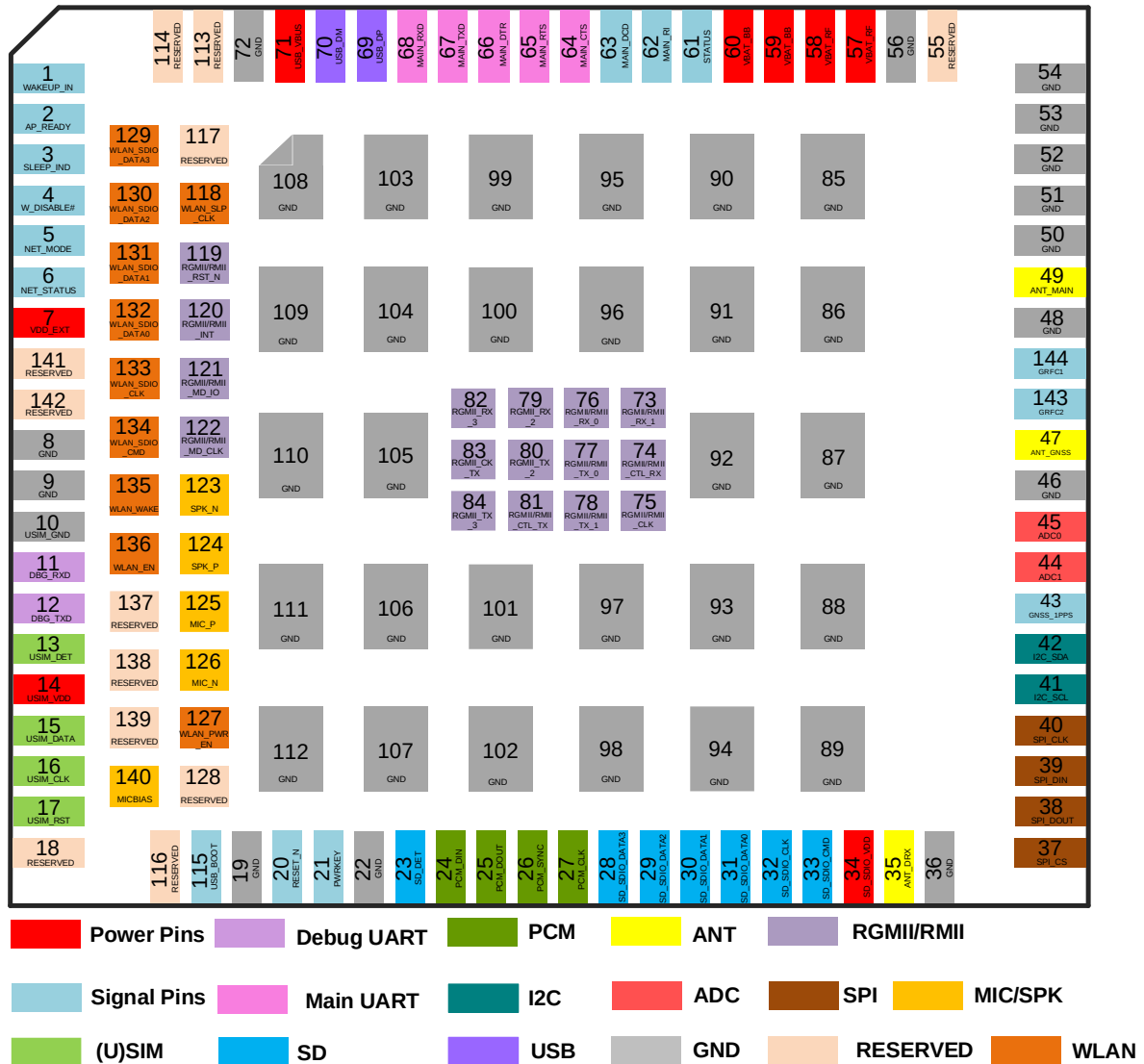


Figure 2: Pin Assignment (Top View)

NOTE

1. Ensure that the pull-up power supply of the module's pins is VDD_EXT or controlled by VDD_EXT, and there is no current sink on the module's pins before the module turns on. For more details, contact Quectel Technical Support.
2. EC200A-CN and EC200A-EL do not support RGMII function, so pins 79–80 and pins 82–84 are reserved. For more information, contact Quectel Technical Support.

3. GNSS is only supported by EC200A-CN and is optional.

2.5. Pin Description

The following table shows the DC characteristics and pin descriptions.

Table 5: Parameter Definition

Parameter	Description
AI	Analog Input
AIO	Analog Input/Output
AO	Analog Output
DI	Digital Input
DIO	Digital Input/Output
DO	Digital Output
OD	Open Drain
PI	Power Input
PO	Power Output

DC characteristics include power domain and rated current.

Table 6: Pin Description

Power Supply					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
VBAT_BB	59, 60	PI	Power supply for the module's BB part	Vmax = 4.5 V Vmin = 3.4 V Vnom = 3.8 V	It must be provided with sufficient current of at least 0.8 A ⁸ .

⁸ The VBAT_BB current of the module with APT function needs to be at least 1.5 A.

VBAT_RF	57, 58	PI	Power supply for the module's RF part		It must be provided with sufficient current of at least 2.0 A.
VDD_EXT	7	PO	Provide 1.8 V for external circuits	Vmin = 1.67 V Vnom = 1.8 V Vmax = 1.93 V Iomax = 50 mA	It can provide a pull-up power to the external GPIO. If unused, keep it open.
GND	8, 9, 19, 22, 36, 46, 48, 50–54, 56, 72, 85–112				

Turn On/Off

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
PWRKEY	21	DI	Turn on/off the module	VILmax = 0.5 V	VBAT power domain. Active low.
RESET_N	20	DI	Reset the module		1.8 V power domain. Active low after turn-on.

Indication Interfaces

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
NET_MODE	5	DO	Indicate the module's network registration mode	VDD_EXT	If unused, keep them open.
NET_STATUS	6	DO	Indicate the module's network activity status		
STATUS	61	OD	Indicate the module's operation status		
SLEEP_IND	3	DO	Indicate the module's sleep mode		

USB Interface

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
USB_VBUS	71	AI	USB connection detect	Vmax = 5.25 V Vmin = 3.0 V Vnom = 5.0 V	If unused, keep it open.

USB_DP	69	AIO	USB 2.0 differential data (+)	90 Ω differential impedance. Compliant with USB 2.0 specification If unused, keep them open.
USB_DM	70	AIO	USB 2.0 differential data (-)	

(U)SIM Interface

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
USIM_VDD	14	PO	(U)SIM card power supply	Low-voltage: Vmin = 1.67 V Vnom = 1.8 V Vmax = 1.93 V High-voltage: Vmin = 2.7 V Vnom = 3.0 V Vmax = 3.3 V	Either 1.8 V or 3.0 V (U)SIM card is supported and can be identified automatically by the module.
USIM_DATA	15	DIO	(U)SIM card data	USIM_VDD	
USIM_CLK	16	DO	(U)SIM card clock		
USIM_RST	17	DO	(U)SIM card reset		
USIM_DET	13	DI	(U)SIM card hot-plug detect	VDD_EXT	If unused, keep it open.
USIM_GND	10		Specified ground for (U)SIM card		Connect to the ground of (U)SIM card.

SD Card Interface

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
SD_SDIO_CLK	32	DO	SD card SDIO clock	SD_SDIO_VDD	If unused, keep them open.
SD_SDIO_CMD	33	DIO	SD card SDIO command		
SD_SDIO_DATA0	31	DIO	SD card SDIO data bit 0		
SD_SDIO_DATA1	30	DIO	SD card SDIO data bit 1		
SD_SDIO_DATA2	29	DIO	SD card SDIO data bit 2		

SD_SDIO_DATA3	28	DIO	SD card SDIO data bit 3		
SD_SDIO_VDD	34	PO	SD card SDIO power supply	Low-voltage: Vmin = 1.67 V Vnom = 1.8 V Vmax = 1.93 V High-voltage: Vmin = 2.7 V Vnom = 2.85 V Vmax = 3.1 V	
SD_DET*	23	DI	SD card hot-plug detect	VDD_EXT	It is recommended to reserve a 10 nF capacitor connected to GND.

Main UART Interface

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
MAIN_RI	62	DO	Main UART ring indication	VDD_EXT	If unused, keep them open.
MAIN_DCD	63	DO	Main UART data carrier detect		
MAIN_CTS	64	DO	Clear to send signal from the module		Connect to the MCU's CTS. If unused, keep it open.
MAIN_RTS	65	DI	Request to send signal to the module		Connect to the MCU's RTS. If unused, keep it open.
MAIN_DTR	66	DI	Main UART data terminal ready		If unused, keep them open.
MAIN_RXD	68	DI	Main UART receive		
MAIN_TXD	67	DO	Main UART transmit		

Debug UART Interface

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
DBG_RXD	11	DI	Debug UART	VDD_EXT	If unused, keep

				receive	them open.
DBG_TXD	12	DO	Debug UART transmit		
SPI Interface					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
SPI_CS	37	DO	SPI chip select	VDD_EXT	If unused, keep them open.
SPI_DOUT	38	DO	SPI data output		
SPI_DIN	39	DI	SPI data input		
SPI_CLK	40	DO	SPI clock		
I2C Interface					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
I2C_SCL	41	OD	I2C serial clock (for external codec)	VDD_EXT	An external 1.8 V pull-up resistor is needed.
I2C_SDA	42	OD	I2C serial data (for external codec)		
PCM Interface					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
PCM_SYNC	26	DIO	PCM data frame sync	VDD_EXT	When the module is in the master mode, this pin is in the output state. when the module is in the slave*mode, it is in the input state. If unused, keep them open.
PCM_CLK	27	DIO	PCM clock		
PCM_DIN	24	DI	PCM data input		If unused, keep them open.
PCM_DOUT	25	DO	PCM data output		
WLAN Interface					

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
WLAN_SLP_CLK	118	DO	WLAN sleep clock	VDD_EXT	If unused, keep them open.
WLAN_PWR_EN	127	DO	WLAN power supply enable control		
WLAN_SDIO_DATA3	129	DIO	WLAN SDIO data bit 3		
WLAN_SDIO_DATA2	130	DIO	WLAN SDIO data bit 2		
WLAN_SDIO_DATA1	131	DIO	WLAN SDIO data bit 1		
WLAN_SDIO_DATA0	132	DIO	WLAN SDIO data bit 0		
WLAN_SDIO_CLK	133	DO	WLAN SDIO clock		
WLAN_SDIO_CMD	134	DIO	WLAN SDIO command		
WLAN_WAKE	135	DI	Wake up the host by an external Wi-Fi module		
WLAN_EN	136	DO	WLAN function enable control		

RF Antenna Interfaces

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
ANT_DRX	35	AI	Diversity antenna interface		50 Ω characteristic impedance.
ANT_MAIN	49	AIO	Main antenna interface		
ANT_GNSS	47	AI	GNSS antenna interface		50 Ω characteristic impedance. If unused, keep it open.

ADC Interfaces

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
ADC0	45	AI	General-purpose ADC interface	Voltage Range: 0 V–VBAT_BB	If unused, keep them open.
ADC1	44	AI			

RGMII*/RMII Interface

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
RGMII/RMII_RX_1	73	DI	RGMII/RMII receive data bit 1	1.8 V RGMII/RMII: V _{ILmin} = -0.3 V V _{ILmax} = 0.54 V V _{IHmin} = 1.26 V V _{IHmax} = 2.0 V V _{OLmax} = 0.2 V V _{OHmin} = 1.6 V	
RGMII/RMII_CTL_RX	74	DI	RGMII/RMII receive control		
RGMII/RMII_CLK	75	DI	RGMII/RMII clock		
RGMII/RMII_RX_0	76	DI	RGMII/RMII receive data bit 0		
RGMII/RMII_TX_0	77	DO	RGMII/RMII transmit data bit 0	3.3 V RMII: V _{ILmin} = -0.3 V V _{ILmax} = 0.8 V V _{IHmin} = 2.0 V V _{IHmax} = 3.6 V V _{OLmax} = 0.4 V V _{OHmin} = 2.4 V	
RGMII/RMII_TX_1	78	DO	RGMII/RMII transmit data bit 1		
RGMII_RX_2	79	DI	RGMII receive data bit 2	VDD_EXT	1.8 V power domain for RGMII.
RGMII_TX_2	80	DO	RGMII transmit data bit 2		
RGMII/RMII_CTL_TX	81	DO	RGMII/RMII transmit control	1.8 V RGMII/RMII: V _{ILmin} = -0.3 V V _{ILmax} = 0.54 V V _{IHmin} = 1.26 V V _{IHmax} = 2.0 V V _{OLmax} = 0.2 V V _{OHmin} = 1.6 V	1.8/3.3 V power domain for RMII. If unused, keep them open.
				3.3 V RMII: V _{ILmin} = -0.3 V V _{ILmax} = 0.8 V V _{IHmin} = 2.0 V V _{IHmax} = 3.6 V V _{OLmax} = 0.4 V V _{OHmin} = 2.4 V	
RGMII_RX_3	82	DI	RGMII receive data bit 3	VDD_EXT	
RGMII_CK_TX	83	DO	RGMII transmit clock		
RGMII_TX_3	84	DO	RGMII transmit data bit 3		

RGMII/RMII_INT	120	DI	RGMII/RMII interrupt input	1.8 V RGMII/RMII: V _{ILmin} = -0.3 V V _{ILmax} = 0.54 V V _{IHmin} = 1.26 V V _{IHmax} = 2.0 V V _{OLmax} = 0.2 V V _{OHmin} = 1.6 V
RGMII/RMII_MD_IO	121	DIO	RGMII/RMII management data input/output	
RGMII/RMII_MD_CLK	122	DO	RGMII/RMII management data clock	3.3 V RMII: V _{ILmin} = -0.3 V V _{ILmax} = 0.8 V V _{IHmin} = 2.0 V V _{IHmax} = 3.6 V V _{OLmax} = 0.4 V V _{OHmin} = 2.4 V
RGMII/RMII_RST_N	119	DO	RGMII/RMII reset external PHY	VDD_EXT Cannot be pulled high before module's successful start-up. If unused, keep it open.

Analog Audio Interfaces

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
SPK_N	123	AO	Analog audio differential output (-)		
SPK_P	124	AO	Analog audio differential output (+)		
MIC_P	125	AI	Microphone analog input (+)		
MIC_N	126	AI	Microphone analog input (-)		
MICBIAS	140	PO	Bias voltage output for microphone		

Other Interfaces

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
USB_BOOT	115	DI	Force the module into download	VDD_EXT	Active High. A test point is

			mode	recommended to be reserved.
WAKEUP_IN*	1	DI	Wake up the module	If unused, keep it open.
AP_READY*	2	DI	Application processor ready	If unused, keep it open.
				Pull-up by default.
W_DISABLE#	4	DI	Airplane mode control	In low voltage level, module can enter airplane mode. If unused, keep it open.
GRFC1	144	DO	Generic RF controller	If unused, keep it open.
GRFC2	143	DO		
GNSS_1PPS	43	DO	GNSS pulse per second output	If unused, keep it open.
RESERVED Pins				
Pin Name	Pin No.			Comment
RESERVED	18, 55, 113, 114, 116, 117, 128, 137–139, 141–142			Keep these pins open.

NOTE

1. The USB_BOOT and RGMII/RMII_RST_N pins cannot be pulled up to high level before the module is turned on successfully.
2. Unused and RESERVED pins are kept open and all GND pins are connected to the ground network.
3. Antenna tuner control interfaces (pins 143 and 144) are optional. For more information, please contact Quectel Technical Support.
4. EC200A-CN and EC200A-EL do not support RGMII function, so pins 79–80 and pins 82–84 are reserved. For more information, contact Quectel Technical Support.
5. GNSS is only supported by EC200A-CN and is optional.

2.6. EVB Kit

Quectel supplies an evaluation board (UMTS<EEVB) with accessories to develop and test the module. For more details, see **document [1]**.

3 Operating Characteristics

3.1. Operating Modes

The table below outlines operating modes of the module.

Table 7: Overview of Operating Modes

Modes	Details	
Full Functionality Mode	Idle	Software is active. The module is registered on the network and ready to send and receive data.
	Voice/Data	Network connection is ongoing. In this mode, the power consumption is decided by network setting and data transmission rate.
Minimum Functionality Mode	AT+CFUN=0 can set the module to a minimum functionality mode. In this case, both RF function and (U)SIM card will be invalid.	
Airplane Mode	AT+CFUN=4 or W_DISABLE# pin can set the module to airplane mode. In this case, RF function will be invalid.	
Sleep Mode	In this mode, power consumption of the module will be reduced to an ultra-low level. In this mode, the module can still receive paging, SMS, voice call and TCP/UDP data from network.	
Shutdown Mode	In this mode, the VBAT power supply is constantly turned on and the software stops working.	

NOTE

For more details about the AT command, see *document [3]*.

3.2. Sleep Mode

In sleep mode, the module can reduce power consumption to an ultra-low level.

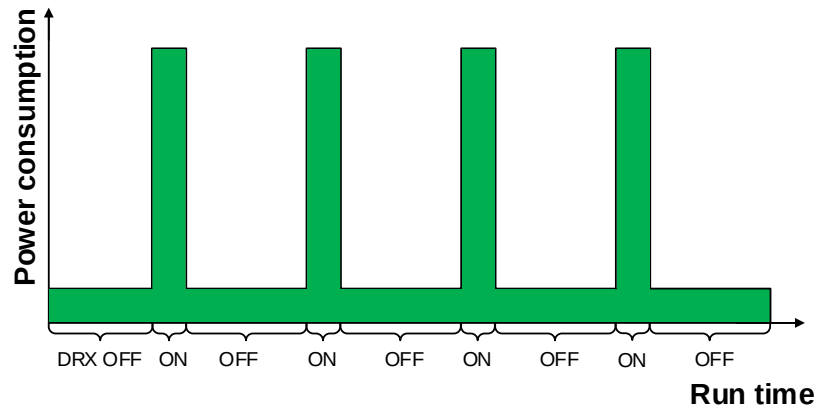


Figure 3: Module Power Consumption in Sleep Mode

NOTE

DRX cycle values are transmitted over the wireless network.

3.2.1. UART Application

If MCU communicates with module via UART interface, the following preconditions should be met to enable the module to enter sleep mode.

- Execute **AT+QSClk=1** to enable sleep mode.
- Drive MAIN_DTR to high level.
- Ensure the USB_VBUS is kept at low level, or kept open.

The following figure shows the connection between the module and MCU.

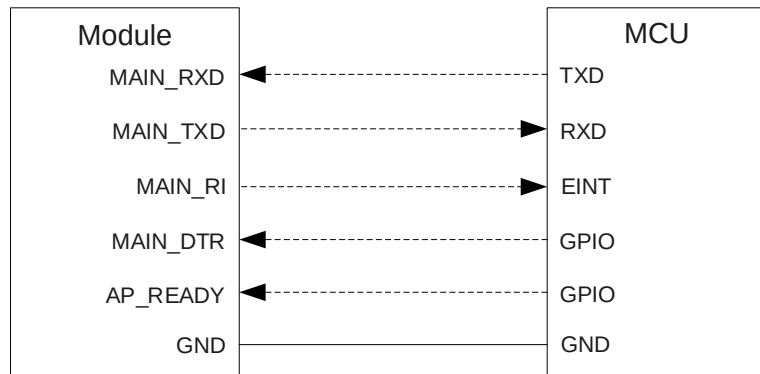


Figure 4: Sleep Mode Application via UART Interface

- Driving MAIN_DTR to low level by MCU will wake up the module.
- When the module has a URC to report, the URC will trigger the behavior of MAIN_RI pin. See **Chapter 4.12** for details about MAIN_RI behavior.

3.2.2. USB Application with USB Remote Wakeup Function

If the host supports USB Suspend/Resume and remote wakeup functions, the following three preconditions must be met to let the module enter sleep mode.

- Execute **AT+QSCCLK=1** to enable the sleep mode.
- Ensure the MAIN_DTR is kept at high level or kept open.
- The host's USB bus, which is connected with the module's USB interface, enters Suspend state.

The following figure shows the connection between the module and the host.

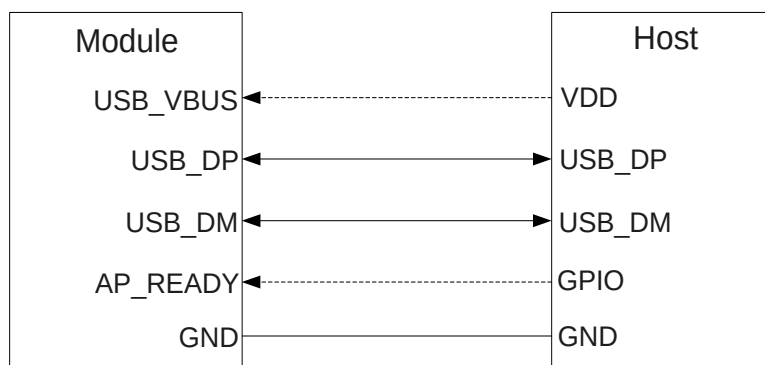


Figure 5: Sleep Mode Application with USB Remote Wakeup

- Sending data to the module through USB will wake up the module.
- When the module has a URC to report, the module will send remote wakeup signals via USB bus to wake up the host.

3.2.3. USB Application Without USB Remote Wakeup Function

If the host supports USB Suspend/Resume, but does not support remote wakeup function, the MAIN_RI signal is needed to wake up the host.

There are three preconditions to let the module enter sleep mode.

- Execute **AT+QSClk=1** to enable the sleep mode.
- Ensure the MAIN_DTR is held at high level or keep it open.
- The host's USB bus, which is connected with the module's USB interface, enters Suspend state.

The following figure shows the connection between the module and the host.

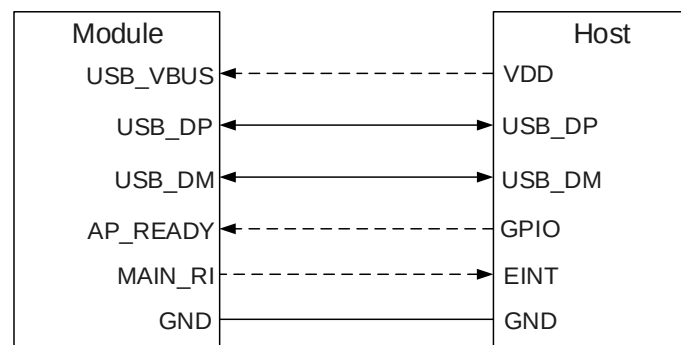


Figure 6: Sleep Mode Application with MAIN_RI

- Sending data to the module through USB will wake up the module.
- When the module has a URC to report, the URC will trigger the behavior of MAIN_RI pin. See **Chapter 4.12** for details about MAIN_RI behavior.

3.2.4. USB Application Without USB Suspend Function

If the host does not support USB Suspend function, disconnect USB_VBUS with additional control circuit to let the module enter sleep mode.

- Execute **AT+QSClk=1** to enable the sleep mode.
- Ensure the MAIN_DTR is held at high level or keep it open.
- Disconnect USB_VBUS.

The following figure shows the connection between the module and the host.

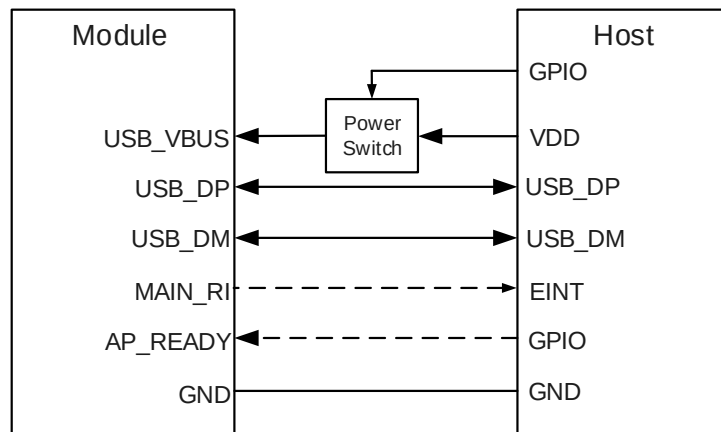


Figure 7: Sleep Mode Application Without Suspend Function

Turn on the power switch and supply power to USB_VBUS will wake up the module.

NOTE

1. Pay attention to the level match shown in dotted line between the module and the host.
2. For more details about the AT command, see **document [3]**.

3.3. Airplane Mode

When the module enters airplane mode, the RF function will be disabled, and all AT commands related to it will be inaccessible. This mode can be set via the following ways.

Hardware:

The W_DISABLE# pin is pulled up by default. Its control function for airplane mode is disabled by default, and **AT+QCFG="airplanecontrol",1** can be used to enable the function. Driving the pin to low level can make the module enter airplane mode.

Software:

AT+CFUN=<fun> provides choices of the functionality level through setting **<fun>** into 0, 1 or 4.

- **AT+CFUN=0:** Minimum functionality mode (Both (U)SIM and RF functions are disabled).
- **AT+CFUN=1:** Full functionality mode (by default).
- **AT+CFUN=4:** Airplane mode (RF function is disabled).

NOTE

For more details about AT command, see *document [3]*.

3.4. Power Supply

3.4.1. Power Supply Pins

The module provides four VBAT pins dedicated to the connection with the external power supply. There are two separate voltage domains for VBAT.

- Two VBAT_RF pins for module's RF part.
- Two VBAT_BB pins for module's BB part.

The following table shows the details of power supply and GND pins.

Table 8: Pin Definition of Power Supply

Pin Name	Pin No.	I/O	Description	Comment
VBAT_BB	59, 60	PI	Power supply for the module's BB part	It must be provided with sufficient current of at least 0.8 A ⁹ .
VBAT_RF	57, 58	PI	Power supply for the module's RF part	It must be provided with sufficient current of at least 2.0 A.
VDD_EXT	7	PO	Provide 1.8 V for external circuits	It can provide a pull-up power to the external GPIO. If unused, keep it open.

3.4.2. Reference Design for Power Supply

The performance of the module largely depends on the power source. For models that support GSM frequency bands, the selected external power supply should be able to provide sufficient current of at least 2.8 A ¹⁰. If the voltage drops between input and output is not too high, it is suggested that an LDO should be used to supply power to the module. If there is a big voltage difference between input

⁹ The VBAT_BB current of the module with APT function needs to be at least 1.5 A.

¹⁰ If using models that do not support the GSM frequency bands, select power supply that can provide at least 2.0 A current capacity in the design.

and the desired output VBAT, a buck converter is preferred as the power supply.

The following figure shows a reference design for +5 V input power source.

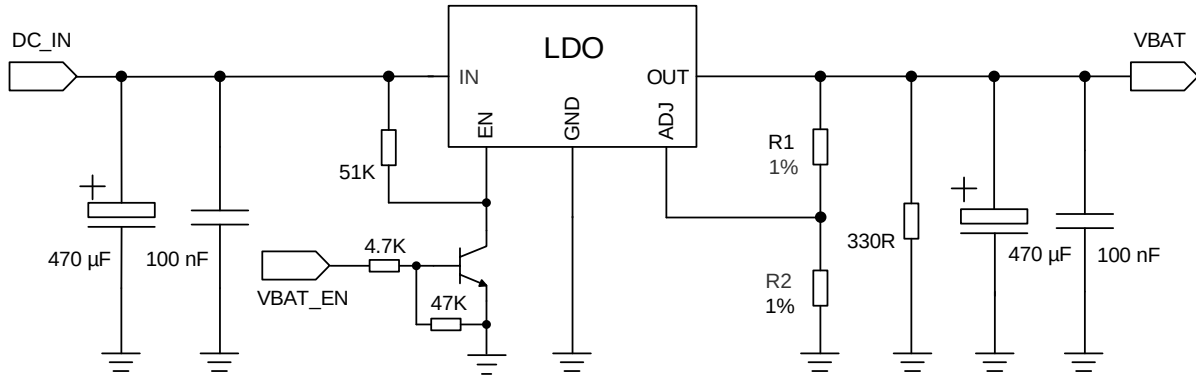


Figure 8: Reference Design of Power Supply

NOTE

It is recommended to add switch control for power supply.

3.4.3. Requirements for Voltage Stability

The power supply range of the module is from 3.4 V to 4.5 V. Make sure the input voltage will never drop below 3.4 V.

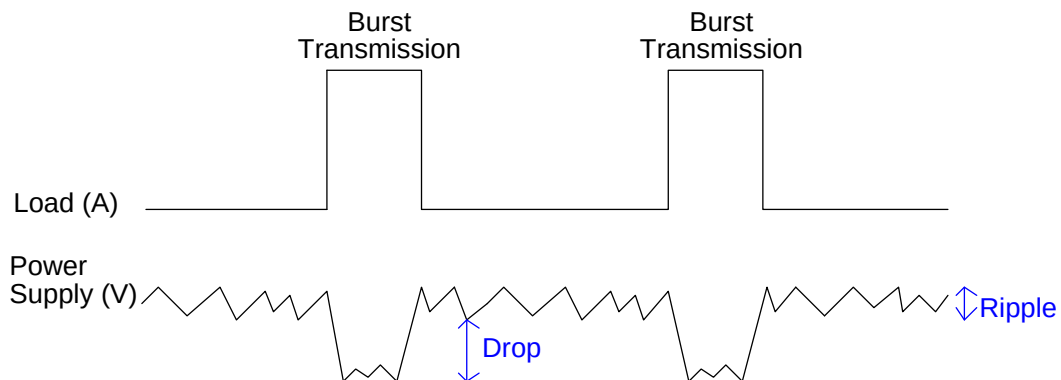


Figure 9: Power Supply Limits During Burst Transmission

To decrease voltage drop, a bypass capacitor of about 100 µF with low ESR ($ESR \leq 0.7 \Omega$) should be

used, and a multi-layer ceramic chip (MLCC) capacitor array should also be reserved due to its ultra-low ESR. It is recommended to use three ceramic capacitors (100 nF, 33 pF, 10 pF) for composing the MLCC array, and place these capacitors close to the VBAT_BB and VBAT_RF pins. The main power supply from an external application has to be a single voltage source and can be expanded to two sub paths with star structure. The width of VBAT_BB trace should be not less than 1.5 mm; and the width of VBAT_RF trace should be not less than 2 mm. In principle, the longer the VBAT trace is, the wider it will be.

In addition, to ensure the stability of power source, it is suggested that a TVS diode of which reverse stand-off voltage is 4.7 V and peak pulse power is up to 2550 W should be used. The following figure shows the star structure of the power supply.

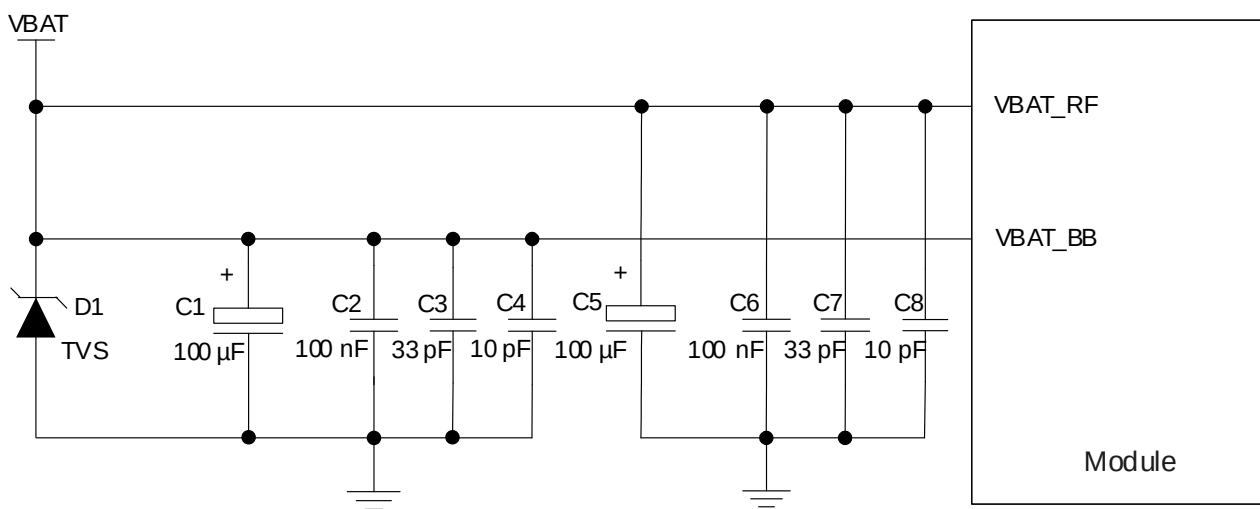


Figure 10: Star Structure of the Power Supply

3.5. Turn-on

3.5.1. Turn-on with PWRKEY

Table 9: Pin Definition of PWRKEY

Pin Name	Pin No.	I/O	Description	Comment
PWRKEY	21	DI	Turn on/off the module	VBAT power domain. Active low.

When the module is in turn-off mode, it can be turned on by driving the PWRKEY pin to a low level for at least 500 ms. It is recommended to use an open drain/collector driver to control the PWRKEY. A simple reference circuit is illustrated in the following figure.

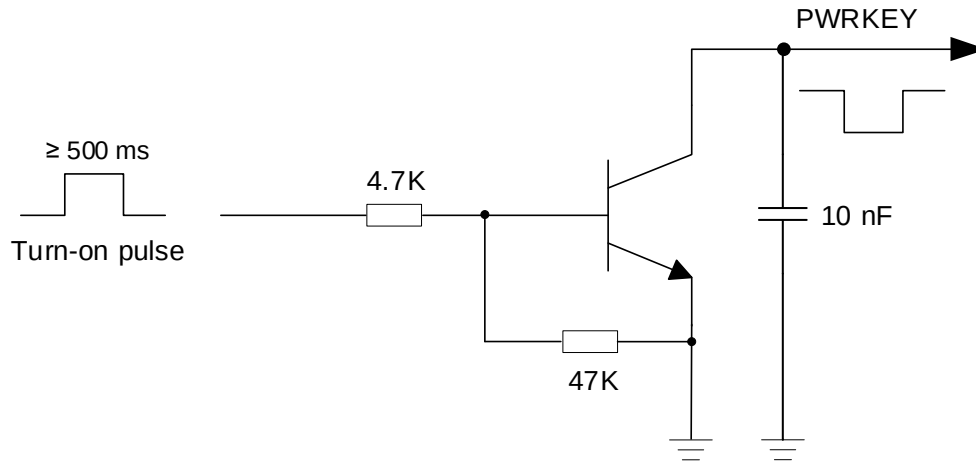


Figure 11: Reference Circuit of Turn-on with Driving Circuit

The other way to control the PWRKEY is using a button directly. When pressing the button, electrostatic strike may generate from finger. Therefore, an ESD protection component should be placed near the button for electrostatic protection. The reference circuit is as follows.

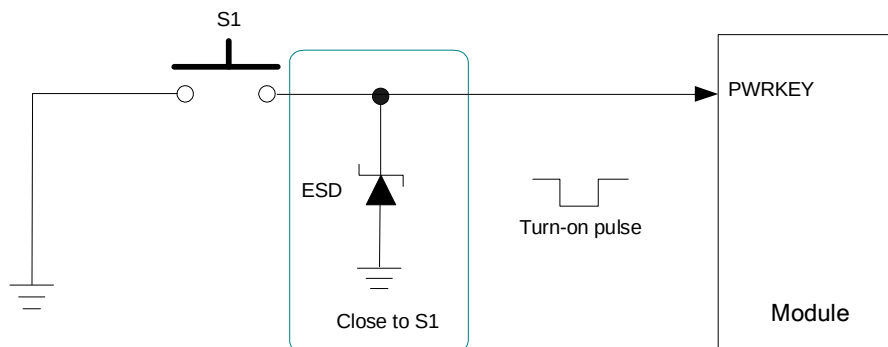


Figure 12: Reference Circuit of Turn-on with Button

The turn-on timing is illustrated in the following figure.

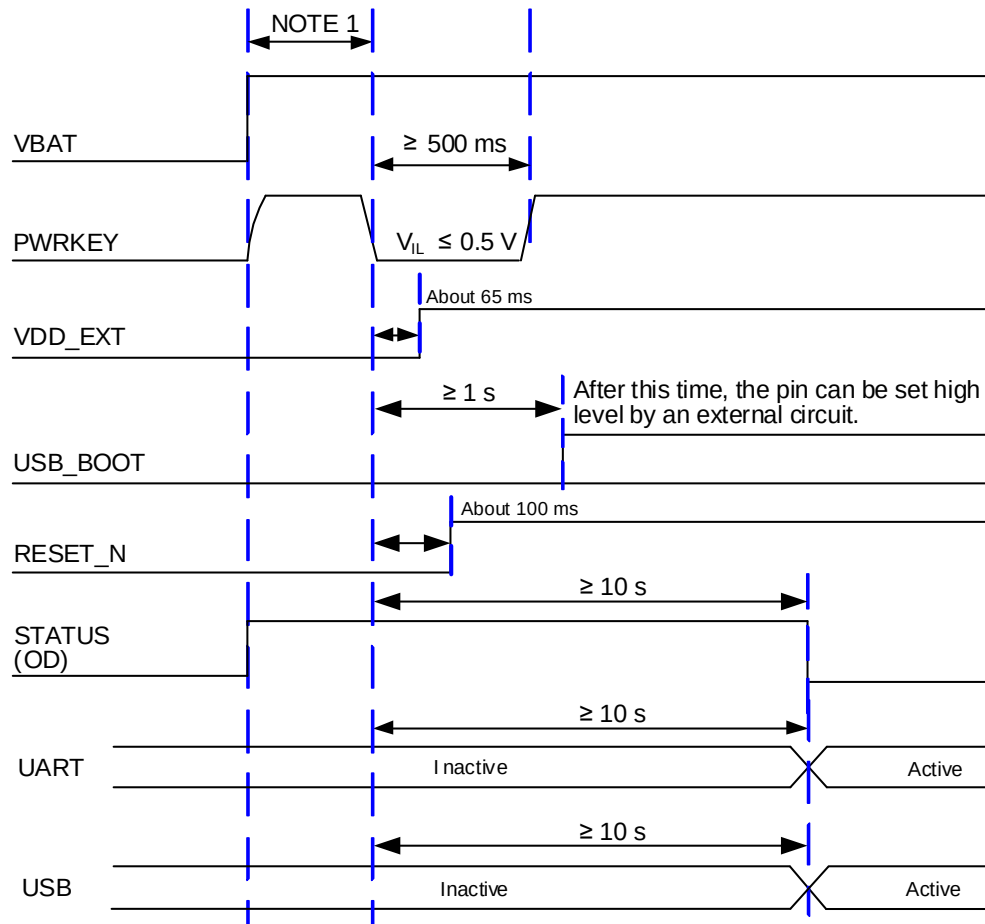


Figure 13: Turn-on Timing

NOTE

1. Make sure that VBAT is stable before pulling down PWRKEY pin. It is recommended that the time difference between powering up VBAT and pulling down PWRKEY pin is not less than 700 ms.
2. PWRKEY can be pulled down directly to GND with a recommended 4.7 k Ω resistor if the module needs to be turned on automatically and shutdown is not needed.
3. The above turn-on timing is applicable only to software versions released in 2023 and later. If you need to use software versions released before 2023, please contact Quectel Technical Support.

3.6. Turn-off

The following procedures can be used to turn-off the module normally.

3.6.1. Turn-off with PWRKEY

Driving the PWRKEY to a low-level voltage for at least 650 ms, then the module will execute turn-off procedure after the PWRKEY is released. The timing of turning off the module is illustrated in the following figure.

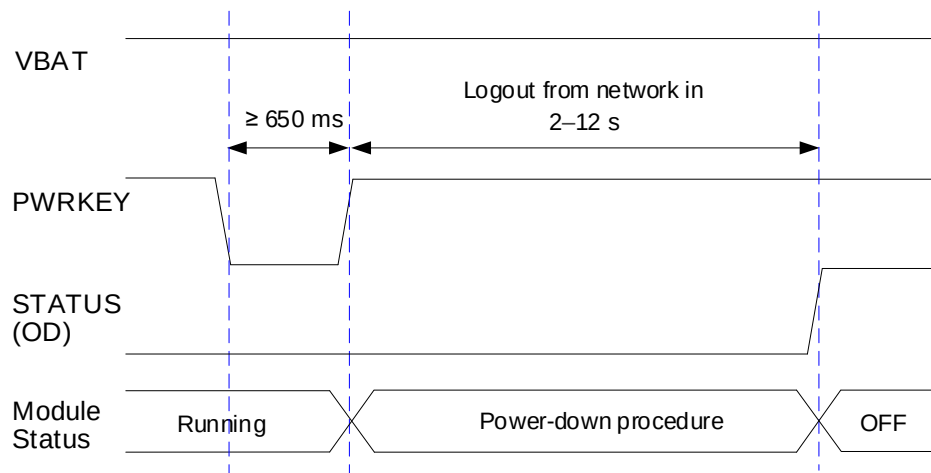


Figure 14: Timing of Turn-off with PWRKEY

During the turn-off process, the module needs to log out from network. The logout time is related to the current network state, which is measured to take about 2–12 s. Therefore, it is recommended to turn off or restart the module after 12 s to ensure that the important software data is saved before completely shutting down.

3.6.2. Turn-off with AT Command

It is safe to use **AT+QPOWD** to turn off the module, which is similar to turning off the module via the PWRKEY pin.

For more details about the AT command, see *document [3]*.

NOTE

1. To avoid corrupting the data of internal flash, do not cut off the power supply when the module works normally. Only after shutting down the module with PWRKEY or AT command can you cut off the power supply.
2. When turning off module with the AT command, keep PWRKEY at high level after the execution of the command. Otherwise, the module will be turned on again after successfully turn-off.

3.7. RESET_N

The module can be reset by driving the RESET_N low for at least 300 ms and then releasing it. The RESET_N signal is sensitive to interference, so it is recommended to route the trace as short as possible and surround it with ground.

Table 10: Pin Definition of RESET_N

Pin Name	Pin No.	I/O	Description	Comment
RESET_N	20	DI	Reset the module	1.8 V power domain. Active low after turn-on.

The recommended circuit is equal to the PWRKEY control circuit. An open drain/collector driver or a button can be used to control the RESET_N.

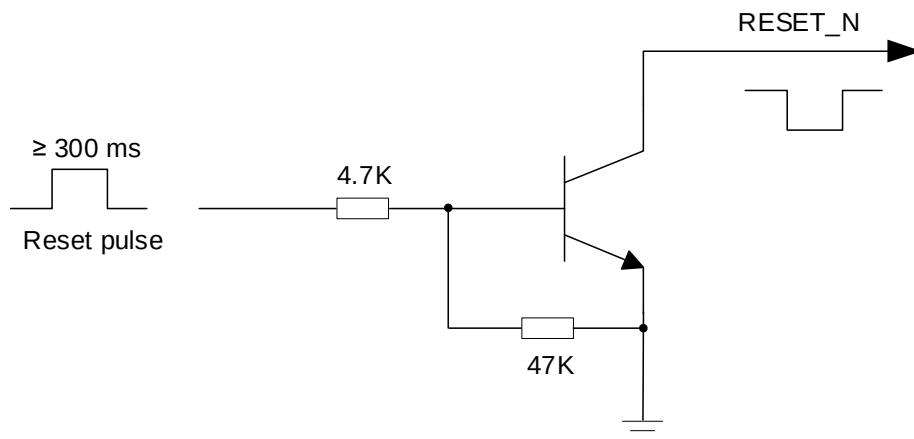


Figure 15: Reference Circuit of RESET_N with Driving Circuit

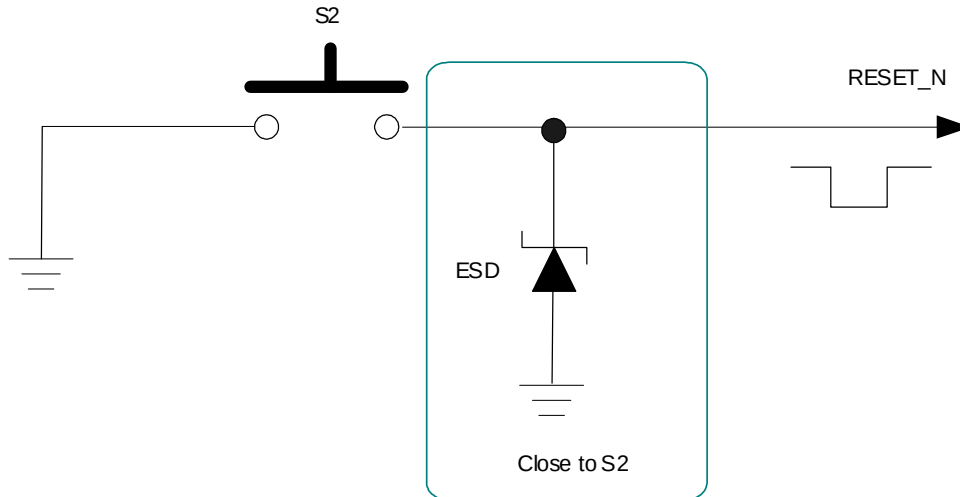


Figure 16: Reference Circuit of RESET_N with Button

The timing of resetting module is illustrated in the following figure.

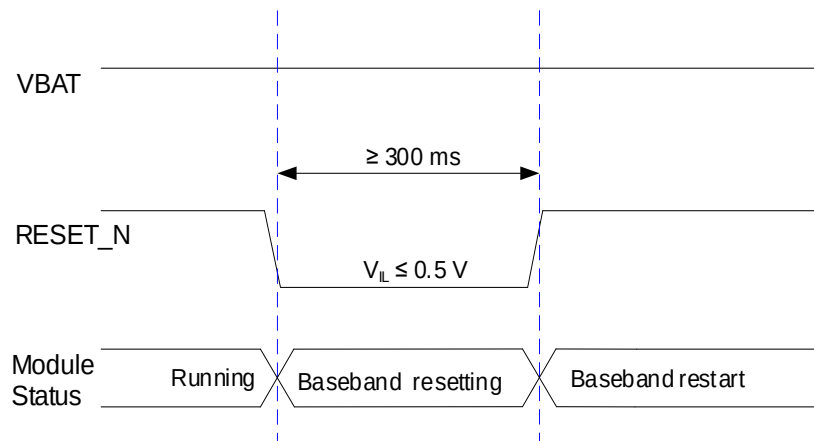


Figure 17: Timing of Resetting the Module

NOTE

1. Ensure that there is no large capacitance with the max value exceeding 10 nF on PWRKEY and RESET_N pins.
2. RESET_N pin only resets the baseband chip inside the module and does not reset the power management chip.

4 Application Interfaces

4.1. USB Interface

EC200A series provides one integrated Universal Serial Bus (USB) interface which complies with the USB 2.0 specification and supports full-speed (12 Mbps) and high-speed (480 Mbps) modes. The USB interface can only serve as a slave device and is used for AT command communication, data transmission, software debugging and firmware upgrade. The following table shows the pin definition of USB interface.

Table 11: Pin Definition of USB Interface

Pin Name	Pin No.	I/O	Description	Comment
USB_VBUS	71	AI	USB connection detect	If unused, keep it open.
USB_DP	69	AIO	USB 2.0 differential data (+)	90 Ω differential impedance. Compliant with USB 2.0 specification
USB_DM	70	AIO	USB 2.0 differential data (-)	If unused, keep them open.

For more details about the USB 2.0 specifications, visit <http://www.usb.org/home>.

Test points must be reserved for debugging and firmware upgrade in your designs. The following figure shows a reference circuit of USB interface.

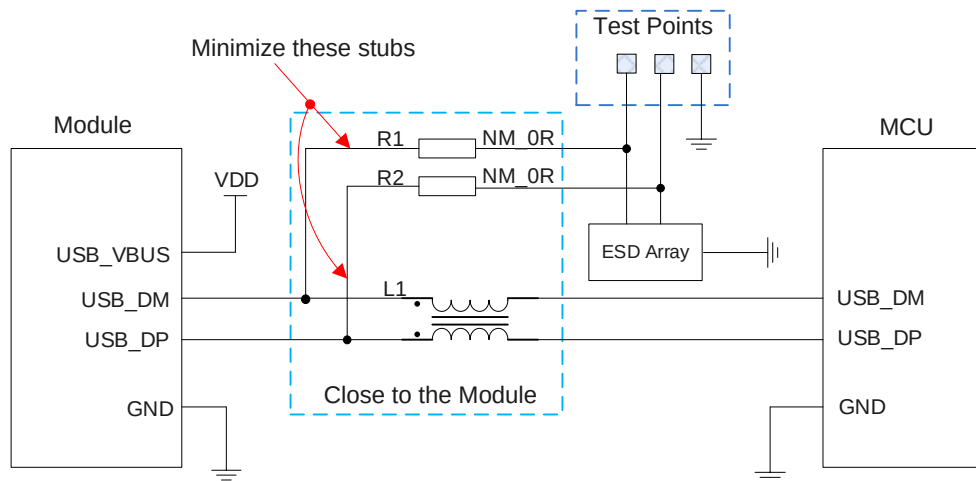


Figure 18: Reference Circuit of USB Interface

A common mode choke L1 is recommended to be added in series between the module and your MCU to suppress EMI. Meanwhile, the 0 Ω resistors (R1 and R2) should be added in series between the module and the test points to facilitate debugging, and the resistors are not mounted by default. To ensure the integrity of USB data trace signal, L1, R1 and R2 components must be placed close to the module, and also resistors R1 and R2 should be placed close to each other. The extra stubs of trace must be as short as possible.

To meet USB specifications, the following principles should be complied with when designing the USB interface.

- It is important to route the USB signal traces as differential pairs with ground surrounded. The impedance of USB differential trace is 90 Ω .
- Do not route signal traces under crystals, oscillators, magnetic devices and RF signal traces. It is important to route the USB differential traces in inner-layer of the PCB, and surround the traces with ground on that layer and ground planes above and below.
- Junction capacitance of the ESD protection component might cause influences on USB data traces, so pay attention to the selection of the device. Typically, the stray capacitance should be less than 2 pF for USB.
- It is recommended to reserve a 0 Ω resistor on USB_DP and USB_DM lines respectively.

For more details about the USB specifications, visit <http://www.usb.org/home>.

4.2. USB_BOOT

The module provides a USB_BOOT pin. You can pull up USB_BOOT to VDD_EXT before turning on the module, and then the module will enter forced download mode. In this mode, the module supports firmware upgrade over USB interface.

Table 12: Pin Definition of USB_BOOT

Pin Name	Pin No.	I/O	Description	Comment
USB_BOOT	115	DI	Force the module into download mode	Active high. A test point is recommended to be reserved.

The following figure shows a reference circuit of USB_BOOT.

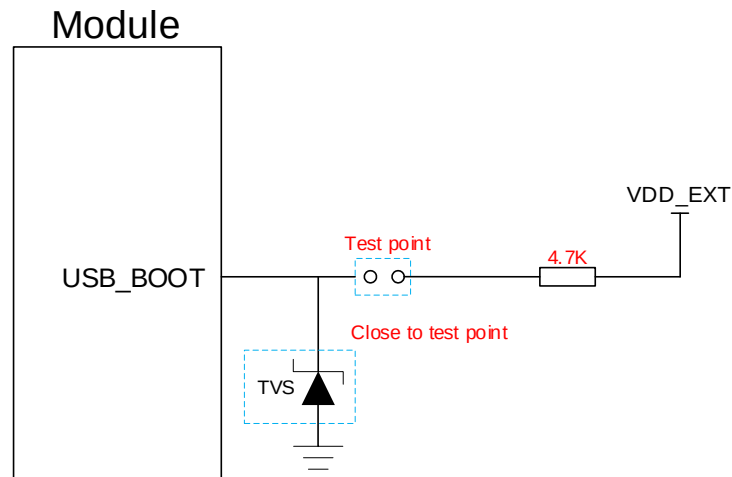


Figure 19: Reference Circuit of USB_BOOT

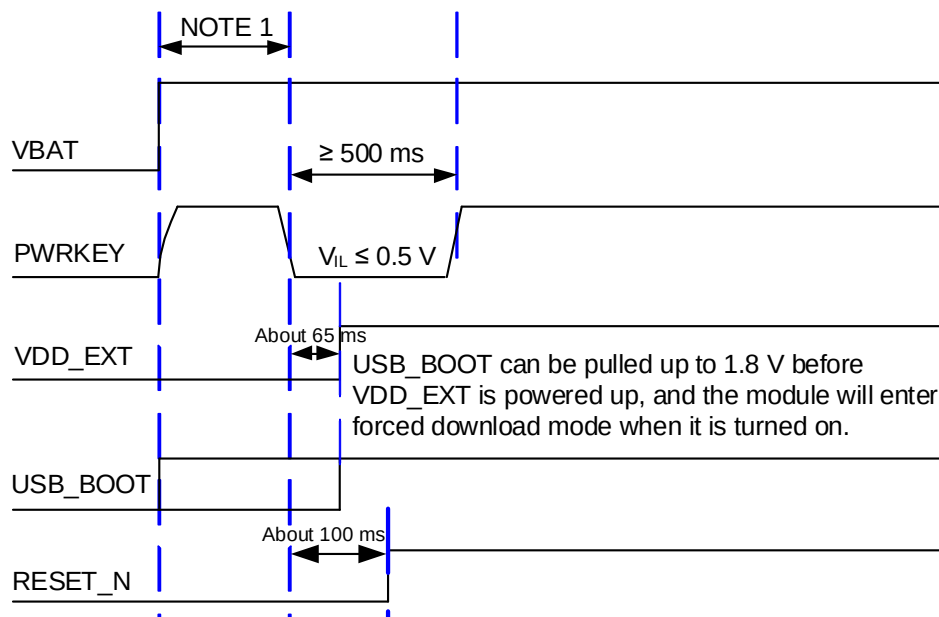


Figure 20: Timing Sequence for Entering Forced Download Mode

NOTE

1. Make sure that VBAT is stable before pulling down PWRKEY pin. It is recommended that the time between powering up VBAT and pulling down PWRKEY pin is not less than 700 ms.
2. When using MCU to control the module to enter the forced download mode, follow the above timing sequence. It is not recommended to pull up USB_BOOT to 1.8 V before powering up VBAT. Directly connect the test points as shown in **Figure 19** can manually force the module into download mode.
3. USB_BOOT cannot be pulled up before module is turned on successfully.
4. The above turn-on timing is only applicable to software versions released in 2023 and later. If you need to use software versions released before 2023, please contact Quectel Technical Support.

4.3. (U)SIM Interface

The (U)SIM interface circuitry meets ETSI and IMT-2000 requirements. Either 1.8 V or 3.0 V (U)SIM card is supported.

Table 13: Pin Definition of (U)SIM Interface

Pin Name	Pin No.	I/O	Description	Comment
USIM_VDD	14	PO	(U)SIM card power supply	Either 1.8 V or 3.0 V (U)SIM card is supported and can be identified automatically by the module.
USIM_DATA	15	DIO	(U)SIM card data	
USIM_CLK	16	DO	(U)SIM card clock	
USIM_RST	17	DO	(U)SIM card reset	
USIM_DET	13	DI	(U)SIM card hot-plug detect	If unused, keep it open.
USIM_GND	10		Specified ground for (U)SIM card	Connect to the ground of (U)SIM card.

The module supports (U)SIM card hot-plug via the USIM_DET pin, and both high-level and low-level detections are supported. By default, it is disabled, and can be configured via **AT+QSIMDET**. See **document [3]** for details about the command.

The reference circuit of the 8-pin (U)SIM interface is as follows.

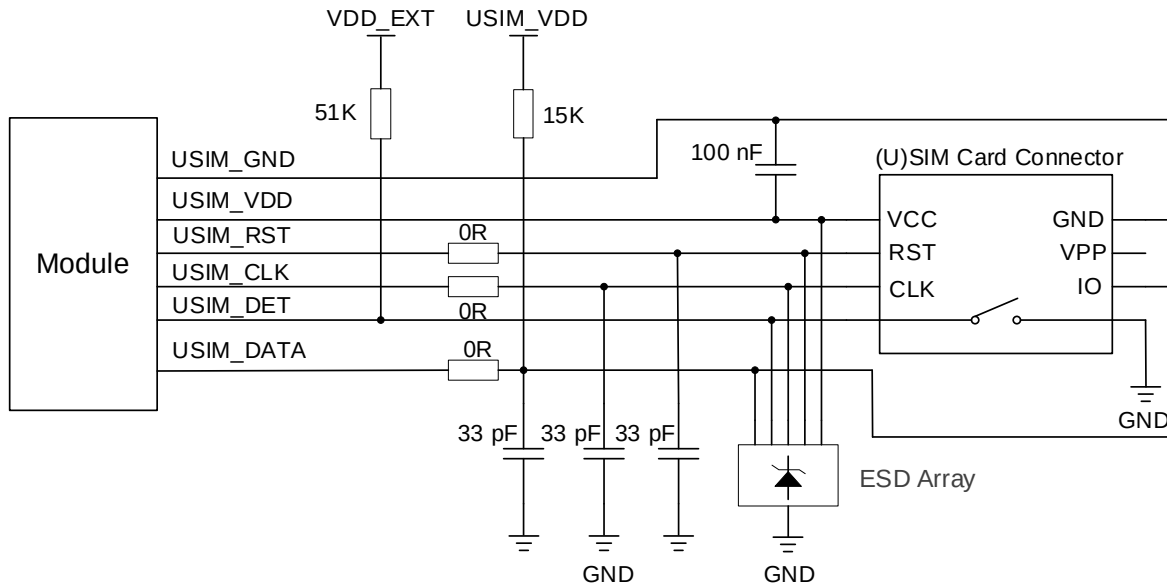


Figure 21: Reference Circuit of (U)SIM Interface with an 8-pin (U)SIM Card Connector

If (U)SIM card detection function is not needed, keep USIM_DET unconnected. A reference circuit for (U)SIM interface with a 6-pin (U)SIM card connector is illustrated in the following figure.

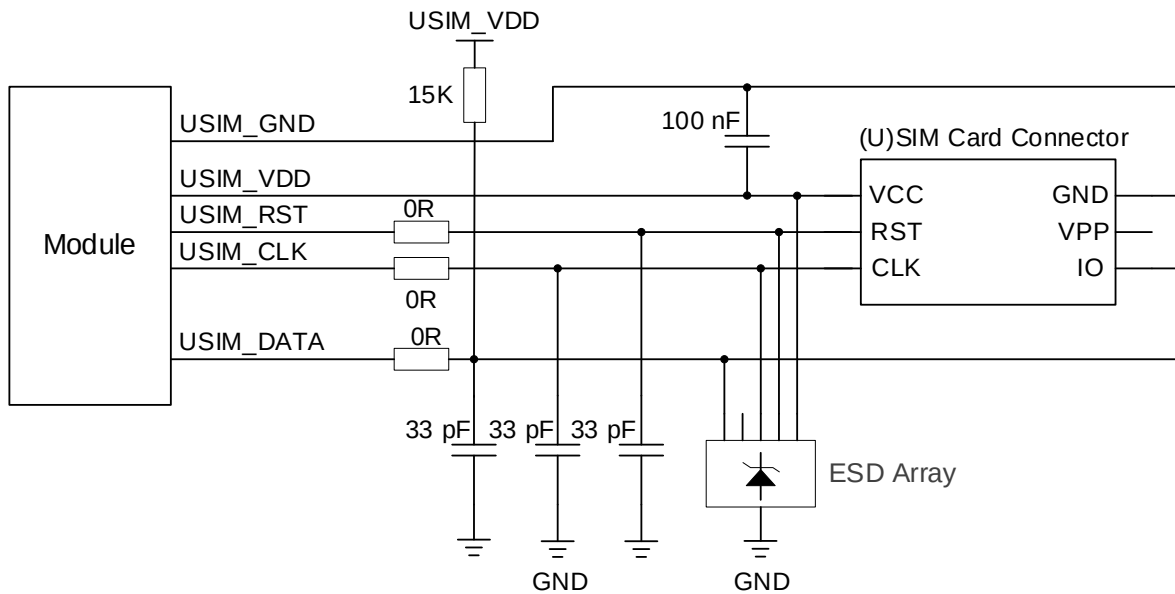


Figure 22: Reference Circuit of (U)SIM Interface with a 6-pin (U)SIM Card Connector

In order to enhance the reliability and availability of the (U)SIM card in applications, follow the criteria below in (U)SIM circuit design.

- Keep (U)SIM card connector as close as possible to the module. Keep the trace length as short as possible, at most 200 mm.

- Keep (U)SIM card signal traces away from RF and power supply traces.
- USIM_VDD's maximum bypass capacitor does not exceed 1 uF.
- Ensure the ground between the module and the (U)SIM card connector is short and wide. Keep the trace width of USIM_GND and USIM_VDD not less than 0.5 mm to maintain the same electric potential.
- To avoid cross-talk between USIM_DATA and USIM_CLK, keep them away from each other and shield them with ground surrounded.
- In order to offer good ESD protection, it is recommended to add an ESD protection component whose parasitic capacitance should not be more than 15 pF. The 0 Ω resistors should be added in series between the module and the (U)SIM card to facilitate debugging. The 33 pF capacitors on the USIM_DATA, USIM_CLK and USIM_RST traces are used for filtering out RF interference. Note that the (U)SIM peripheral circuit should be close to the (U)SIM card connector.
- The pull-up resistor on USIM_DATA can improve anti-jamming capability of the (U)SIM card. If the (U)SIM card traces are too long, or the interference source is relatively close, it is recommended to add a pull-up resistor near the (U)SIM card connector.

4.4. PCM and I2C Interfaces

The module provides one Pulse Code Modulation (PCM) digital interface for audio design, which supports the primary mode (short frame synchronization) and the module works as both master and slave* device.

The module can only be used as primary devices in applications related to I2C interfaces and does not support multi-host mode. It conforms to the I2C bus protocol specification (100/400 kHz).

In short frame mode, the data is sampled on the falling edge of the PCM_CLK and transmitted on the rising edge. The PCM_SYNC falling edge represents the MSB. In this mode, the PCM interface supports 256 kHz, 512 kHz, 1024 kHz, 2048 kHz PCM_CLK at 8 kHz PCM_SYNC, and also supports 4069 kHz PCM_CLK at 16 kHz PCM_SYNC.

The module supports a 16-bit linear data format. The following figure shows the sequence diagram of short frame mode (PCM_SYNC = 8 kHz, PCM_CLK = 2048 kHz).

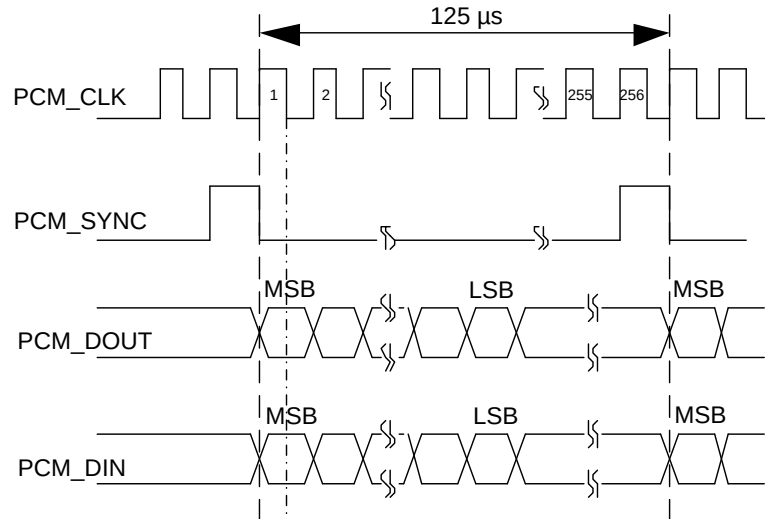


Figure 23: Timing Sequence for Short Frame Mode

Table 14: Pin Definition of PCM Interface

Pin Name	Pin No.	I/O	Description	Comment
PCM_SYNC	26	DIO	PCM data frame sync	When the module is in the master mode, this pin is in the output state. when the module is in the slave*mode, it is in the input state. If unused, keep them open.
PCM_CLK	27	DIO	PCM clock	
PCM_DIN	24	DI	PCM data input	
PCM_DOUT	25	DO	PCM data output	

Table 15: Pin Definition of I2C Interface

Pin Name	Pin No.	I/O	Description	Comment
I2C_SCL	41	OD	I2C serial clock (for external codec)	Used for external codec. An external 1.8 V pull-up resistor is needed.
I2C_SDA	42	OD	I2C serial data (for external codec)	

Clock and mode can be configured by software, and the default configuration is short frame synchronization format with 2048 kHz PCM_CLK and 8 kHz PCM_SYNC.

The following is a reference design for the PCM and I2C interfaces with external Codec chip.

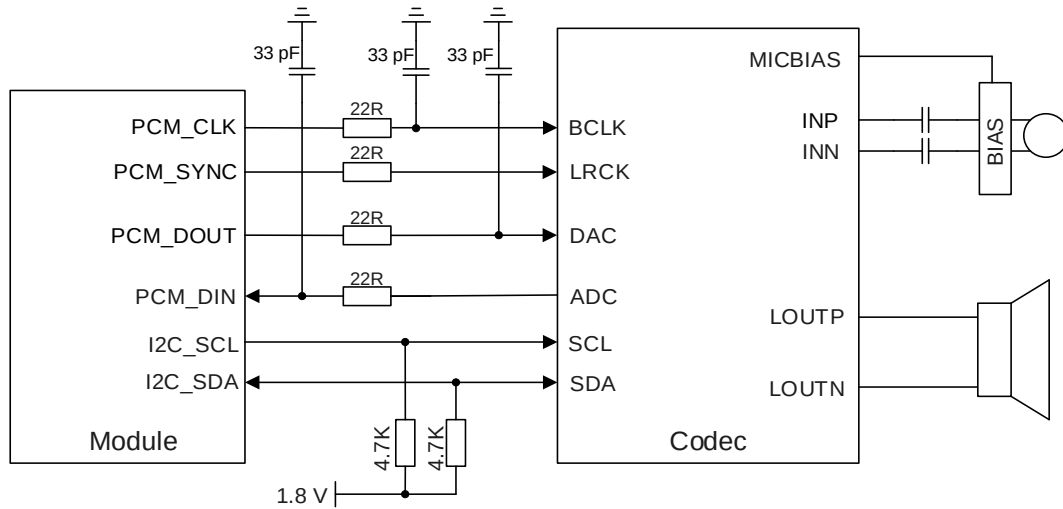


Figure 24: Reference Design of PCM and I2C Interfaces

NOTE

It is recommended to reserve the RC ($R = 22\ \Omega$, $C = 33\ \text{pF}$) circuit on the PCM signal trace and the capacitor should be placed close to the module, especially on PCM_CLK.

4.5. SPI Interface

The module provides one SPI interface that supports master mode with a maximum clock frequency of 52 MHz.

Table 16: Pin Definition of SPI Interface

Pin Name	Pin No.	I/O	Description	Comment
SPI_CLK	40	DO	SPI clock	
SPI_CS	37	DO	SPI chip select	If unused, keep them open.
SPI_DIN	39	DI	SPI data input	
SPI_DOUT	38	DO	SPI data output	

4.6. Analog Audio Interfaces

The module provides one analog audio input channel and one analog audio output channel. Pin definition is shown in the following table.

Table 17: Pin Definition of Analog Audio Interfaces

Channel	Pin Name	Pin No.	I/O	Description
POUT	MICBIAS	140	PO	Bias voltage output for microphone
AIN	MIC_P	125	AI	Microphone analog input (+)
	MIC_N	126		Microphone analog input (-)
AOUT	SPK_P	124	AO	Analog audio differential output (+)
	SPK_N	123		Analog audio differential output (-)

- The POUT channel is a bias voltage output and typically used to provide DC bias current to the microphone.
- AIN channels are differential inputs and are used for microphone input. Electret microphones are usually used.
- AOUT channels are differential outputs and usually used for earpiece.

4.6.1. Audio Interface Design Considerations

It is recommended to use the electret microphone with dual built-in capacitors (e.g., 10 pF and 33 pF) to filter out RF interference, thus reducing TDD noise. Without this capacitor, TDD noise could be heard during the call. Note that the resonant frequency point of a capacitor largely depends on the material and production technique. Therefore, you need to discuss with your capacitor vendors to choose the most suitable capacitor to filter out high-frequency noises.

For model(s) that support(s) GSM, the severity degree of the RF interference in the voice channel during GSM transmitting largely depends on the application design. Therefore, a suitable capacitor can be selected based on the test results. Sometimes, even no RF filtering capacitor is required. The filter capacitor on the PCB should be placed near the audio device or audio interface as close as possible, and the trace should be as short as possible. The filter capacitor should be passed before reaching other connection points.

To decrease signal interferences, RF antennas should be placed away from audio interfaces and audio traces. Power traces and audio traces should not be parallel, and they should be far away from each other.

The differential audio traces must be routed according to the differential signal layout rule.

4.6.2. Microphone Interface Circuit

The microphone interface reference circuit is shown below

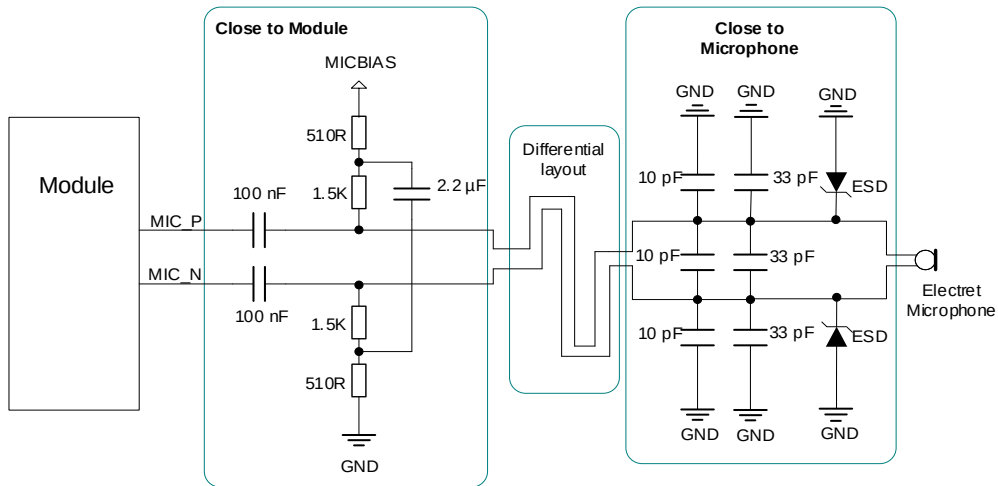


Figure 25: Microphone Interface Reference Circuit

NOTE

Since MIC channels are sensitive to ESD, it is recommended to reserve the ESD protection components.

4.6.3. Earpiece Interface Circuit

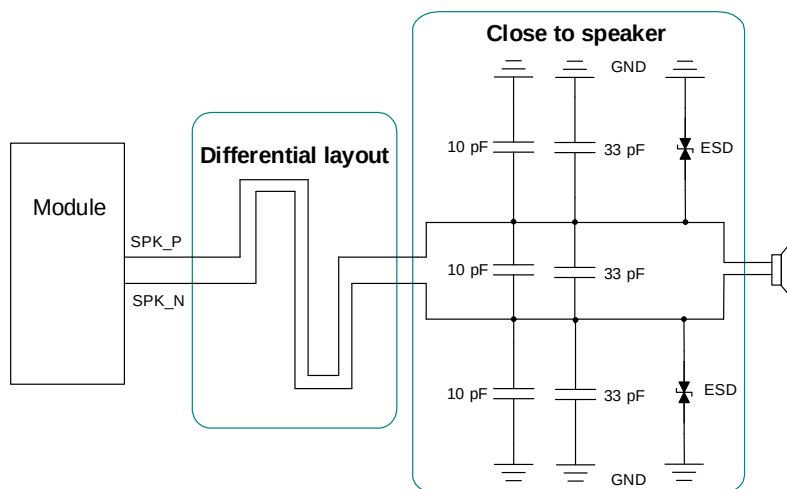


Figure 26: Earpiece Interface Reference Circuit

4.7. UARTs

The module provides two UARTs: the main UART and the debug UART. The following shows their features.

- The main UART supports 4800 bps, 9600 bps, 19200 bps, 38400 bps, 57600 bps, 115200 bps, 230400 bps, 460800 bps, 921600 bps baud rates, and the baud rate is 115200 bps by default. This interface is used for data transmission and AT command communication. Also, it supports RTS and CTS hardware flow control.
- The debug UART supports 115200 bps baud rate. It is used for the output of partial logs.

Table 18: Pin Definition of Main UART

Pin Name	Pin No.	I/O	Description	Comment
MAIN_RI	62	DO	Main UART ring indication	If unused, keep them open.
MAIN_DCD	63	DO	Main UART data carrier detect	
MAIN_CTS	64	DO	Clear to send signal from the module	Connect to the MCU's CTS. If unused, keep it open.
MAIN_RTS	65	DI	Request to send signal to the module	Connect to the MCU's CTS. If unused, keep it open.
MAIN_DTR	66	DI	Main UART data terminal ready	If unused, keep them open.
MAIN_RXD	68	DI	Main UART receive	
MAIN_TXD	67	DO	Main UART transmit	

Table 19: Pin Definition of Debug UART

Pin Name	Pin No.	I/O	Description	Comment
DBG_RXD	11	DI	Debug UART receive	If unused, keep them open.
DBG_TXD	12	DO	Debug UART transmit	

The module provides a 1.8 V UART interface. A voltage-level translator should be used if MCU is equipped with a 3.3 V UART interface. A voltage-level translator TXS0108EPWR provided by Texas

Instruments is recommended. The following figure shows a reference design.

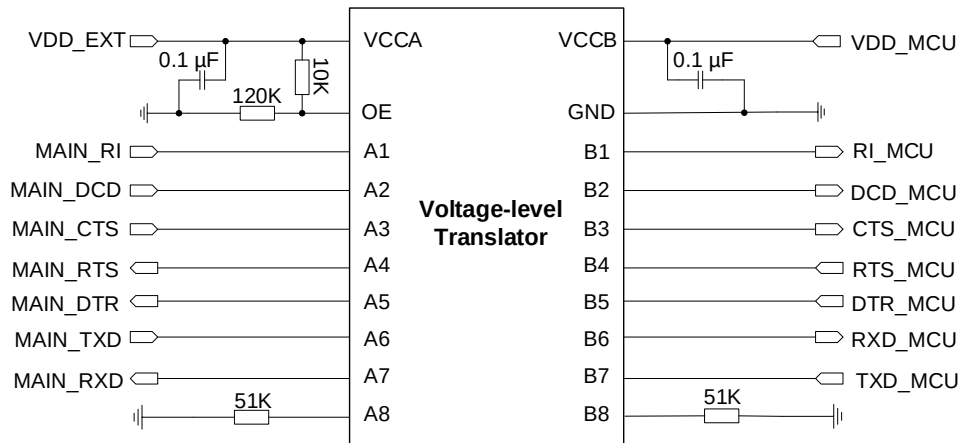


Figure 27: Reference Design of UART Interface with a Voltage-level Translator (Main UART)

Visit <http://www.ti.com> for more information.

Another example with transistor circuit is shown as below. For the design of circuits shown in dotted lines, see that shown in solid lines, but pay attention to the direction of connection.

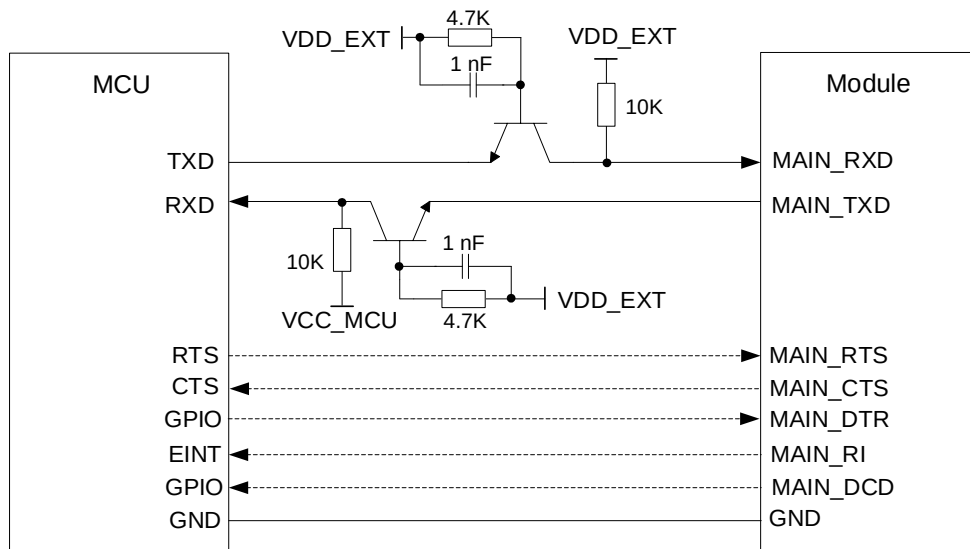


Figure 28: Reference Design of UART Interface with Transistor Circuit (Main UART)

NOTE

1. Transistor circuit solution is not suitable for applications with baud rates exceeding 460 kbps.
2. Please note that the module's CTS is connected to the MCU's CTS, and the module's RTS is

connected to the MCU's RTS.

3. To increase the stability of UART communication, it is recommended to add UART hardware flow control design.
4. The level-shifting circuits (**Figure 27** and **Figure 28**) take the main UART as an example. The circuits of the debug UART is connected in the same way as the main UART.

4.8. SD Card Interface

The module provides one SD card interface which supports SD 3.0 protocol.

Table 20: Pin Definition of SD Card Interface

Pin Name	Pin No.	I/O	Description	Comment
SD_SDIO_CLK	32	DO	SD card SDIO clock	
SD_SDIO_CMD	33	DIO	SD card SDIO command	
SD_SDIO_DATA0	31	DIO	SD card SDIO data bit 0	If unused, keep them open.
SD_SDIO_DATA1	30	DIO	SD card SDIO data bit 1	
SD_SDIO_DATA2	29	DIO	SD card SDIO data bit 2	
SD_SDIO_DATA3	28	DIO	SD card SDIO data bit 3	
SD_SDIO_VDD	34	PO	SD card SDIO power supply	
SD_DET*	23	DI	SD card hot-plug detect	It is recommended to reserve a 10 nF capacitor connected to GND.

The following figure illustrates a reference design of SD card interface with the module.

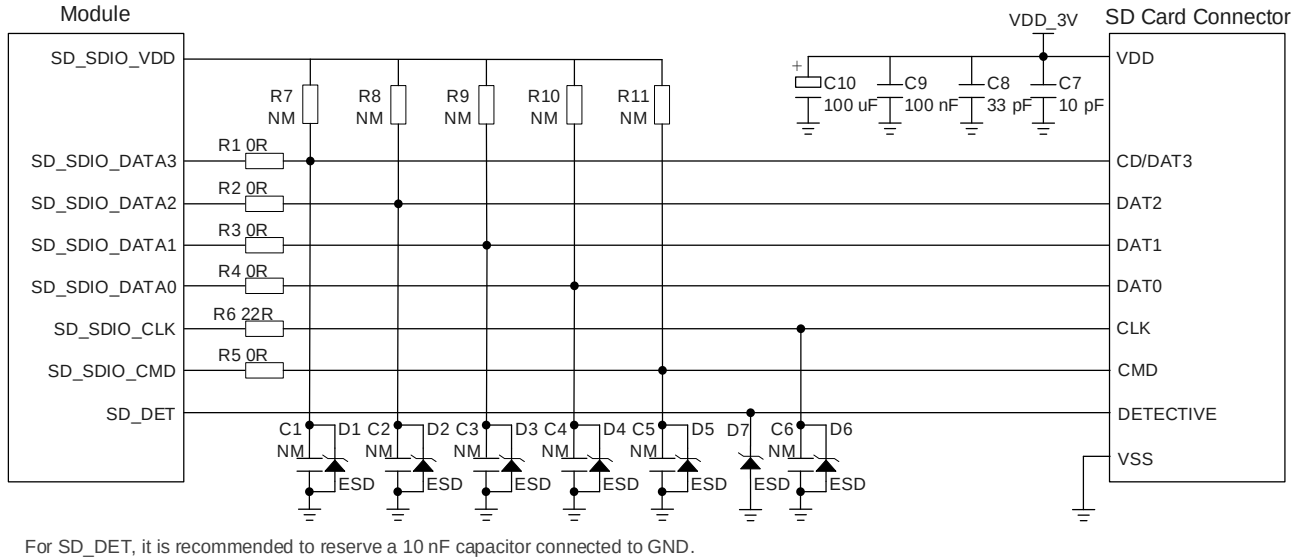


Figure 29: Reference Circuit of SD Card Interface

In SD card interface design, in order to ensure good communication performance with SD card, the following design principles should be complied with:

- The voltage range of SD card power supply VDD_3V is 2.7–3.6 V and a sufficient current of at least 0.8 A should be provided. The maximum output current of SD_SDIO_VDD is 50 mA which can only be used for SDIO pull-up resistors, an external power supply is needed for SD card.
- To avoid jitter of bus, resistors R7–R11 are not mounted by default and needed to pull up the SDIO to SD_SDIO_VDD. Value of these resistors is 10 kΩ to 100 kΩ and the recommended value is 100 kΩ. SD_SDIO_VDD should be used as the pull-up power.
- In order to improve signal quality, it is recommended to add 0 Ω resistors R1 to R5 in series between the module and the SD card and add a 15–24 Ω resistor in series to SD_SDIO_CLK. The trace distance between the SD_SDIO_CLK pin and the resistor must be less than 5 mm. The bypass capacitors C1 to C6 are reserved and not mounted by default. All resistors and bypass capacitors should be placed close to the module.
- In order to offer good ESD protection, it is recommended to add an ESD protection component on SD card pins near the SD card connector with junction capacitance less than 8 pF.
- It is important to route the SDIO signal traces with ground surrounded. The impedance of SDIO data trace is 50 Ω (±10 %).
- Keep SDIO signals far away from other sensitive circuits/signals such as RF circuits, analog signals, as well as noisy signals such as clock signals, DC-DC signals.
- It is recommended to keep the traces of SD_SDIO_CLK, SD_SDIO_DATA [0:3] and SD_SDIO_CMD with equal length (the difference among them is less than 1 mm) and the total routing length needs to be less than 50 mm.
- Make sure the adjacent trace spacing is twice of the trace width and the load capacitance of SDIO Bus should be less than 15 pF.

4.9. ADC Interfaces

The module provides two Analog-to-Digital Converter (ADC) interfaces. In order to improve the accuracy of ADC, the trace of ADC interfaces should be surrounded by ground.

Table 21: Pin Definition of ADC Interface

Pin Name	Pin No.	I/O	Description	Comment
ADC0	45	AI	General-purpose ADC interface	If unused, keep them open.
ADC1	44	AI		

The voltage value on ADC pins can be read via **AT+QADC=<port>**:

- **AT+QADC=0**: read the voltage value on ADC0
- **AT+QADC=1**: read the voltage value on ADC1

For more details about the AT command, see *document [3]*.

The following table describes the characteristic of the ADC interfaces.

Table 22: Characteristics of ADC Interfaces

Parameter	Min.	Typ.	Max.	Unit
ADC0 Input Voltage Range	0	-	VBAT_BB	V
ADC1 Input Voltage Range	0	-	VBAT_BB	V
ADC Resolution	-	12	-	bits

NOTE

1. The input voltage of ADC should not exceed its corresponding voltage range.
2. It is prohibited to supply any voltage to ADC pin when VBAT is removed.
3. If ADC interface uses a resistor divider circuit design, the resistance of the external voltage-dividing resistor must be less than 100 kΩ, otherwise the measurement accuracy of the ADC will be significantly reduced.

4.10. RGMII*/RMII Interface

The module provides one RGMII/RMII interface that can be used to connect 100/1000 Mbps Ethernet PHYs.

Table 23: Pin Definition of RGMII/RMII Interface

Pin Name	Pin No.	I/O	Description	Comment
RGMII/RMII_RX_1	73	DI	RGMII/RMII receive data bit 1	
RGMII/RMII_CTL_RX	74	DI	RGMII/RMII receive control	
RGMII/RMII_CLK	75	DI	RGMII/RMII clock	
RGMII/RMII_RX_0	76	DI	RGMII/RMII receive data bit 0	
RGMII/RMII_TX_0	77	DO	RGMII/RMII transmit data bit 0	
RGMII/RMII_TX_1	78	DO	RGMII/RMII transmit data bit 1	
RGMII_RX_2	79	DI	RGMII receive data bit 2	1.8 V power domain for RGMII. 1.8/3.3 V power domain for RMII. If unused, keep them open.
RGMII_TX_2	80	DO	RGMII transmit data bit 2	
RGMII/RMII_CTL_TX	81	DO	RGMII/RMII transmit control	
RGMII_RX_3	82	DI	RGMII receive data bit 3	
RGMII_CK_TX	83	DO	RGMII transmit clock	
RGMII_TX_3	84	DO	RGMII transmit data bit 3	
RGMII/RMII_INT	120	DI	RGMII/RMII interrupt input	
RGMII/RMII_MD_IO	121	DIO	RGMII/RMII management data input/output	
RGMII/RMII_MD_CLK	122	DO	RGMII/RMII management data clock	
RGMII/RMII_RST_N	119	DO	RGMII/RMII reset external PHY	Cannot be pulled high before module's successful turn-on. If unused, keep it open.

The following figure shows a reference circuit of RMII MAC (3.3 V power domain) to PHY interface (3.3 V power domain).

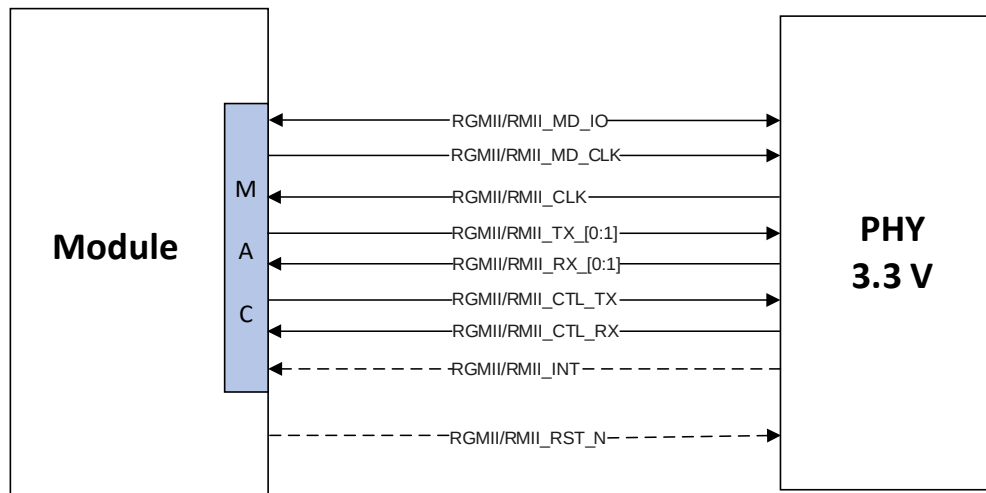


Figure 30: Reference Circuit of RMII to PHY Interface

The following figure shows a reference circuit of RGMII MAC (1.8 V power domain) to PHY (1.8 V power domain) interface.

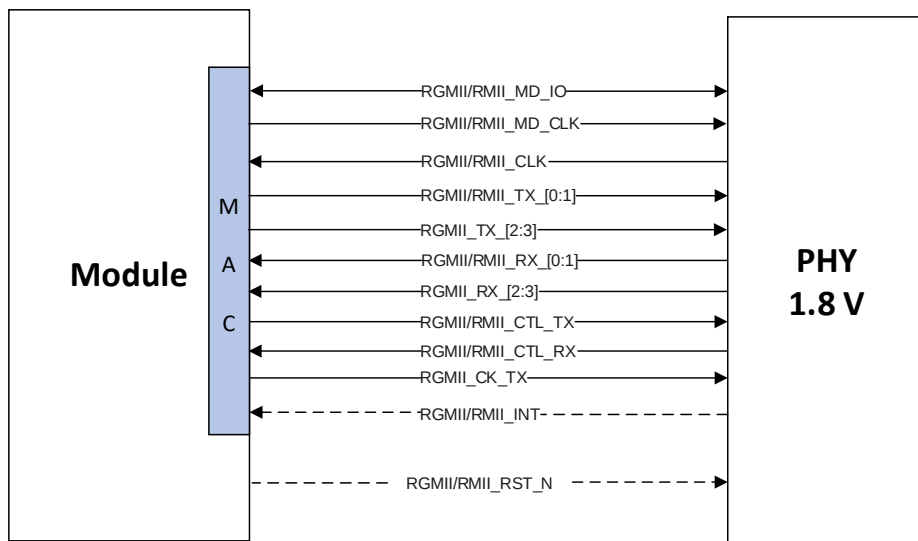


Figure 31: Reference Circuit of RGMII to PHY Interface

To enhance the reliability and availability of application designs, follow the criteria below for RGMII/RMII circuit design:

- Keep RMII and RGMII data and control signals away from power supply traces, crystals, oscillators,

magnetic devices and other sensitive signals such as RF circuits, analog signals, as well as noisy signals such as clock signals, DC-DC signals.

- The single-ended impedance of RGMII/RMII data trace is $50\ \Omega \pm 10\%$.
- The length difference of RGMII/RMII_TX_[0:1], RGMII_TX_[2:3], RGMII/RMII_CTL_TX and RGMII_CK_TX should not exceed 0.25 mm, and the space between the signal traces should be larger than twice of trace width. Similarly, The length difference of RGMII/RMII_RX_[0:1], RGMII_RX_[2:3], RGMII/RMII_CTL_RX and RGMII/RMII_CLK should not exceed 0.25 mm, and the space between the signal traces should be larger than twice of trace width.
- Spacing between Tx bus and Rx bus is larger than 2.5 times of the trace width.
- Spacing between Tx/Rx bus and other signals bus is larger than 3 times of the trace width.

NOTE

1. Pay attention to the level match shown in dotted line.
2. Pin 119 is used for the reset function and is connected to the external PHY chip. Note that it cannot be pulled high before module's successful turn-on. If an alternative pin is required to reset the PHY chip, please contact Quectel Technical Support.
3. If only RMII is used, the length difference between RGMII/RMII_TX_[0:1] and RGMII/RMII_CTL_TX needs to be less than 2 mm, and the length difference between RGMII/RMII_RX_[0:1], RGMII/RMII_CTL_RX, and RGMII/RMII_CLK needs to be less than 2 mm.
4. EC200A-CN and EC200A-EL do not support RGMII function, so pins 79–80 and pins 82–84 are reserved. For more information, contact Quectel Technical Support.

4.11. Indication Signals

The pin definition of indication signals is as follows:

Table 24: Pin Definition of Indication Signals

Pin Name	Pin No.	I/O	Description	Comment
STATUS	61	OD	Indicate the module's operation status	If unused, keep them open.
NET_MODE	5	DO	Indicate the module's network registration mode	
NET_STATUS	6	DO	Indicate the module's network activity status	
SLEEP_IND	3	DO	Indicate the module's sleep mode	

4.11.1. Network Status Indication

The network indication pins can be used to drive network status indication LEDs. The module provides two network indication pins: NET_MODE and NET_STATUS. The following tables describe pin definition and logic level changes in different network status.

Table 25: Network Status Indication Pin Level and Module Network Status

Pin Name	Status	Description
NET_MODE	Always High	Registered on LTE network
	Always Low	Others
NET_STATUS	Blink slowly (200 ms High/1800 ms Low)	Network searching
	Blink slowly (1800 ms High/200 ms Low)	Idle
	Blink quickly (125 ms High/125 ms Low)	Data transmission is ongoing
	Always High	Voice calling

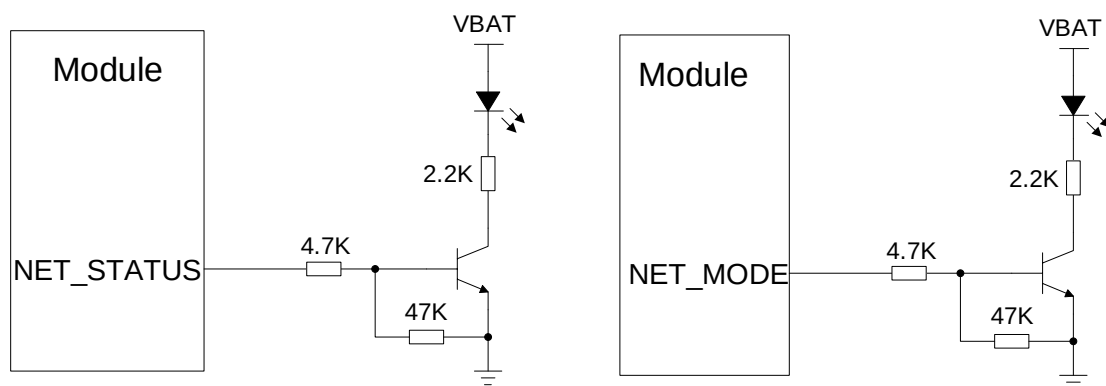


Figure 32: Reference Circuit of the Network Status Indication

4.11.2. STATUS

The STATUS pin is an open drain output for module's operation status indication. It can be connected to a GPIO of MCU with a pulled-up resistor, or as an LED indication circuit as below. When the module is turned on normally, the STATUS will present the low state. Otherwise, the STATUS will present high-impedance state.

The following figure shows different circuit designs of STATUS, and you can choose either one

according to the application demands.

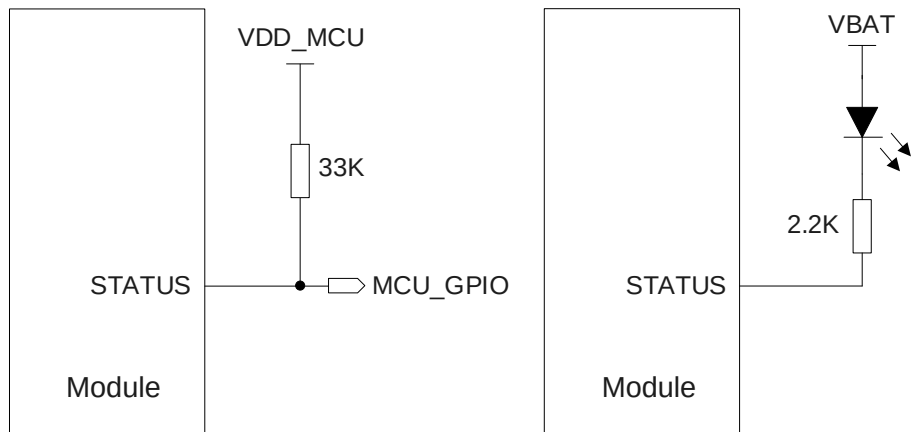


Figure 33: Reference Circuits of STATUS

NOTE

The STATUS pin cannot be used as indication of the module's shutdown status when VBAT is removed.

4.11.3. SLEEP_IND

SLEEP_IND is used for indicating the module's sleep mode. The pin will output high level when the module is in sleep mode.

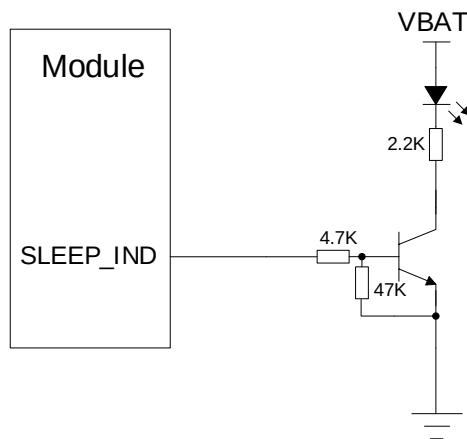


Figure 34: Reference Design of SLEEP_IND

4.12. MAIN_RI

AT+QCFG="risignaltpe","physical" can be used to configure MAIN_RI behavior. No matter on which port a URC is presented, the URC will trigger the behavior of MAIN_RI pin.

NOTE

The URC can be outputted via UART port, USB AT port and USB modem port, which can be set by **AT+QURCCFG**. The default port is USB AT port.

In addition, MAIN_RI behavior can be configured flexibly. The default behavior of the MAIN_RI is shown as below.

Table 26: MAIN_RI

Module State	MAIN_RI Level Status
Idle	MAIN_RI keeps at high level
URC	MAIN_RI outputs 120 ms low pulse when a new URC returns

The MAIN_RI behavior can be changed via **AT+QCFG**. See *document [3]* for details.

4.13. WLAN SDIO Interface

The module provides one low-power SDIO 3.0 interface and one control interface for WLAN design.

Table 27: Pin Definition of WLAN Interface

Pin Name	Pin No.	I/O	Description	Comment
WLAN_SLP_CLK	118	DO	WLAN sleep clock	
WLAN_PWR_EN	127	DO	WLAN power supply enable control	
WLAN_SDIO_DATA3	129	DIO	WLAN SDIO data bit 3	If unused, keep them open.
WLAN_SDIO_DATA2	130	DIO	WLAN SDIO data bit 2	
WLAN_SDIO_DATA1	131	DIO	WLAN SDIO data bit 1	

WLAN_SDIO_DATA0	132	DIO	WLAN SDIO data bit 0
WLAN_SDIO_CLK	133	DO	WLAN SDIO clock
WLAN_SDIO_CMD	134	DIO	WLAN SDIO command
WLAN_WAKE	135	DI	Wake up the host by an external Wi-Fi module
WLAN_EN	136	DO	WLAN function enable control

To ensure optimal performance in WLAN interface design, the following principles should be complied with:

- It is important to route the SDIO signal traces with total grounding. The impedance of SDIO signal trace is $50\ \Omega \pm 10\%$.
- To avoid bus jitter, it is necessary to reserve a pull-up resistor on the SDIO signal trace, with a resistance range of 10–100 k Ω and a recommended value of 100 k Ω . It is recommended to select VDD_EXT for pull-up power supply.
- Keep SDIO signals far away from other sensitive circuits/signals such as RF circuits, analog signals, as well as noisy signals such as clock signals, DC-DC signals.
- It is recommended to keep the trace length difference between WLAN_SDIO_CLK, WLAN_SDIO_DATA and WLAN_SDIO_CMD less than 1 mm and the total routing length less than 50 mm.
- In order to improve the signal quality, the WLAN_SDIO_CLK signal trace needs to be connected with 15–24 Ω resistors near the module in series, and the distance from the WLAN_SDIO_CLK pin to the resistor needs to be less than 5 mm.
- The spacing between SDIO signals and other signals needs to be greater than twice the trace width, and the load capacitance of SDIO bus is less than 15 pF.

5 RF Specifications

Appropriate antenna type and design should be used with matched antenna parameters according to specific application. It is required to perform a comprehensive functional test for the RF design before mass production of terminal products. The entire content of this chapter is provided for illustration only. Analysis, evaluation and determination are still necessary when designing target products.

5.1. Cellular Network

5.1.1. Antenna Interface & Frequency Bands

Table 28: Pin Definition of Cellular Network Interface

Pin Name	Pin No.	I/O	Description	Comment
ANT_DRX	35	AI	Diversity antenna interface	50 Ω characteristic impedance.
ANT_MAIN	49	AIO	Main antenna interface	

Table 29: Operating Frequency of EC200A-CN

Operating Frequency	Transmit (MHz)	Receive (MHz)
EGSM900	880–915	925–960
DCS1800	1710–1785	1805–1880
WCDMA B1	1920–1980	2110–2170
WCDMA B5	824–849	869–894
WCDMA B8	880–915	925–960
LTE-FDD B1	1920–1980	2110–2170
LTE-FDD B3	1710–1785	1805–1880
LTE-FDD B5	824–849	869–894

LTE-FDD B8	880–915	925–960
LTE-TDD B34	2010–2025	2010–2025
LTE-TDD B38	2570–2620	2570–2620
LTE-TDD B39	1880–1920	1880–1920
LTE-TDD B40	2300–2400	2300–2400
LTE-TDD B41	2535–2655	2535–2655

Table 30: Operating Frequency of EC200A-AU

Operating Frequency	Transmit (MHz)	Receive (MHz)
GSM850	824–849	869–894
EGSM900	880–915	925–960
DCS1800	1710–1785	1805–1880
PCS1900	1850–1910	1930–1990
WCDMA B1	1920–1980	2110–2170
WCDMA B2	1850–1910	1930–1990
WCDMA B4	1710–1755	2110–2155
WCDMA B5	824–849	869–894
WCDMA B8	880–915	925–960
LTE-FDD B1	1920–1980	2110–2170
LTE FDD B2	1850–1910	1930–1990
LTE-FDD B3	1710–1785	1805–1880
LTE FDD B4	1710–1755	2110–2155
LTE-FDD B5	824–849	869–894
LTE-FDD B7	2500–2570	2620–2690
LTE-FDD B8	880–915	925–960

LTE-FDD B28	703–748	758–803
LTE-FDD B66	1710–1780	2110–2180
LTE-TDD B40	2300–2400	2300–2400

Table 31: Operating Frequency of EC200A-EU

Operating Frequency	Transmit (MHz)	Receive (MHz)
EGSM900	880–915	925–960
DCS1800	1710–1785	1805–1880
WCDMA B1	1920–1980	2110–2170
WCDMA B5	824–849	869–894
WCDMA B8	880–915	925–960
LTE-FDD B1	1920–1980	2110–2170
LTE-FDD B3	1710–1785	1805–1880
LTE-FDD B5	824–849	869–894
LTE-FDD B7	2500–2570	2620–2690
LTE-FDD B8	880–915	925–960
LTE-FDD B20	832–862	791–821
LTE-FDD B28	703–748	758–803
LTE-TDD B38	2570–2620	2570–2620
LTE-TDD B40	2300–2400	2300–2400
LTE-TDD B41	2535–2675	2535–2675

Table 32: Operating Frequency of EC200A-EL

Operating Frequency	Transmit (MHz)	Receive (MHz)
WCDMA B1	1920–1980	2110–2170
WCDMA B5	824–849	869–894
WCDMA B8	880–915	925–960
LTE-FDD B1	1920–1980	2110–2170
LTE-FDD B3	1710–1785	1805–1880
LTE-FDD B5	824–849	869–894
LTE-FDD B7	2500–2570	2620–2690
LTE-FDD B8	880–915	925–960
LTE-FDD B20	832–862	791–821
LTE-FDD B28	703–748	758–803
LTE-TDD B38	2570–2620	2570–2620
LTE-TDD B40	2300–2400	2300–2400
LTE-TDD B41	2535–2675	2535–2675

NOTE

B41 supports 120 MHz (2535–2655) by default for EC200A-CN, and B41 supports 140 MHz (2535–2675 MHz) by default for EC200A-EU/-EL.

5.1.2. Antenna Tuner Control Interfaces

The module can use GRFC (generic RF control) interfaces to control external antenna tuner.

Table 33: Pin Definition of GRFC Interfaces

Pin Name	Pin No.	I/O	Description	Comment
GRFC1	144	DO	Generic RF controller	If unused, keep them open.
GRFC2	143	DO		

5.1.3. Transmitting Power

The following table shows the RF transmitting power of the module.

Table 34: Transmitting Power

Frequency	Max. Transmitting Power	Min. Transmitting Power
GSM850	33 dBm $\pm$ 2 dB	5 dBm $\pm$ 5 dB
EGSM900	33 dBm $\pm$ 2 dB	5 dBm $\pm$ 5 dB
DCS1800	30 dBm $\pm$ 2 dB	0 dBm $\pm$ 5 dB
PCS1900	30 dBm $\pm$ 2 dB	0 dBm $\pm$ 5 dB
GSM850 (8-PSK)	27 dBm $\pm$ 3 dB	5 dBm $\pm$ 5 dB
EGSM900 (8-PSK)	27 dBm $\pm$ 3 dB	5 dBm $\pm$ 5 dB
DCS1800 (8-PSK)	26 dBm $\pm$ 3 dB	0 dBm $\pm$ 5 dB
PCS1900 (8-PSK)	26 dBm $\pm$ 3 dB	0 dBm $\pm$ 5 dB
WCDMA B1/B2/B4/B5/B8	23 dBm $\pm$ 2 dB	< -49 dBm
LTE-FDD B1/B2/B3/B4/B5/B7/B8/B20/B28/B66	23 dBm $\pm$ 2 dB	< -39 dBm
LTE-TDD B34/B38/B39/B40/B41	23 dBm $\pm$ 2 dB	< -39 dBm

NOTE

For GPRS transmission on 4 uplink timeslots, the maximum output power reduction is 4.0 dB. The design conforms to 3GPP TS 51.010-1 **subclause 13.16**.

5.1.4. Receiver Sensitivity

The following table shows conducted Rx sensitivity of the module.

Table 35: Conducted RF Receiver Sensitivity of EC200A-CN (Unit: dBm)

Frequency	Receiving Sensitivity (Typ.)			3GPP Requirement (SIMO)
	Primary	Diversity	SIMO ¹¹	
EGSM900	-108.1	-	-	-102.4
DCS1800	-107	-	-	-102.4
WCDMA B1	-109.7	-	-	-106.7
WCDMA B5	-111.1	-	-	-104.7
WCDMA B8	-110.3	-	-	-103.7
LTE-FDD B1	-97.7	-98.4	-101	-96.3
LTE-FDD B3	-97.3	-97.5	-99.5	-93.3
LTE-FDD B5	-98.1	-98.9	-101	-94.3
LTE-FDD B8	-97.7	-98.7	-101	-93.3
LTE-TDD B34	-96.8	-97.6	-100	-96.3
LTE-TDD B38	-96.8	-97.4	-100	-96.3
LTE-TDD B39	-97.1	-97.6	-100.3	-96.3
LTE-TDD B40	-97.4	-98.8	-100.9	-96.3
LTE-TDD B41	-96.7	-97	-99.6	-94.3

¹¹ For the SIMO receiving sensitivity, LTE bands are tested with 2 Rx antennas.

Table 36: Conducted RF Receiver Sensitivity of EC200A-AU (Unit: dBm)

Frequency	Receiving Sensitivity (Typ.)			3GPP Requirement (SIMO)
	Primary	Diversity	SIMO ¹¹	
GSM850	-109.7	-	-	-102.4
EGSM900	-109.4	-	-	-102.4
DCS1800	-107.1	-	-	-102.4
PCS1900	-107	-	-	-102.4
WCDMA B1	-109.6	-	-	-106.7
WCDMA B2	-108.6	-	-	-104.7
WCDMA B4	-108.2	-	-	-106.7
WCDMA B5	-111.2	-	-	-104.7
WCDMA B8	-110.1	-	-	-103.7
LTE-FDD B1	-97.8	-97.9	-100.8	-96.3
LTE FDD B2	-96.7	-97.8	-99.8	-94.3
LTE-FDD B3	-96.6	-97.3	-99.7	-93.3
LTE FDD B4	-97	-97.7	-100.1	-96.3
LTE-FDD B5	-98.1	-99	-100.7	-94.3
LTE-FDD B7	-96.1	-97.8	-100	-94.3
LTE-FDD B8	-96.8	-98.6	-100.1	-93.3
LTE-FDD B28	-98.7	-99.6	-102	-94.8
LTE-FDD B66	-97.1	-97.9	-100.4	-96.5
LTE-TDD B40	-97.3	-98.8	-101.1	-96.3

Table 37: Conducted RF Receiver Sensitivity of EC200A-EU (Unit: dBm)

Frequency	Receiving Sensitivity (Typ.)			3GPP Requirement (SIMO)
	Primary	Diversity	SIMO ¹¹	
EGSM900	-108.4	-	-	-102.4
DCS1800	-106.7	-	-	-102.4
WCDMA B1	-110.1	-	-	-106.7
WCDMA B5	-110.8	-	-	-104.7
WCDMA B8	-110.1	-	-	-103.7
LTE-FDD B1	-97.9	-98.1	-101	-96.3
LTE-FDD B3	-96.7	-97.3	-100	-93.3
LTE-FDD B5	-98.4	-99.1	-101.8	-94.3
LTE-FDD B7	-95	-96.8	-99	-94.3
LTE-FDD B8	-97.8	-98.7	-101.3	-93.3
LTE-FDD B20	-96.7	-98.2	-100.5	-93.3
LTE-FDD B28	-98.3	-99	-101.7	-94.8
LTE-TDD B38	-96.8	-97.1	-99.8	-96.3
LTE-TDD B40	-97.1	-98.3	-100.8	-96.3
LTE-TDD B41	-96.2	-97.1	-99.7	-94.3

Table 38: Conducted RF Receiver Sensitivity of EC200A-EL (Unit: dBm)

Frequency	Receiving Sensitivity (Typ.)			3GPP Requirement (SIMO)
	Primary	Diversity	SIMO ¹¹	
WCDMA B1	-109.3	-	-	-106.7
WCDMA B5	-110.4	-	-	-104.7
WCDMA B8	-111	-	-	-103.7
LTE-FDD B1	-97.8	-98.8	-101.2	-96.3

LTE-FDD B3	-97.4	-96.9	-99.9	-93.3
LTE-FDD B5	-97.7	-98.4	-100.9	-94.3
LTE-FDD B7	-96.9	-97	-99.6	-94.3
LTE-FDD B8	-98.3	-98.9	-101.8	-93.3
LTE-FDD B20	-97.8	-97	-100.4	-93.3
LTE-FDD B28	-99.1	-98.5	-101.8	-94.8
LTE-TDD B38	-96.8	-97.2	-99.2	-96.3
LTE-TDD B40	-98	-97.7	-100.4	-96.3
LTE-TDD B41	-97.4	-96.8	-99.8	-94.3

5.1.5. Reference Design

The module provides two RF antenna interfaces for antenna connection.

It is recommended to reserve a dual L-type circuit for better RF performance. The capacitors C1–C4 are not mounted by default.

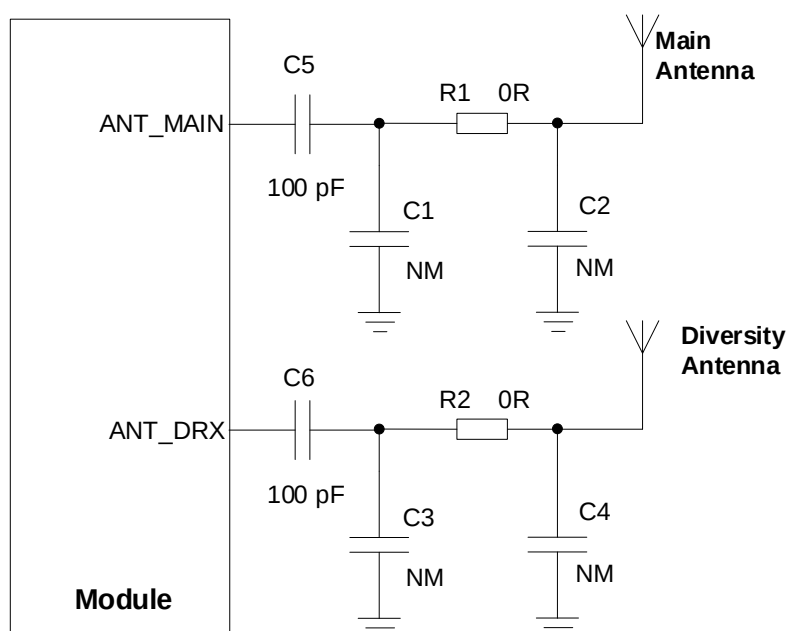


Figure 35: Reference Circuit for RF Antenna Interfaces

NOTE

1. Place the dual-L type components (R1, R2, C1–C6) to antennas as close as possible.
2. Notes on C5 and C6:
 - 1) If there is DC power at the antenna ports, place a capacitor on C5 and C6 to prevent short circuit to ground. The capacitance value is recommended to be 100 pF, which can be adjusted according to the debugging results.
 - 2) If there is no DC power in the peripheral design:
 - Do not reserve C5 and C6.
 - If C5 and C6 have already been reserved, they should be mounted with components, and it is recommended to use 0 Ω resistors. You can also match the components according to the debugging results.

5.2. GNSS

GNSS information of the module is as follows:

- Supports GPS, BDS positioning system.
- Supports NMEA 0183 protocol and outputs NMEA sentences via USB interface (data update rate for positioning: 1 Hz by default).
- The module's GNSS function is OFF by default. It must be ON via AT commands.

For more details about GNSS technology and configurations, see the document of GNSS application note.

NOTE

GNSS is only supported by EC200A-CN and is optional.

5.2.1. Antenna Interface & Frequency Bands

Table 39: Pin Definition of GNSS Antenna Interface

Pin Name	Pin No.	I/O	Description	Comment
ANT_GNSS	47	AI	GNSS antenna interface	50 Ω characteristic impedance. If unused, keep it open.

NOTE

For EC200A-CN with built-in GNSS, pin 47 is the GNSS antenna interface. For EC200A-CN without built-in GNSS and EC200A-AU/-EU/-EL, pin 47 is RESERVED. For more information, contact Quectel Technical Support.

Table 40: GNSS Frequency (Unit: MHz)

Type	Frequency
GPS	1575.42 ±1.023 (L1)
BDS	1561.098 ±2.046 (B1I)

5.2.2. GNSS Performance

Table 41: GNSS Frequency (Unit: MHz)

Parameters	Modes	Conditions	Typ.	Units
Sensitivity	Acquisition	Autonomous	-143	dBm
	Reacquisition		-155	
	Tracking		-157	
TTFF	Cold start @ open sky	Autonomous	26.98	s
		AGNSS start	2.06	
	Warm start @ open sky	Autonomous	26.2	
		AGNSS start	1.42	
	Hot start @ open sky	Autonomous	4.47	
		AGNSS start	1.2	
Accuracy	CEP-50	Autonomous @ open sky	1.6	m

NOTE

- Tracking sensitivity: the minimum GNSS signal power at which the module can maintain lock of navigation signals (keep positioning for at least 3 minutes continuously).

2. Reacquisition sensitivity: the minimum GNSS signal power required for the module to maintain lock of navigation signals within 3 minutes after loss of lock.
3. Acquisition sensitivity: the minimum GNSS signal power at which the module can fix position of navigation signals successfully within 3 minutes after executing cold start command.

5.2.3. Reference Design

GNSS antenna connection reference circuit is shown in the figure below.

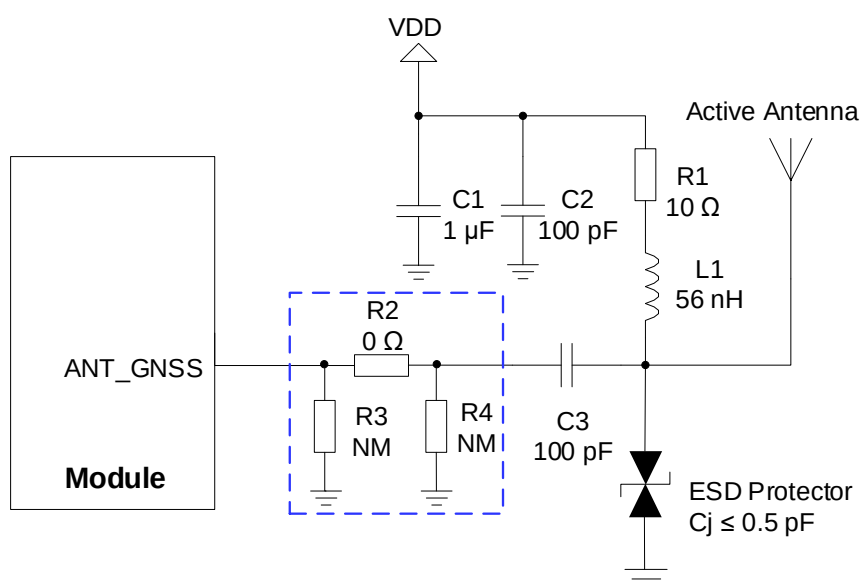


Figure 36: Reference Circuit of GNSS Active Antenna

NOTE

1. You can select an external LDO according to the active antenna types.
2. If you select passive antenna for your design, the VDD circuit is not needed.
3. Notes on C3:
 - 3) If there is DC power at the antenna ports, place a capacitor on C3 to prevent short circuit to ground. The capacitance value is recommended to be 100 pF, which can be adjusted according to the debugging results.
 - 4) If there is no DC power in the peripheral design:
 - Do not reserve C3.
 - If C3 have already been reserved, it should be mounted with components, and it is recommended to use 0 Ω resistors. You can also match the components according to the debugging results.

5.3. RF Routing Guidelines

For user's PCB, the characteristic impedance of all RF traces should be controlled to $50\ \Omega$. The impedance of the RF traces is usually determined by the trace width (W), the materials' dielectric constant, the height from the reference ground to the signal layer (H), and the spacing between RF traces and grounds (S). Microstrip or coplanar waveguide is typically used in RF layout to control characteristic impedance. The following are reference designs of microstrip or coplanar waveguide with different PCB structures.

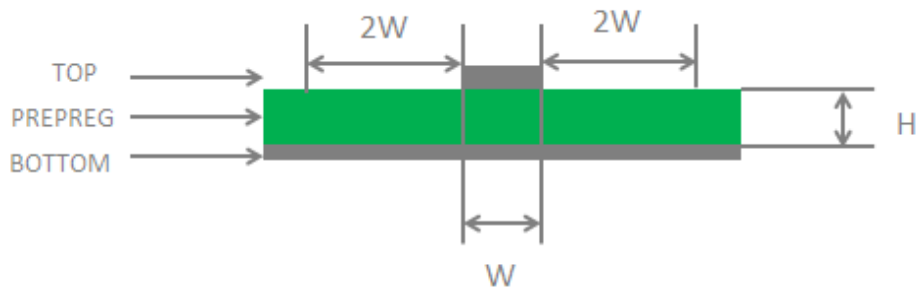


Figure 37: Microstrip Design on a 2-layer PCB

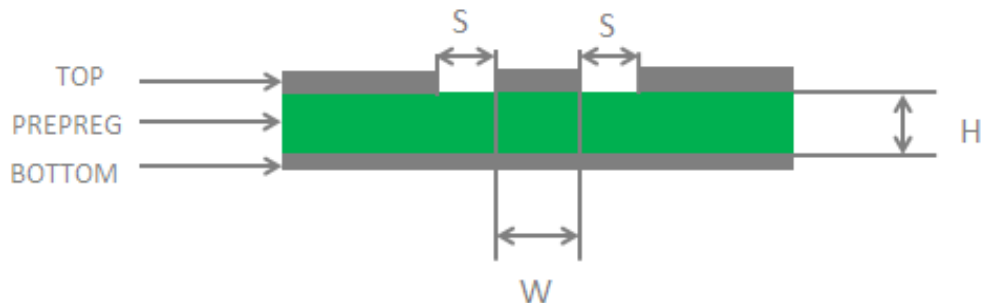


Figure 38: Coplanar Waveguide Design on a 2-layer PCB

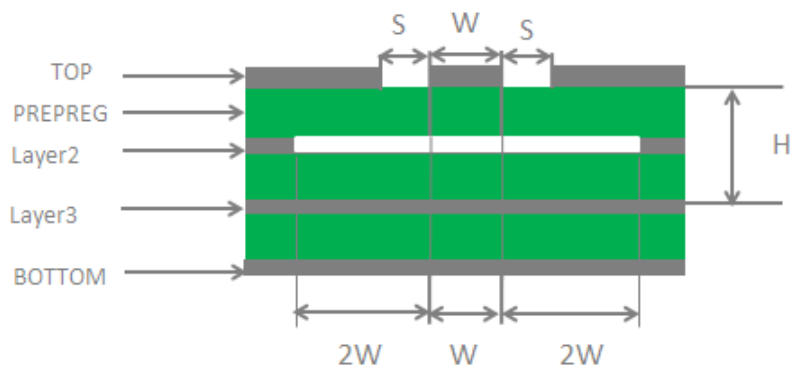


Figure 39: Coplanar Waveguide Design on a 4-layer PCB (Layer 3 as Reference Ground)

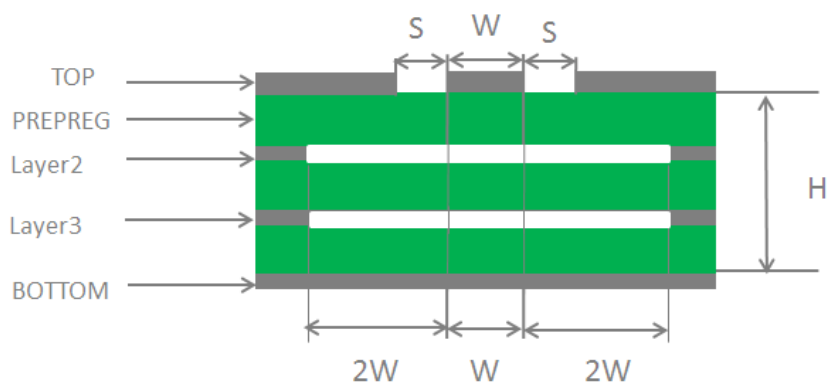


Figure 40: Coplanar Waveguide Design on a 4-layer PCB (Layer 4 as Reference Ground)

In order to ensure RF performance and reliability, the following principles should be complied with in RF layout design:

- Use an impedance simulation tool to accurately control the characteristic impedance of RF traces to 50 Ω .
- The GND pins adjacent to RF pins should not be designed as thermal relief pads, and should be fully connected to ground.
- The distance between the RF pins and the RF connector should be as short as possible, and all the right-angle traces should be changed to curved ones.
- There should be clearance under the signal pin of the antenna connector or solder joint.
- The reference ground of RF traces should be complete. Meanwhile, ground vias around RF traces and the reference ground improves RF performance. The distance between the ground vias and RF traces should be at least twice the width of RF signal traces ($2 \times W$).
- Keep RF traces away from interference sources (such as DC-DC, (U)SIM/USB/SDIO high frequency digital signals, display signals, and clock signals), and avoid intersection and paralleling between traces on adjacent layers.

For more details about RF layout, see *document [4]*.

5.4. Requirements for Antenna Design

Table 42: Requirements for Antenna Design

Antenna Type	Requirements
GNSS ¹²	● Frequency range: 1559–1577 MHz ● Polarization: RHCP or linear ● VSWR: ≤ 2 (Typ.)
	For passive antenna usage: Passive antenna gain: > 0 dBi
	For active antenna usage: Active antenna noise figure: < 1.5 dB (Typ.) Active antenna embedded LNA gain: < 26 dB (Typ.) ¹³
Cellular	● VSWR: ≤ 2 ● Efficiency: > 30 % ● Gain: 1 dBi ● Max. input power: 50 W ● Input impedance: 50Ω ● Polarization: vertical direction ● Cable insertion loss: < 1 dB: LB (< 1 GHz) < 1.5 dB: MB (1–2.3 GHz) < 2 dB: HB (> 2.3 GHz)

¹² GNSS is only supported by EC200A-CN and is optional.

¹³ If you need active antenna with an embedded LNA gain greater than 26 dB, contact Quectel Technical Support.

5.5. RF Connector Recommendation

If RF connector is used for antenna connection, it is recommended to use U.FL-R-SMT connector provided by Hirose.

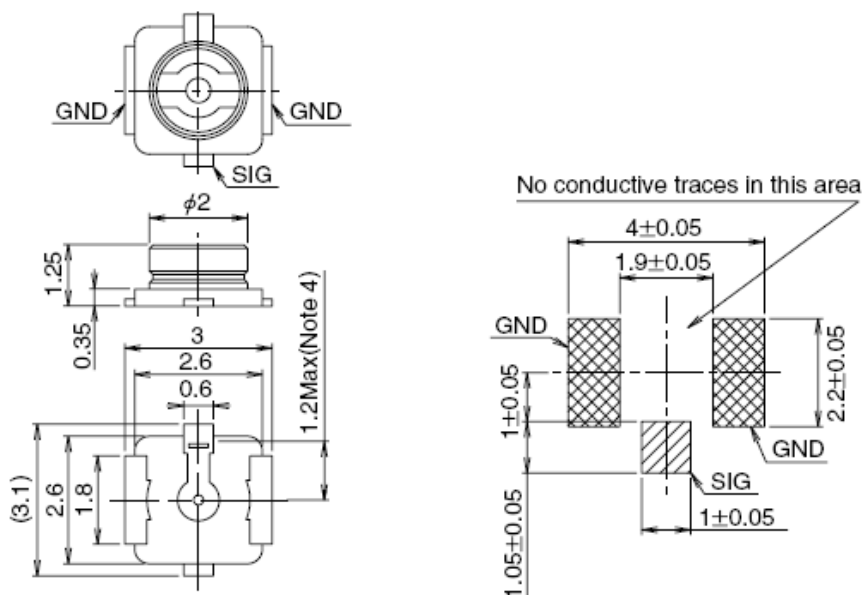


Figure 41: Dimensions of the Receptacle (Unit: mm)

U.FL-LP series mated plugs listed in the following figure can be used to match the U.FL-R-SMT connector.

Part No.	U.FL-LP-040	U.FL-LP-066	U.FL-LP(V)-040	U.FL-LP-062	U.FL-LP-088
Mated Height	2.5mm Max. (2.4mm Nom.)	2.5mm Max. (2.4mm Nom.)	2.0mm Max. (1.9mm Nom.)	2.4mm Max. (2.3mm Nom.)	2.4mm Max. (2.3mm Nom.)
Applicable cable	Dia. 0.81mm Coaxial cable	Dia. 1.13mm and Dia. 1.32mm Coaxial cable	Dia. 0.81mm Coaxial cable	Dia. 1mm Coaxial cable	Dia. 1.37mm Coaxial cable
Weight (mg)	53.7	59.1	34.8	45.5	71.7
RoHS	YES				

Figure 42: Specifications of Mated Plugs

The following figure describes the space factor of mated connector.

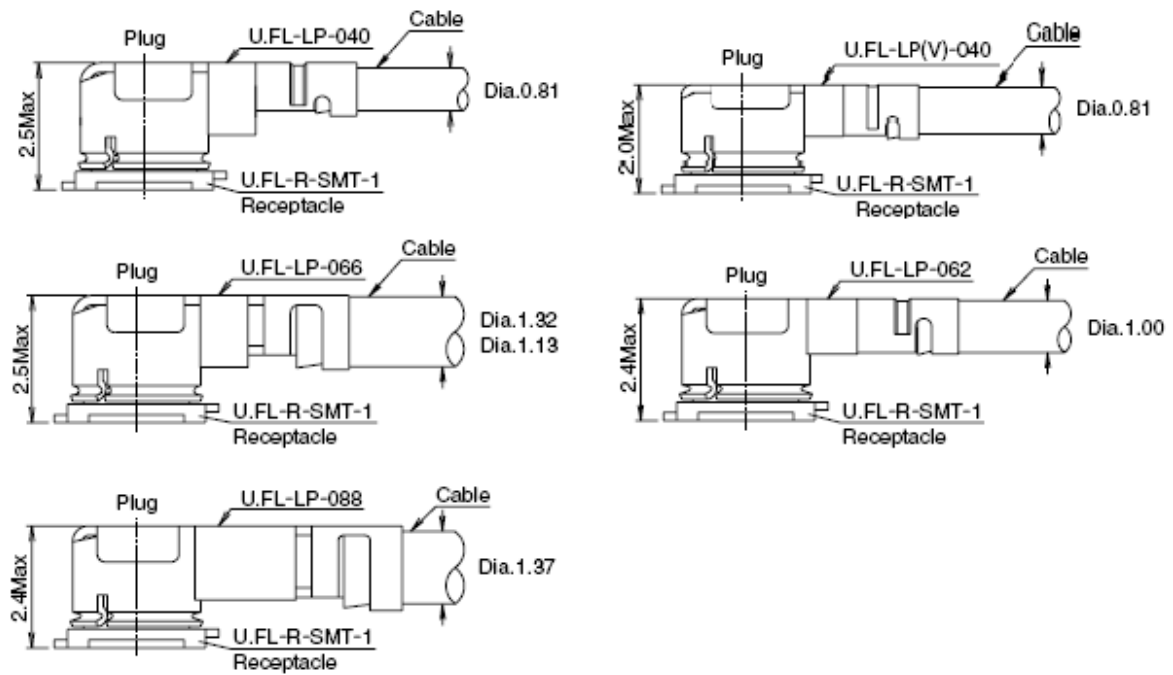


Figure 43: Space Factor of Mated Connector (Unit: mm)

For more details, visit <http://www.hirose.com>.

6 Electrical Characteristics and Reliability

6.1. Absolute Maximum Ratings

Absolute maximum ratings for power supply and voltage on digital and analog pins of the module are listed in the following table.

Table 43: Absolute Maximum Ratings

Parameter	Min.	Max.	Unit
VBAT_RF/VBAT_BB	-0.3	5.5	V
USB_VBUS	-0.3	5.5	V
Peak Current of VBAT_BB	-	0.8	A
Peak Current of VBAT_RF	-	2.0	A
Voltage on Digital Pins	-0.3	2.3	V
Input Voltage at ADC0	0	VBAT_BB	V
Input Voltage at ADC1	0	VBAT_BB	V

6.2. Power Supply Ratings

Table 44: Power Supply Ratings

Parameter	Description	Conditions	Min.	Typ.	Max.	Unit
VBAT	VBAT_BB and VBAT_RF	The actual input voltages must stay between the minimum and maximum values	3.4	3.8	4.5	V
	Voltage drop during transmitting burst	Maximum power control level	0	0	400	mV
I _{VBAT_RF}	Peak supply current	Maximum power control level	-	-	2.0	A
USB_VBUS	USB connection detection		3.0	5.0	5.25	V

6.3. Power Consumption

Table 45: EC200A-CN Power Consumption

Description	Conditions	Typ.	Unit
OFF State	Turn off	12	μA
	AT+CFUN=0 (USB disconnected)	0.91	mA
	EGSM900 @ DRX = 2 (USB disconnected)	1.74	mA
	EGSM900 @ DRX = 5 (USB disconnected)	1.31	mA
	EGSM900 @ DRX = 5 (USB Suspend)	1.46	mA
Sleep State	EGSM900 @ DRX = 9 (USB disconnected)	1.18	mA
	DCS1800 @ DRX = 2 (USB disconnected)	1.74	mA
	DCS1800 @ DRX = 5 (USB disconnected)	1.31	mA
	DCS1800 @ DRX = 5 (USB Suspend)	1.45	mA
	DCS1800 @ DRX = 9 (USB disconnected)	1.17	mA

	WCDMA @ PF = 64 (USB disconnected)	2.40	mA
	WCDMA @ PF = 64 (USB Suspend)	2.57	mA
	WCDMA @ PF = 128 (USB disconnected)	1.70	mA
	WCDMA @ PF = 256 (USB disconnected)	1.35	mA
	WCDMA @ PF = 512 (USB disconnected)	1.19	mA
	LTE-FDD @ PF = 32 (USB disconnected)	2.06	mA
	LTE-FDD @ PF = 64 (USB disconnected)	1.46	mA
	LTE-FDD @ PF = 64 (USB Suspend)	1.62	mA
	LTE-FDD @ PF = 128 (USB disconnected)	1.20	mA
	LTE-FDD @ PF = 256 (USB disconnected)	1.07	mA
	LTE-TDD @ PF = 32 (USB disconnected)	2.05	mA
	LTE-TDD @ PF = 64 (USB disconnected)	1.47	mA
	LTE-TDD @ PF = 64 (USB Suspend)	1.62	mA
	LTE-TDD @ PF = 128 (USB disconnected)	1.20	mA
	LTE-TDD @ PF = 256 (USB disconnected)	1.07	mA
Idle State	EGSM900 @ DRX = 5 (USB disconnected)	19.51	mA
	EGSM900 @ DRX = 5 (USB connected)	34.16	mA
	WCDMA @ PF = 64 (USB disconnected)	20.17	mA
	WCDMA @ PF = 64 (USB connected)	34.79	mA
	LTE-FDD @ PF = 64 (USB disconnected)	20.03	mA
	LTE-FDD @ PF = 64 (USB connected)	34.70	mA
	LTE-TDD @ PF = 64 (USB disconnected)	19.90	mA
	LTE-TDD @ PF = 64 (USB connected)	34.72	mA
GPRS Data Transmission	EGSM900 4DL/1UL @ 32.31 dBm	243	mA
	EGSM900 3DL/2UL @ 32.35 dBm	424	mA

	EGSM900 2DL/3UL @ 30.81 dBm	516	mA
	EGSM900 1DL/4UL @ 28.95 dBm	552	mA
	DCS1800 4DL/1UL @ 29.53 dBm	171	mA
	DCS1800 3DL/2UL @ 29.52 dBm	277	mA
	DCS1800 2DL/3UL @ 28.01 dBm	324	mA
	DCS1800 1DL/4UL @ 26.13 dBm	337	mA
EDGE Data Transmission	EGSM900 4DL/1UL @ 26.68 dBm	172	mA
	EGSM900 3DL/2UL @ 26.29 dBm	275	mA
	EGSM900 2DL/3UL @ 24.65 dBm	347	mA
	EGSM900 1DL/4UL @ 22.42 dBm	403	mA
	DCS1800 4DL/1UL @ 25.00 dBm	145	mA
	DCS1800 3DL/2UL @ 24.96 dBm	226	mA
	DCS1800 2DL/3UL @ 23.86 dBm	298	mA
	DCS1800 1DL/4UL @ 22.04 dBm	360	mA
WCDMA Data Transmission	WCDMA B1 HSDPA @ 21.40 dBm	566	mA
	WCDMA B5 HSDPA @ 21.34 dBm	491	mA
	WCDMA B8 HSDPA @ 21.53 dBm	477	mA
	WCDMA B1 HSUPA @ 20.86 dBm	546	mA
	WCDMA B5 HSUPA @ 20.81 dBm	470	mA
	WCDMA B8 HSUPA @ 21.34 dBm	470	mA
LTE Data Transmission	LTE-FDD B1 @ 22.72 dBm	618	mA
	LTE-FDD B3 @ 22.14 dBm	543	mA
	LTE-FDD B5 @ 22.57 dBm	543	mA
	LTE-FDD B8 @ 22.29 dBm	569	mA
	LTE-TDD B34 @ 22.62 dBm	264	mA

	LTE-TDD B38 @ 22.81 dBm	381	mA
	LTE-TDD B39 @ 22.85 dBm	270	mA
	LTE-TDD B40 @ 22.74 dBm	342	mA
	LTE-TDD B41 @ 23.09 dBm	383	mA
GSM Voice Call	EGSM900 PCL = 5 @ 32.23 dBm	240	mA
	EGSM900 PCL = 12 @ 18.94 dBm	102	mA
	EGSM900 PCL = 19 @ 4.67 dBm	72	mA
	DCS1800 PCL = 0 @ 29.50 dBm	166	mA
	DCS1800 PCL = 7 @ 15.78 dBm	86	mA
	DCS1800 PCL = 15 @ 0.59 dBm	73	mA
WCDMA Voice Call	WCDMA B1 @ 23.05 dBm	578	mA
	WCDMA B5 @ 22.42 dBm	510	mA
	WCDMA B8 @ 22.48 dBm	513	mA

Table 46: EC200A-AU Power Consumption

Description	Conditions	Typ.	Unit
OFF State	Turn off	12	μA
Sleep State	AT+CFUN=0 (USB disconnected)	0.84	mA
	EGSM900 @ DRX = 2 (USB disconnected)	1.77	mA
	EGSM900 @ DRX = 5 (USB disconnected)	1.22	mA
	EGSM900 @ DRX = 5 (USB Suspend)	1.46	mA
	EGSM900 @ DRX = 9 (USB disconnected)	1.10	mA
	DCS1800 @ DRX = 2 (USB disconnected)	1.78	mA
	DCS1800 @ DRX = 5 (USB disconnected)	1.23	mA
	DCS1800 @ DRX = 5 (USB Suspend)	1.35	mA

	DCS1800 @ DRX = 9 (USB disconnected)	1.13	mA
	WCDMA @ PF = 64 (USB disconnected)	2.28	mA
	WCDMA @ PF = 64 (USB Suspend)	2.48	mA
	WCDMA @ PF = 128 (USB disconnected)	1.63	mA
	WCDMA @ PF = 256 (USB disconnected)	1.29	mA
	WCDMA @ PF = 512 (USB disconnected)	1.13	mA
	LTE-FDD @ PF = 32 (USB disconnected)	2.02	mA
	LTE-FDD @ PF = 64 (USB disconnected)	1.46	mA
	LTE-FDD @ PF = 64 (USB Suspend)	1.62	mA
	LTE-FDD @ PF = 128 (USB disconnected)	1.18	mA
	LTE-FDD @ PF = 256 (USB disconnected)	1.05	mA
	LTE-TDD @ PF = 32 (USB disconnected)	2.04	mA
	LTE-TDD @ PF = 64 (USB disconnected)	1.45	mA
	LTE-TDD @ PF = 64 (USB Suspend)	1.61	mA
	LTE-TDD @ PF = 128 (USB disconnected)	1.19	mA
	LTE-TDD @ PF = 256 (USB disconnected)	1.06	mA
Idle State	EGSM900 @ DRX = 5 (USB disconnected)	18.30	mA
	EGSM900 @ DRX = 5 (USB connected)	32.78	mA
	WCDMA @ PF = 64 (USB disconnected)	18.91	mA
	WCDMA @ PF = 64 (USB connected)	33.35	mA
	LTE-FDD @ PF = 64 (USB disconnected)	18.60	mA
	LTE-FDD @ PF = 64 (USB connected)	33.17	mA
	LTE-TDD @ PF = 64 (USB disconnected)	18.64	mA
	LTE-TDD @ PF = 64 (USB connected)	33.18	mA
GPRS Data	GSM850 4DL/1UL @ 32.06 dBm	221	mA

Transmission	GSM850 3DL/2UL @ 32.06 dBm	392	mA
	GSM850 2DL/3UL @ 30.60 dBm	484	mA
	GSM850 1DL/4UL @ 28.68 dBm	521	mA
	EGSM900 4DL/1UL @ 32.39 dBm	227	mA
	EGSM900 3DL/2UL @ 32.34 dBm	406	mA
	EGSM900 2DL/3UL @ 30.84 dBm	497	mA
	EGSM900 1DL/4UL @ 28.96 dBm	536	mA
	DCS1800 4DL/1UL @ 29.55 dBm	160	mA
	DCS1800 3DL/2UL @ 29.56 dBm	269	mA
	DCS1800 2DL/3UL @ 28.11 dBm	315	mA
	DCS1800 1DL/4UL @ 26.24 dBm	329	mA
	PCS1900 4DL/1UL @ 29.53 dBm	158	mA
	PCS1900 3DL/2UL @ 29.52 dBm	266	mA
	PCS1900 2DL/3UL @ 28.02 dBm	314	mA
	PCS1900 1DL/4UL @ 26.18 dBm	330	mA
EDGE Data Transmission	GSM850 4DL/1UL @ 26.32 dBm	153	mA
	GSM850 3DL/2UL @ 26.09 dBm	252	mA
	GSM850 2DL/3UL @ 24.27 dBm	315	mA
	GSM850 1DL/4UL @ 22.30 dBm	370	mA
	EGSM900 4DL/1UL @ 26.13 dBm	155	mA
	EGSM900 3DL/2UL @ 25.95 dBm	254	mA
	EGSM900 2DL/3UL @ 24.42 dBm	321	mA
	EGSM900 1DL/4UL @ 22.33 dBm	376	mA
	DCS1800 4DL/1UL @ 25.72 dBm	135	mA
	DCS1800 3DL/2UL @ 25.52 dBm	219	mA

	DCS1800 2DL/3UL @ 23.99 dBm	281	mA
	DCS1800 1DL/4UL @ 22.16 dBm	339	mA
	PCS1900 4DL/1UL @ 25.95 dBm	140	mA
	PCS1900 3DL/2UL @ 25.84 dBm	230	mA
	PCS1900 2DL/3UL @ 24.13 dBm	300	mA
	PCS1900 1DL/4UL @ 22.13 dBm	365	mA
WCDMA Data Transmission	WCDMA B1 HSDPA @ 22.08 dBm	552	mA
	WCDMA B2 HSDPA @ 22.24 dBm	528	mA
	WCDMA B4 HSDPA @ 22.12 dBm	548	mA
	WCDMA B5 HSDPA @ 21.63 dBm	490	mA
	WCDMA B8 HSDPA @ 21.89 dBm	486	mA
	WCDMA B1 HSUPA @ 21.52 dBm	505	mA
	WCDMA B2 HSUPA @ 21.27 dBm	508	mA
	WCDMA B4 HSUPA @ 20.50 dBm	487	mA
	WCDMA B5 HSUPA @ 20.15 dBm	432	mA
	WCDMA B8 HSUPA @ 21.19 dBm	451	mA
LTE Data Transmission	LTE-FDD B1 @ 23.07 dBm	579	mA
	LTE-FDD B2 @ 23.36 dBm	581	mA
	LTE-FDD B3 @ 23.25 dBm	530	mA
	LTE-FDD B4 @ 22.60 dBm	527	mA
	LTE-FDD B5 @ 22.74 dBm	525	mA
	LTE-FDD B7 @ 23.22 dBm	678	mA
	LTE-FDD B8 @ 22.52 dBm	544	mA
	LTE-FDD B28A @ 22.20 dBm	519	mA
	LTE-FDD B28B @ 22.59 dBm	492	mA

GSM Voice Call	LTE-FDD B66 @ 22.68 dBm	573	mA
	LTE-TDD B40 @ 23.30 dBm	352	mA
	GSM850 PCL = 5 @ 32.07 dBm	229	mA
	GSM850 PCL = 12 @ 18.92 dBm	97	mA
	GSM850 PCL = 19 @ 4.68 dBm	69	mA
	EGSM900 PCL = 5 @ 32.26 dBm	234	mA
	EGSM900 PCL = 12 @ 18.97 dBm	101	mA
	EGSM900 PCL = 19 @ 4.78 dBm	71	mA
	DCS1800 PCL = 0 @ 29.57 dBm	163	mA
	DCS1800 PCL = 7 @ 16.06 dBm	85	mA
	DCS1800 PCL = 15 @ 0.71 dBm	71	mA
	PCS1900 PCL = 0 @ 29.56 dBm	162	mA
	PCS1900 PCL = 7 @ 16.08 dBm	83	mA
	PCS1900 PCL = 15 @ 0.63 dBm	70	mA
	WCDMA B1 @ 23.59 dBm	567	mA
	WCDMA B2 @ 23.48 dBm	568	mA
	WCDMA B4 @ 23.61 dBm	573	mA
	WCDMA B5 @ 22.96 dBm	520	mA
	WCDMA B8 @ 23.11 dBm	506	mA

Table 47: EC200A-EU Power Consumption

Description	Conditions	Typ.	Unit
OFF State	Turn off	11	μA
Sleep State	AT+CFUN=0 (USB disconnected)	0.94	mA
	EGSM900 @ DRX = 2 (USB disconnected)	1.67	mA

	EGSM900 @ DRX = 5 (USB disconnected)	1.24	mA
	EGSM900 @ DRX = 5 (USB Suspend)	1.40	mA
	EGSM900 @ DRX = 9 (USB disconnected)	1.09	mA
	DCS1800 @ DRX = 2 (USB disconnected)	1.68	mA
	DCS1800 @ DRX = 5 (USB disconnected)	1.26	mA
	DCS1800 @ DRX = 5 (USB Suspend)	1.39	mA
	DCS1800 @ DRX = 9 (USB disconnected)	1.14	mA
	WCDMA @ PF = 64 (USB disconnected)	2.26	mA
	WCDMA @ PF = 64 (USB Suspend)	2.41	mA
	WCDMA @ PF = 128 (USB disconnected)	1.59	mA
	WCDMA @ PF = 256 (USB disconnected)	1.25	mA
	WCDMA @ PF = 512 (USB disconnected)	1.07	mA
	LTE-FDD @ PF = 32 (USB disconnected)	1.98	mA
	LTE-FDD @ PF = 64 (USB disconnected)	1.41	mA
	LTE-FDD @ PF = 64 (USB Suspend)	1.55	mA
	LTE-FDD @ PF = 128 (USB disconnected)	1.15	mA
	LTE-FDD @ PF = 256 (USB disconnected)	1.02	mA
	LTE-TDD @ PF = 32 (USB disconnected)	2.02	mA
	LTE-TDD @ PF = 64 (USB disconnected)	1.42	mA
	LTE-TDD @ PF = 64 (USB Suspend)	1.58	mA
	LTE-TDD @ PF = 128 (USB disconnected)	1.16	mA
	LTE-TDD @ PF = 256 (USB disconnected)	1.02	mA
Idle State	EGSM900 @ DRX = 5 (USB disconnected)	18.27	mA
	EGSM900 @ DRX = 5 (USB connected)	32.85	mA
	WCDMA @ PF = 64 (USB disconnected)	18.89	mA

	WCDMA @ PF = 64 (USB connected)	33.43	mA
	LTE-FDD @ PF = 64 (USB disconnected)	18.45	mA
	LTE-FDD @ PF = 64 (USB connected)	32.99	mA
	LTE-TDD @ PF = 64 (USB disconnected)	18.44	mA
	LTE-TDD @ PF = 64 (USB connected)	33.02	mA
GPRS Data Transmission	EGSM900 4DL/1UL @ 32.77 dBm	234	mA
	EGSM900 3DL/2UL @ 32.70 dBm	424	mA
	EGSM900 2DL/3UL @ 31.26 dBm	517	mA
	EGSM900 1DL/4UL @ 29.39 dBm	558	mA
	DCS1800 4DL/1UL @ 29.50 dBm	160	mA
	DCS1800 3DL/2UL @ 29.50 dBm	269	mA
	DCS1800 2DL/3UL @ 28.01 dBm	313	mA
	DCS1800 1DL/4UL @ 26.13 dBm	329	mA
EDGE Data Transmission	EGSM900 4DL/1UL @ 27.41 dBm	163	mA
	EGSM900 3DL/2UL @ 27.23 dBm	272	mA
	EGSM900 2DL/3UL @ 25.40 dBm	339	mA
	EGSM900 1DL/4UL @ 22.99 dBm	386	mA
	DCS1800 4DL/1UL @ 24.91 dBm	135	mA
	DCS1800 3DL/2UL @ 24.78 dBm	217	mA
	DCS1800 2DL/3UL @ 23.41 dBm	281	mA
WCDMA Data Transmission	DCS1800 1DL/4UL @ 21.58 dBm	341	mA
	WCDMA B1 HSDPA @ 21.83 dBm	528	mA
	WCDMA B5 HSDPA @ 21.61 dBm	461	mA
	WCDMA B8 HSDPA @ 21.88 dBm	439	mA
	WCDMA B1 HSUPA @ 20.37 dBm	481	mA

LTE Data Transmission	WCDMA B5 HSUPA @ 20.18 dBm	415	mA
	WCDMA B8 HSUPA @ 20.45 dBm	395	mA
	LTE-FDD B1 @ 22.65 dBm	558	mA
	LTE-FDD B3 @ 22.70 dBm	587	mA
	LTE-FDD B5 @ 22.88 dBm	520	mA
	LTE-FDD B7 @ 22.44 dBm	667	mA
	LTE-FDD B8 @ 22.52 dBm	532	mA
	LTE-FDD B20 @ 22.09 dBm	515	mA
	LTE-FDD B28 @ 23.02 dBm	534	mA
	LTE-TDD B38 @ 22.53 dBm	366	mA
	LTE-TDD B40 @ 22.37 dBm	347	mA
	LTE-TDD B41 @ 22.95 dBm	377	mA
	EGSM900 PCL = 5 @ 32.69 dBm	243	mA
	EGSM900 PCL = 12 @ 19.48 dBm	103	mA
GSM Voice Call	EGSM900 PCL = 19 @ 5.48 dBm	71	mA
	DCS1800 PCL = 0 @ 29.51 dBm	163	mA
	DCS1800 PCL = 7 @ 16.03 dBm	84	mA
	DCS1800 PCL = 15 @ 0.45 dBm	71	mA
	WCDMA B1 @ 22.59 dBm	545	mA
WCDMA Voice Call	WCDMA B5 @ 22.53 dBm	487	mA
	WCDMA B8 @ 22.60 dBm	471	mA

Table 48: EC200A-EL Power Consumption

Description	Conditions	Typ.	Unit
OFF State	Turn off	9.5	μA
Sleep State	AT+CFUN=0 (USB disconnected)	0.78	mA
	WCDMA @ PF = 64 (USB disconnected)	2.62	mA
	WCDMA @ PF = 64 (USB Suspend)	2.74	mA
	WCDMA @ PF = 128 (USB disconnected)	1.95	mA
	WCDMA @ PF = 256 (USB disconnected)	1.65	mA
	WCDMA @ PF = 512 (USB disconnected)	1.42	mA
	LTE-FDD @ PF = 32 (USB disconnected)	2.50	mA
	LTE-FDD @ PF = 64 (USB disconnected)	1.89	mA
	LTE-FDD @ PF = 64 (USB Suspend)	2.04	mA
	LTE-FDD @ PF = 128 (USB disconnected)	1.57	mA
	LTE-FDD @ PF = 256 (USB disconnected)	1.42	mA
	LTE-TDD @ PF = 32 (USB disconnected)	2.49	mA
	LTE-TDD @ PF = 64 (USB disconnected)	1.87	mA
	LTE-TDD @ PF = 64 (USB Suspend)	2.12	mA
	LTE-TDD @ PF = 128 (USB disconnected)	1.53	mA
	LTE-TDD @ PF = 256 (USB disconnected)	1.37	mA
Idle State	WCDMA @ PF = 64 (USB disconnected)	17.93	mA
	WCDMA @ PF = 64 (USB connected)	32.35	mA
	LTE-FDD @ PF = 64 (USB disconnected)	17.66	mA
	LTE-FDD @ PF = 64 (USB connected)	31.97	mA
	LTE-TDD @ PF = 64 (USB disconnected)	17.66	mA
	LTE-TDD @ PF = 64 (USB connected)	32.02	mA

WCDMA Data Transmission	WCDMA B1 HSDPA @ 22.32 dBm	562	mA
	WCDMA B5 HSDPA @ 21.94 dBm	537	mA
	WCDMA B8 HSDPA @ 22.02 dBm	607	mA
	WCDMA B1 HSUPA @ 22.86 dBm	551	mA
	WCDMA B5 HSUPA @ 21.43 dBm	498	mA
	WCDMA B8 HSUPA @ 21.55 dBm	538	mA
LTE Data Transmission	LTE-FDD B1 @ 23.40 dBm	600	mA
	LTE-FDD B3 @ 23.33 dBm	652	mA
	LTE-FDD B5 @ 22.51 dBm	550	mA
	LTE-FDD B7 @ 23.04 dBm	658	mA
	LTE-FDD B8 @ 22.47 dBm	642	mA
	LTE-FDD B20 @ 22.76 dBm	593	mA
	LTE-FDD B28 @ 23.13 dBm	583	mA
	LTE-TDD B38 @ 22.76 dBm	399	mA
	LTE-TDD B40 @ 23.11 dBm	396	mA
WCDMA Voice Call	WCDMA B1 @ 23.18 dBm	575	mA
	WCDMA B5 @ 22.60 dBm	548	mA
	WCDMA B8 @ 22.64 dBm	624	mA

NOTE

The above power consumption data are tested under 50 Ω impedance.

6.4. Digital I/O Characteristics

Table 49: VDD_EXT I/O Requirements

Parameter	Description	Min.	Max.	Unit
V _{IH}	High-level input voltage	$0.7 \times VDD_EXT$	$VDD_EXT + 0.2$	V
V _{IL}	Low-level input voltage	-0.3	$0.3 \times VDD_EXT$	V
V _{OH}	High-level output voltage	$VDD_EXT - 0.2$	VDD_EXT	V
V _{OL}	Low-level output voltage	0	0.2	V

Table 50: (U)SIM High/Low-voltage I/O Requirements

Parameter	Description	Min.	Max.	Unit
V _{IH}	High-level input voltage	$0.8 \times USIM_VDD$	$USIM_VDD$	V
V _{IL}	Low-level input voltage	-0.3	$0.12 \times USIM_VDD$	V
V _{OH}	High-level output voltage	$0.7 \times USIM_VDD$	$USIM_VDD$	V
V _{OL}	Low-level output voltage	0	$0.15 \times USIM_VDD$	V

Table 51: SDIO Low-voltage I/O Requirements

Parameter	Description	Min.	Max.	Unit
V _{IH}	High-level input voltage	$0.7 \times SD_SDIO_VDD$	$SD_SDIO_VDD + 0.2$	V
V _{IL}	Low-level input voltage	-0.3	$0.3 \times SDIO_VDD$	V
V _{OH}	High-level output voltage	$SDIO_VDD - 0.2$	SD_SDIO_VDD	V
V _{OL}	Low-level output voltage	0	0.2	V

Table 52: SDIO High-voltage I/O Requirements

Parameter	Description	Min.	Max.	Unit
V _{IH}	High-level input voltage	2.0	SD_SDIO_VDD + 0.3	V
V _{IL}	Low-level input voltage	-0.3	0.8	V
V _{OH}	High-level output voltage	2.4	SD_SDIO_VDD	V
V _{OL}	Low-level output voltage	0	0.4	V

6.5. ESD Protection

Static electricity occurs naturally and it may damage the module. Therefore, applying proper ESD countermeasures and handling methods is imperative. For example, wear anti-static gloves during the development, production, assembly and testing of the module; add ESD protection components to the ESD sensitive interfaces and points in the product design.

Table 53: Electrostatic Discharge Characteristics (Temperature: 25–30 °C, Humidity: 40 ±5 %)

Tested Interfaces	Contact Discharge	Air Discharge	Unit
VBAT, GND	±8	±10	kV
Antenna Interfaces	±8	±10	kV
Other Interfaces	±0.5	±1	kV

6.6. Operating and Storage Temperatures

Table 54: Operating and Storage Temperatures

Parameter	Min.	Typ.	Max.	Unit
Normal Operating Temperature Range ¹⁴	-35	+25	+75	°C
Extended Operating Temperature Range ¹⁵	-40	-	+85	°C
Storage Temperature Range	-40	-	+90	°C

¹⁴ Within this range, the module's indicators comply with 3GPP specification requirements.

¹⁵ Within this range, the module retains the ability to establish and maintain functions such as voice, SMS, data transmission and emergency call, without any unrecoverable malfunction. Radio spectrum and radio network remain uninfluenced, whereas the value of one or more parameters, such as P_{out} , may decrease and fall below the range of the 3GPP specified tolerances. When the temperature returns to the normal operating temperature range, the module's indicators will comply with 3GPP specification requirements again.

7 Mechanical Information

This chapter describes the mechanical dimensions of the module. All dimensions are measured in millimeter (mm), and the dimensional tolerances are ± 0.2 mm unless otherwise specified.

7.1. Mechanical Dimensions

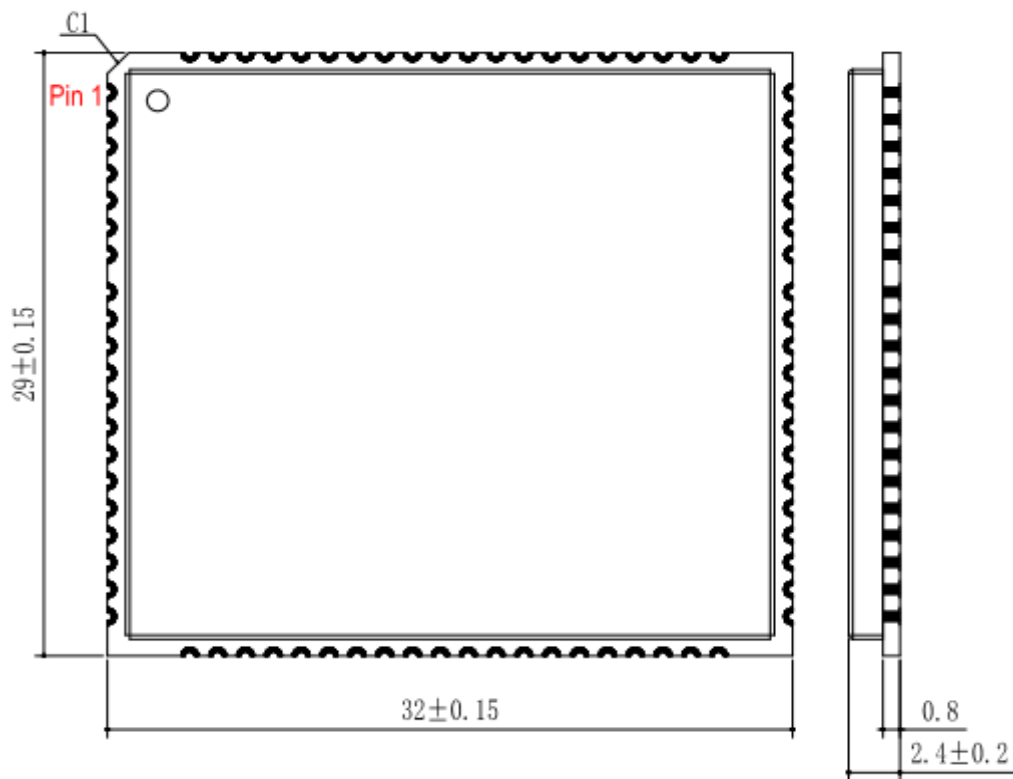


Figure 44: Module Top and Side Dimensions

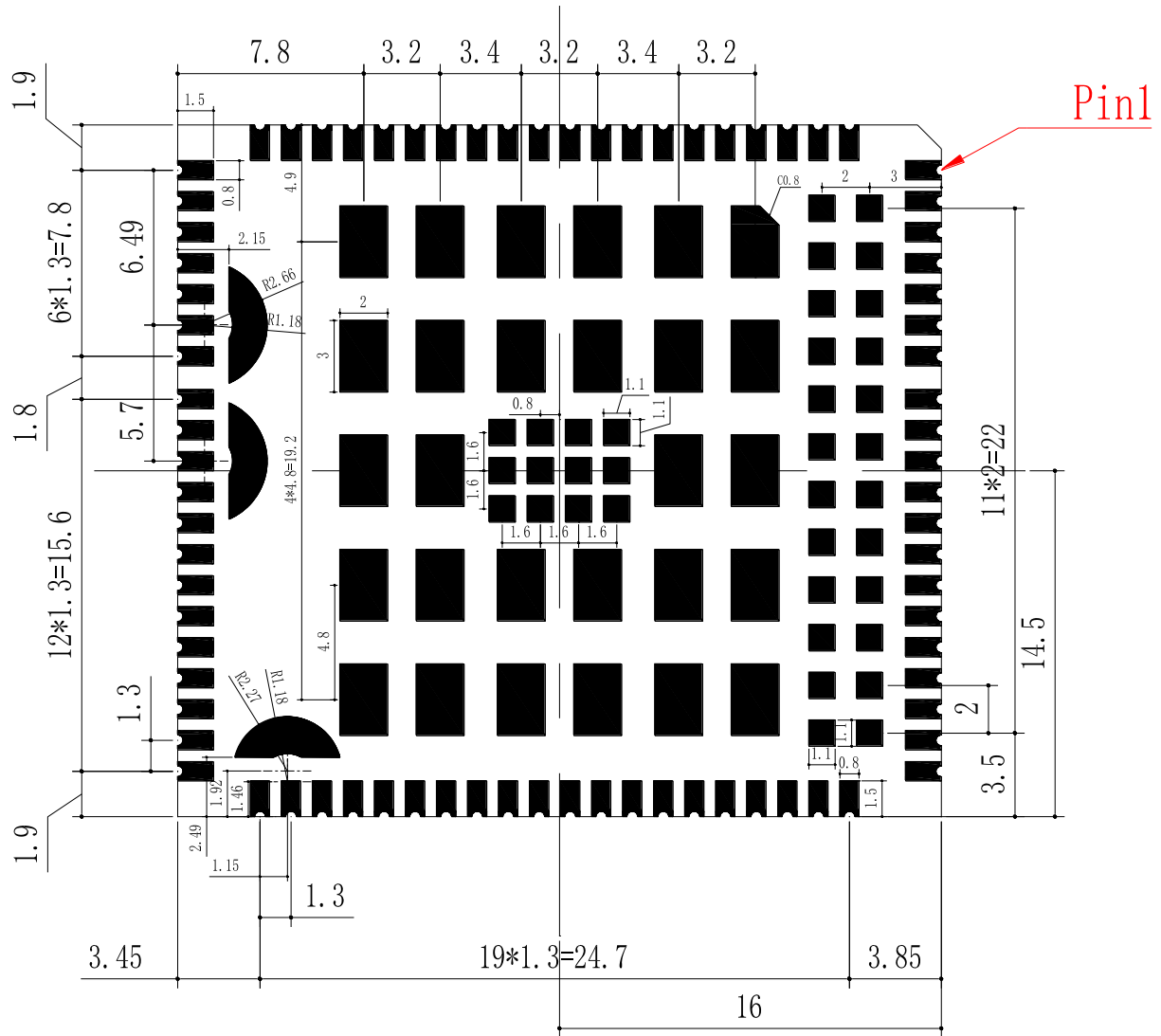


Figure 45: Bottom Dimension (Bottom View)

NOTE

The module's coplanarity standard: ≤ 0.13 mm.

7.2. Recommended Footprint

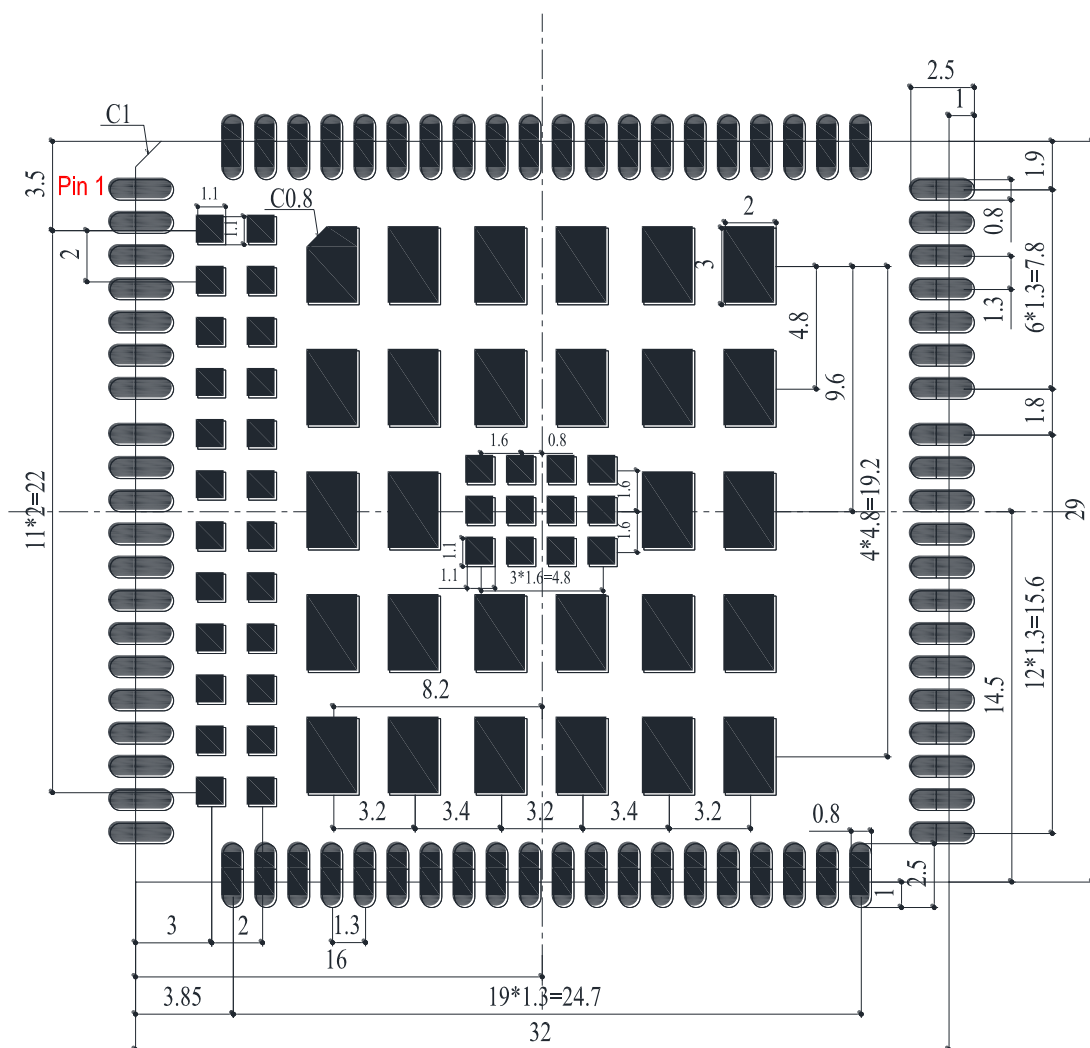


Figure 46: Recommended Footprint

NOTE

Keep at least 3 mm between the module and other components on the motherboard to improve soldering quality and maintenance convenience.

7.3. Top and Bottom Views

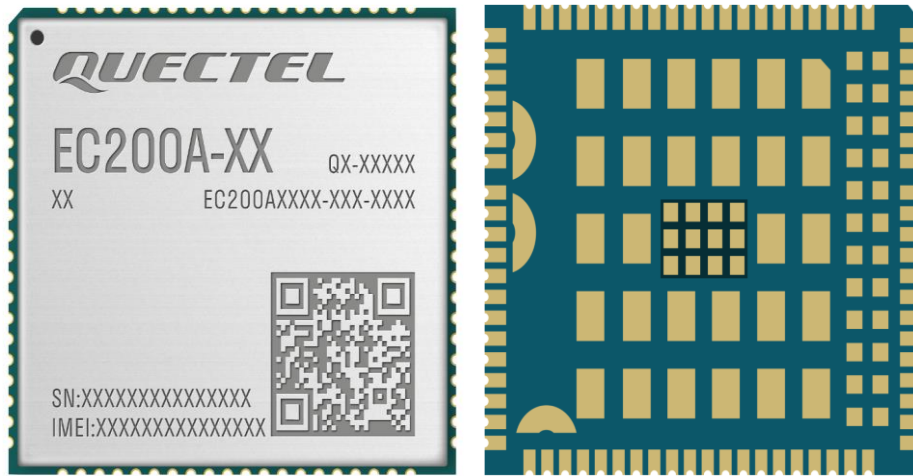


Figure 47: Top and Bottom Views of the Module

NOTE

Images above are for illustration purpose only and may differ from the actual module. For authentic appearance and label, refer to the module received from Quectel.

8 Storage, Manufacturing and Packaging

8.1. Storage Conditions

The module is provided with vacuum-sealed packaging. MSL of the module is rated as 3. The storage requirements are shown below.

1. Recommended Storage Condition: the temperature should be 23 ± 5 °C and the relative humidity should be 35–60 %.
2. Shelf life (in a vacuum-sealed packaging): 12 months in Recommended Storage Condition.
3. Floor life: 168 hours ¹⁶ in a factory where the temperature is 23 ± 5 °C and relative humidity is below 60 %. After the vacuum-sealed packaging is removed, the module must be processed in reflow soldering or other high-temperature operations within 168 hours. Otherwise, the module should be stored in an environment where the relative humidity is less than 10 % (e.g., a dry cabinet).
4. The module should be pre-baked to avoid blistering, cracks and inner-layer separation in PCB under the following circumstances:
 - The module is not stored in Recommended Storage Condition;
 - Violation of the third requirement mentioned above;
 - Vacuum-sealed packaging is broken, or the packaging has been removed for over 24 hours;
 - Before module repairing.
5. If needed, the pre-baking should follow the requirements below:
 - The module should be baked for 24 hours at 120 ± 5 °C;
 - The module must be soldered to PCB within 24 hours after the baking, otherwise it should be put in a dry environment such as in a dry cabinet.

¹⁶ This floor life is only applicable when the environment conforms to *IPC/JEDEC J-STD-033*. It is recommended to start the solder reflow process within 24 hours after the package is removed if the temperature and moisture do not conform to, or are not sure to conform to *IPC/JEDEC J-STD-033*. And do not unpack the modules in large quantities until they are ready for soldering.

NOTE

1. To avoid blistering, layer separation and other soldering issues, extended exposure of the module to the air is forbidden.
2. Take out the module from the package and put it on high-temperature-resistant fixtures before baking. If shorter baking time is desired, see *IPC/JEDEC J-STD-033* for the baking procedure.
3. Pay attention to ESD protection, such as wearing anti-static gloves, when touching the modules.

8.2. Manufacturing and Soldering

Push the squeegee to apply the solder paste on the surface of stencil, thus making the paste fill the stencil openings and then penetrate to the PCB. Apply proper force on the squeegee to produce a clean stencil surface on a single pass. To guarantee module soldering quality, the thickness of stencil for the module is recommended to be 0.15–0.20 mm. For more details, see **document [5]**.

The recommended peak reflow temperature should be 235–246 °C, with 246 °C as the absolute maximum reflow temperature. To avoid damage to the module caused by repeated heating, it is recommended that the module should be mounted only after reflow soldering for the other side of PCB has been completed. The recommended reflow soldering thermal profile (lead-free reflow soldering) and related parameters are shown below.

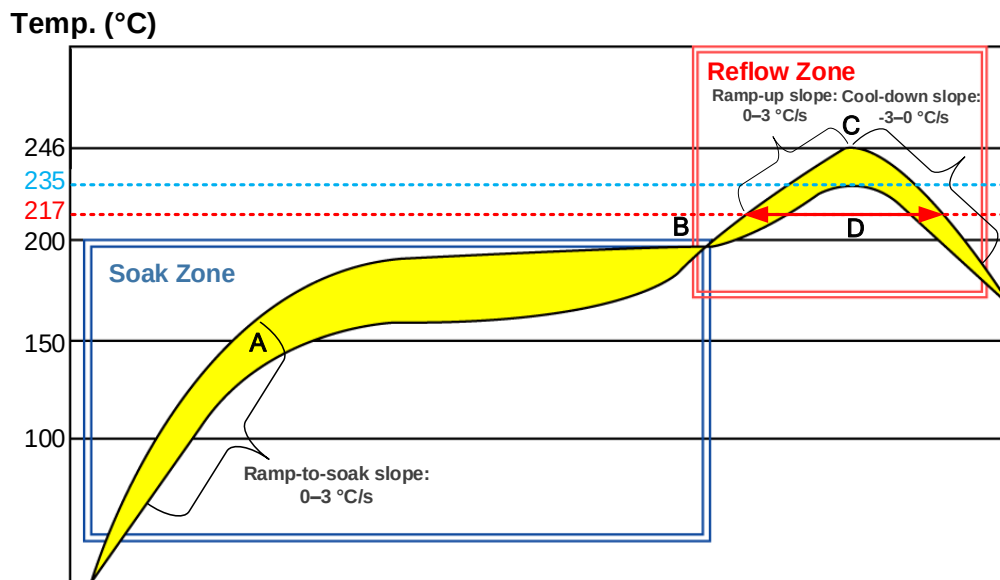


Figure 48: Recommended Reflow Soldering Thermal Profile

Table 55: Recommended Thermal Profile Parameters

Factor	Recommended Value
Soak Zone	
Ramp-to-soak slope	0–3 °C/s
Soak time (between A and B: 150 °C and 200 °C)	70–120 s
Reflow Zone	
Ramp-up slope	0–3 °C/s
Reflow time (D: over 217°C)	40–70 s
Max temperature	235–246 °C
Cool-down slope	-3–0 °C/s
Reflow Cycle	
Max reflow cycle	1

NOTE

1. The above profile parameter requirements are for the measured temperature of the solder joints. Both the hottest and coldest spots of solder joints on the PCB should meet the above requirements.
2. During manufacturing and soldering, or any other processes that may contact the module directly, NEVER wipe the module's shielding can with organic solvents, such as acetone, ethyl alcohol, isopropyl alcohol, trichloroethylene. Otherwise, the shielding can may become rusted.
3. The shielding can for the module is made of Cupro-Nickel base material. It is tested that after 12 hours' Neutral Salt Spray test, the laser engraved label information on the shielding can is still clearly identifiable and the QR code is still readable, although white rust may be found.
4. If a conformal coating is necessary for the module, do NOT use any coating material that may chemically react with the PCB or shielding cover, and prevent the coating material from flowing into the module.
5. Avoid using ultrasonic technology for module cleaning since it can damage crystals inside the module.
6. Avoid using materials that contain mercury (Hg), such as adhesives, for module processing, even if the materials are RoHS compliant and their mercury content is below 1000 ppm (0.1 %).
7. Corrosive gases may corrode the electronic components inside the module, affecting their reliability and performance, and potentially leading to a shortened service life that fails to meet the designed lifespan. Therefore, do not store or use unprotected modules in environments containing corrosive gases such as hydrogen sulfide, sulfur dioxide, chlorine, and ammonia.
8. Due to the complexity of the SMT process, contact Quectel Technical Supports in advance for any situation that you are not sure about, or any process (e.g. selective soldering, ultrasonic soldering) that is not mentioned in **document [6]**.

8.3. Packaging Specification

This chapter outlines the key packaging parameters and processes. All figures below are for reference purposes only, as the actual appearance and structure of packaging materials may vary in delivery.

The modules are packed in a tape and reel packaging as specified in the sub-chapters below.

8.3.1. Carrier Tape

Carrier tape dimensions are illustrated in the following figure and table:

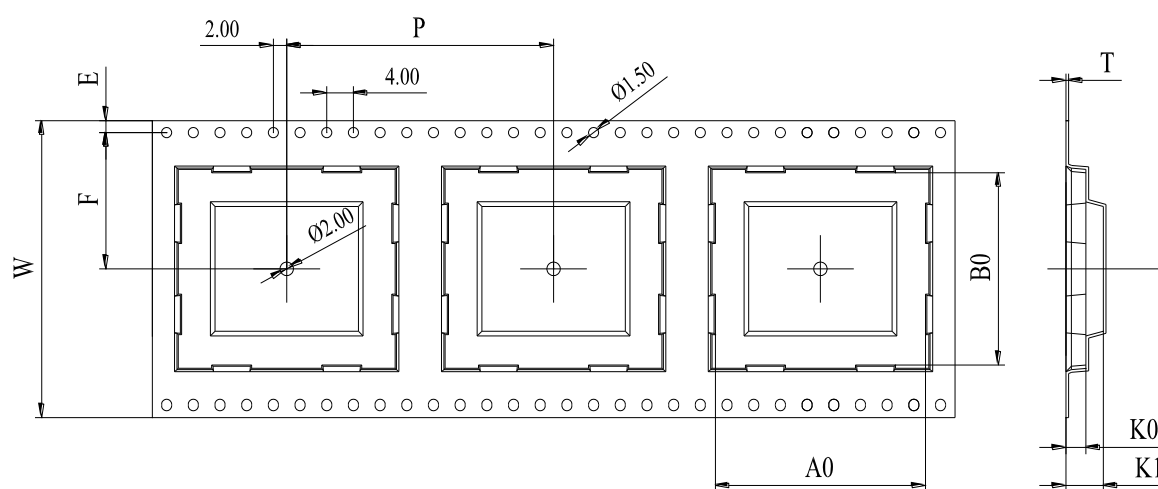


Figure 49: Carrier Tape Dimension Drawing (Unit: mm)

Table 56: Carrier Tape Dimension Table (Unit: mm)

W	P	T	A0	B0	K0	K1	F	E
44	44	0.35	32.5	29.5	3	3.8	20.2	1.75

8.3.2. Plastic Reel

Plastic reel dimensions are illustrated in the following figure and table:

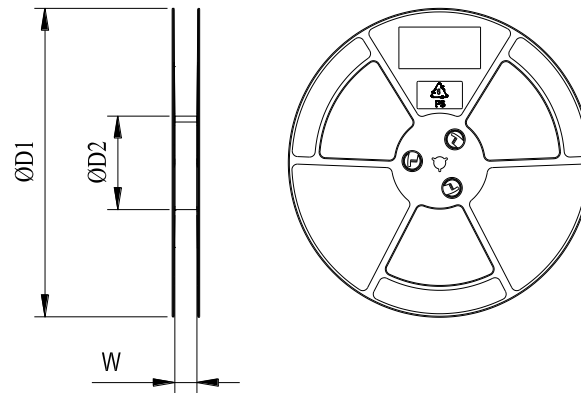


Figure 50: Plastic Reel Dimension Drawing

Table 57: Plastic Reel Dimension Table (Unit: mm)

ØD1	ØD2	W
330	100	44.5

8.3.3. Mounting Direction

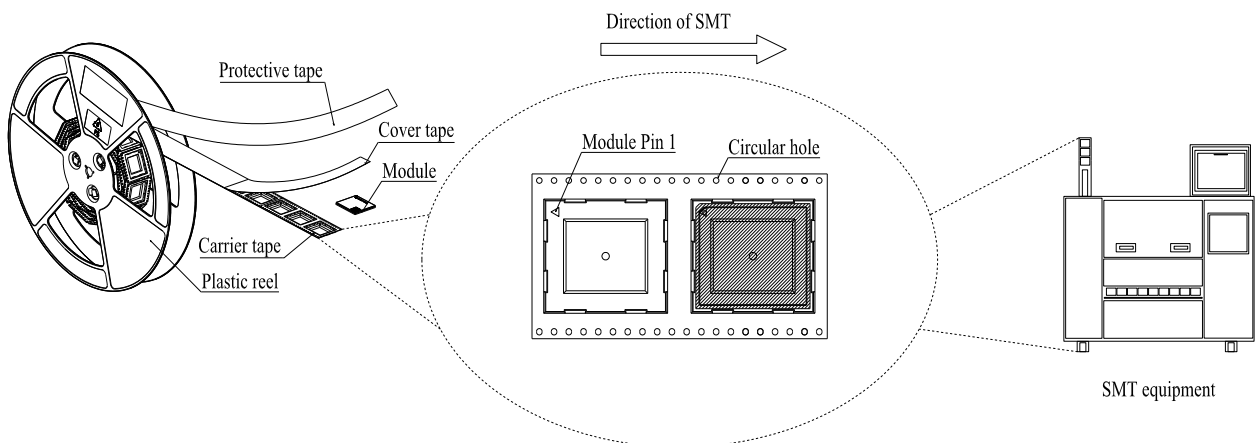
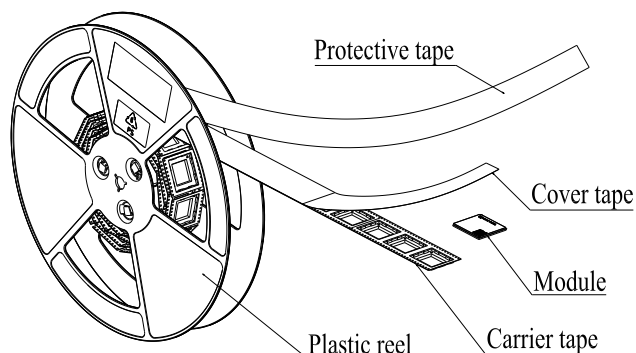


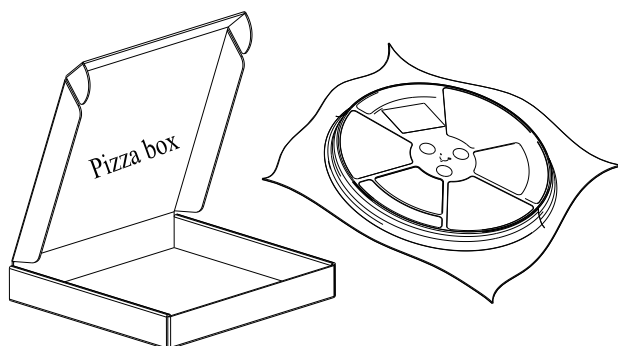
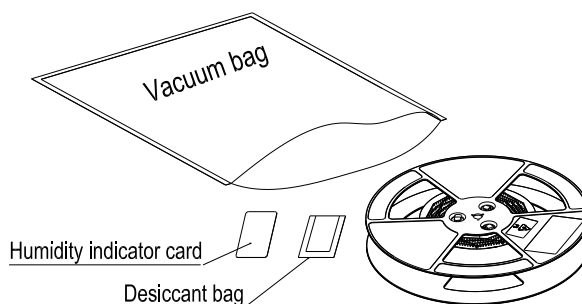
Figure 51: Mounting Direction

8.3.4. Packaging Process



Place the modules onto the carrier tape cavity and cover them securely with cover tape. Wind the heat-sealed carrier tape onto a plastic reel and apply a protective tape for additional protection. 1 plastic reel can pack 250 modules.

Place the packaged plastic reel, humidity indicator card and desiccant bag into a vacuum bag, and vacuumize it.



Place the vacuum-packed plastic reel into a pizza box.

Place the 4 packaged pizza boxes into 1 carton and seal it. 1 carton can pack 1000 modules.

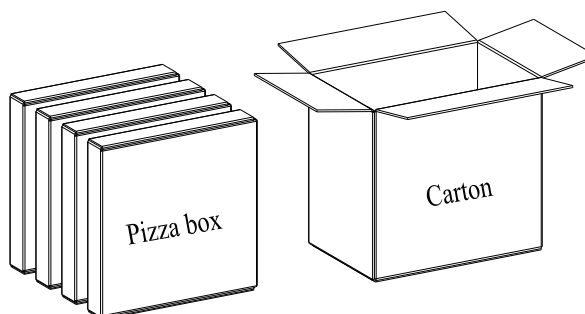


Figure 52: Packaging Process

9 Appendix References

Table 58: Related Documents

Document Name
[1] Quectel_ECx00x&EG800K&EG810M&EG915N&EG950A_Series_GNSS_Application_Note
[2] Quectel_UMTS<E_EVB_User_Guide
[3] Quectel_ECx00x&EG800K&EG810M&EG91xN&EG915K&EG950A_Series_AT_Commands_Manual
[4] Quectel_RF_Layout_Application_Note
[5] Quectel_Module_Stencil_Design_Requirements
[6] Quectel_Module_SMT_Application_Note

Table 59: Terms and Abbreviations

Abbreviation	Description
AMR	Adaptive Multi-Rate
APT	Average Power Tracking
BDS	BeiDou Navigation Satellite System
bps	Bytes per second
CDMA	Code Division Multiple Access
C _j	junction capacitance
CS	Coding Scheme
CTS	Clear To Send
DRX	Discontinuous Reception
DTE	Data Terminal Equipment

EFR	Enhanced Full Rate
EGSM	Enhanced GSM
ESD	Electrostatic Discharge
EVB	Evaluation Board
FDD	Frequency Division Duplexing
FR	Full Rate
FTP	File Transfer Protocol
FTPS	FTP over SSL
GMSK	Gaussian Filtered Minimum Shift Keying
GND	Ground
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GSM	Global System for Mobile Communications
HR	Half Rate
HSDPA	High Speed Downlink Packet Access
HTTPS	Hypertext Transfer Protocol Secure
LGA	Land Grid Array
LTE	Long Term Evolution
MCS	Modulation and Coding Scheme
MMS	Multimedia Messaging Service
NTP	Network Time Protocol
PAP	Password Authentication Protocol
PCB	Printed Circuit Board
PCM	Pulse Code Modulation
PHY	Physical Layer Transceiver

PING	Packet Internet Groper
PPP	Point-to-Point Protocol
PSK	Phase Shift Keying
QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Shift Keying
RF	Radio Frequency
RoHS	Restriction of Hazardous Substances
RTS	Request To Send
SDIO	Secure Digital Input and Output Card
SLIC	Subscriber Line Interface Circuit
SMS	Short Message Service
SMTP	Simple Mail Transfer Protocol
SMTPS	Simple Mail Transfer Protocol Secure
SSL	Secure Sockets Layer
TCP	Transmission Control Protocol
TDD	Time Division Duplexing
UDP	User Datagram Protocol
UMTS	Universal Mobile Telecommunications System
USB	Universal Serial Bus
(U)SIM	(Universal) Subscriber Identity Module
V _{max}	Maximum Voltage Value
V _{nom}	Nominal Voltage Value
V _{min}	Minimum Voltage Value
V _{IHmax}	Maximum Input High Level Voltage Value
V _{IHmin}	Minimum Input High Level Voltage Value

V_{ILmax}	Maximum Input Low Level Voltage Value
V_{ILmin}	Minimum Input Low Level Voltage Value
V_{OHmin}	Minimum Output High Level Voltage Value
V_{OLmax}	Maximum Output Low Level Voltage Value
VSWR	Voltage Standing Wave Ratio
WCDMA	Wideband Code Division Multiple Access
